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Field Application Engineer

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1-PHASE FILTER DESIGN

Related Appnote: ANP015

Application Note

1-Phasen Netzfilter Design



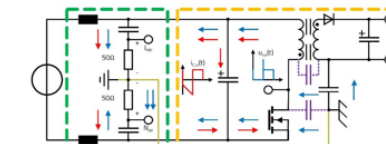
ANP015b // ANDREAS NADLER

1 Einleitung

Ziel dieser Appnote ist es, dem Leser so kompakt wie möglich einen umfassenden Überblick der notwendigen Schritte hin zum passend dimensionierten Netzfilter zu geben. Hierbei wird ein diskreter 1-Stufen mit einem diskreten 2-Stufen Netzfilter mittels Berechnung, Simulation und Messung verglichen. Im weiteren Verlauf werden die unterschiedlichen Kernmaterialien von Stromkompensierten Drosseln und deren Eigenschaften erläutert. Zudem widmet sich diese Appnote der Berechnung von: Varistoren, Leckströmen und Entladewiderständen. Diese Appnote setzt gewisse Grundkenntnisse von passiven Bauelementen, Filtern sowie EMV Messtechniken voraus.

2 Precompliance Messaufbau

Grundlegend ist zwischen zwei verschiedenen Störstrompfaden zu unterscheiden: Gleichtakt (Common Mode, CM) sowie Gegentakt (Differential Mode, DM). In einer EMV Abnahmemessung werden grundsätzlich beide Störstrompfade gleichzeitig gemessen. Um einen Netzfilter auszulegen ist es vorteilhaft im Vorfeld beide Strömpfade, CM und DM, messen zu können. Dazu wird eine LISN (Line Impedance Stabilization Network) benötigt, bei der die zwei Messausgänge gleichzeitig nutzbar sind. In der LISN sind zwei 50Ω Messwiderstände verbaut. In der DM Messung liegen diese in Reihe (100Ω), wohingegen sie in der CM Messung als parallel zu betrachten sind (25Ω). Das Blockschaltbild in Abbildung 1 zeigt die DM- und CM-Störstrompfade zwischen einem Sperrwandler (Störquelle) und der LISN



$$U_{DM} = \frac{U_{L,RF} - U_{N,RF}}{2}$$

$$U_{CM} = \frac{U_{L,RF} + U_{N,RF}}{2}$$

Für die Precompliance Messung wird ein Rohde&Schwarz RTA4004 mit 500MHz analoger Bandbreite in Kombination mit der Desktop Software R&S EMI Debug Tool und einer CISPR16 LISN (Eigenbau) verwendet (Aufbau in Abbildung 2).

Zu beachten ist, dass die maximale vertikale Auflösung der Messeingänge des Oszilloskops optimal ausgenutzt wird, der Amplitudenmessbereich aber auch nicht übersteuert wird. Hat die erfasste Störspannung im Zeitbereich z.B. einen Peak to Peak Pegel von 85mV, so sollte die vertikale Einstellung auf einen Endwert von 100mV gesetzt werden, um die Empfindlichkeit des Oszilloskops maximal zu nutzen (Abbildung 3).

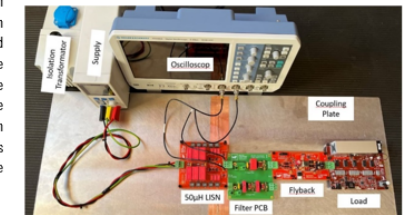
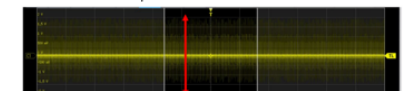


Abb. 2: Precompliance Messaufbau mit Koppelplatte, LISN, DUT, Trenntrafo & Oszilloskop



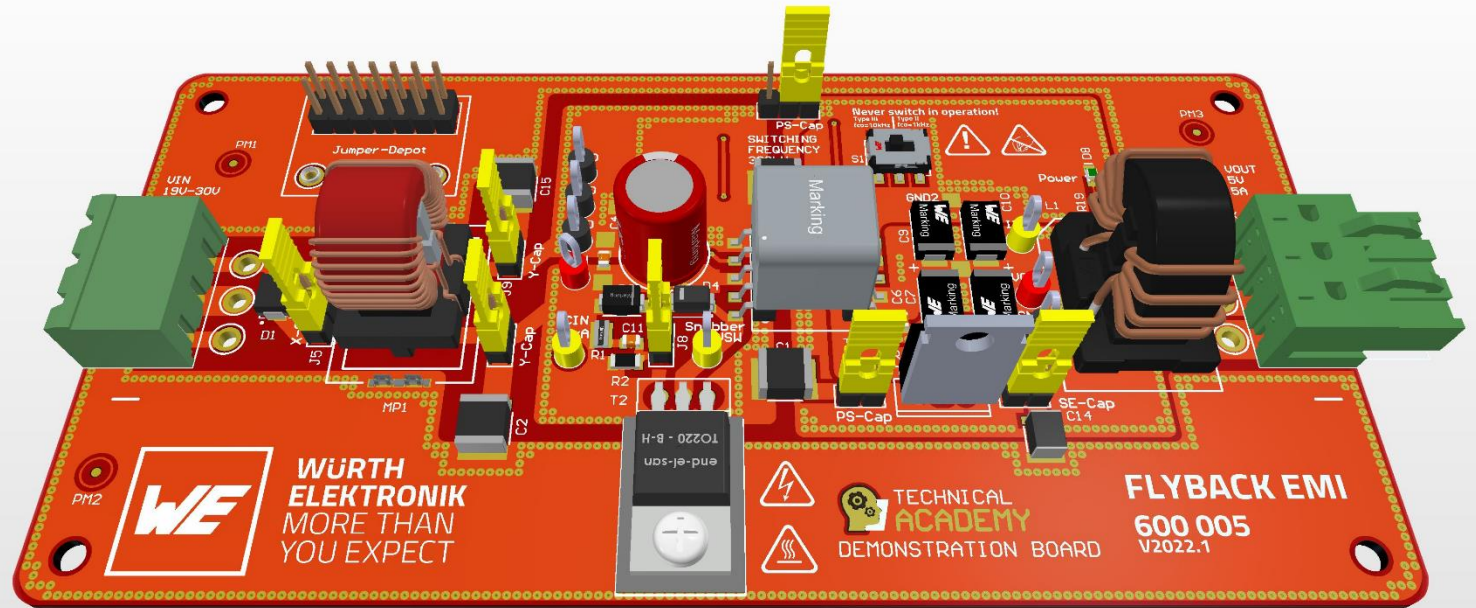
AGENDA

- Sources of interference
- Components for filtering
- 1-stage filter design
- 2-stage filter design
- Calculation and simulation
- Measurements of interference suppression
- Y-Cap placement and PE connection style
- Varistor & TVSP calculation
- Bleed/Discharge resistor for X-Caps



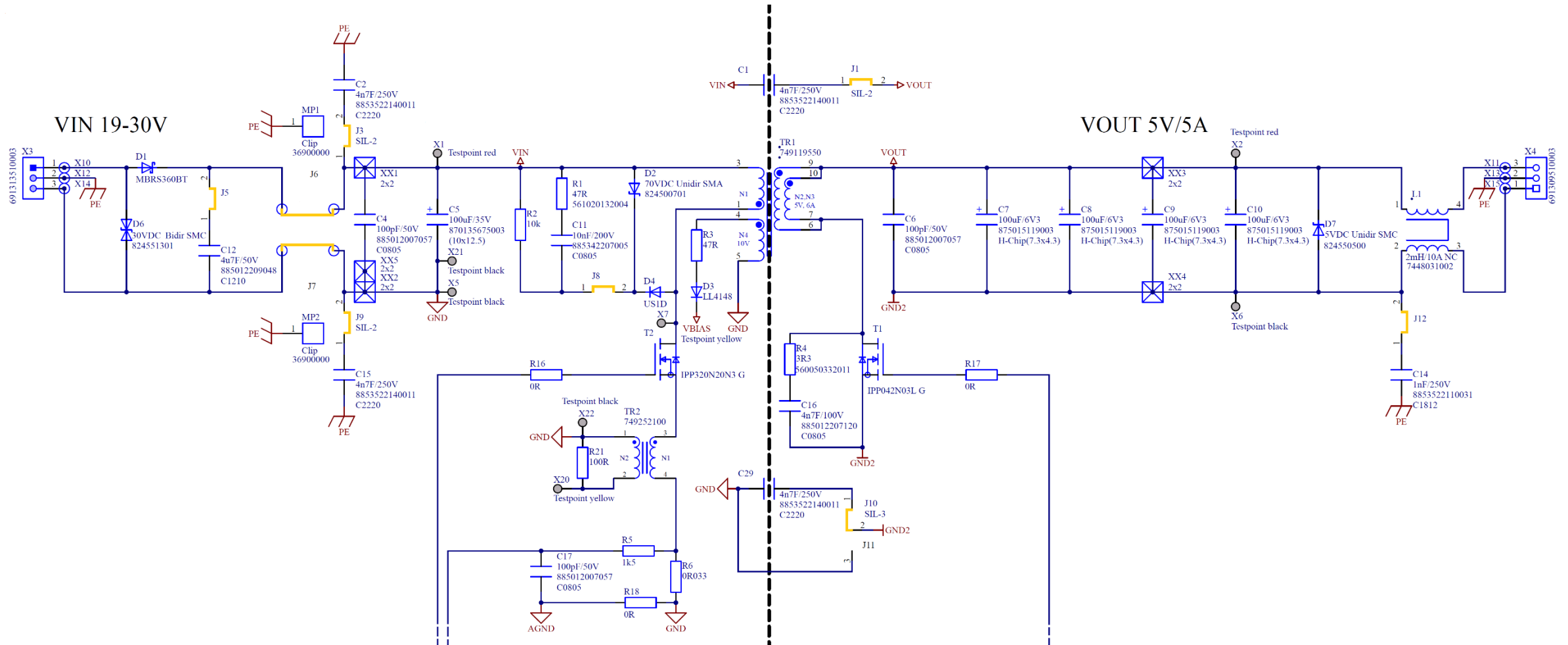
Technical specification noise source

- DC/DC Flyback-Converter CCM (Forced Continuous Conduction Mode)
 - $U_{in} = 24V$ (19-30V)
 - $U_{out} = 5V$
 - $I_{out,max} = 5A$ (25W)
 - $f_{sw} \approx 300kHz$
 - Efficiency $\approx 90\%$
- IC: ADP1071-2 (Analog Devices)
 - with synchronous rectifier
- Transformer: 749119550
- MOSFETs in TO220-package



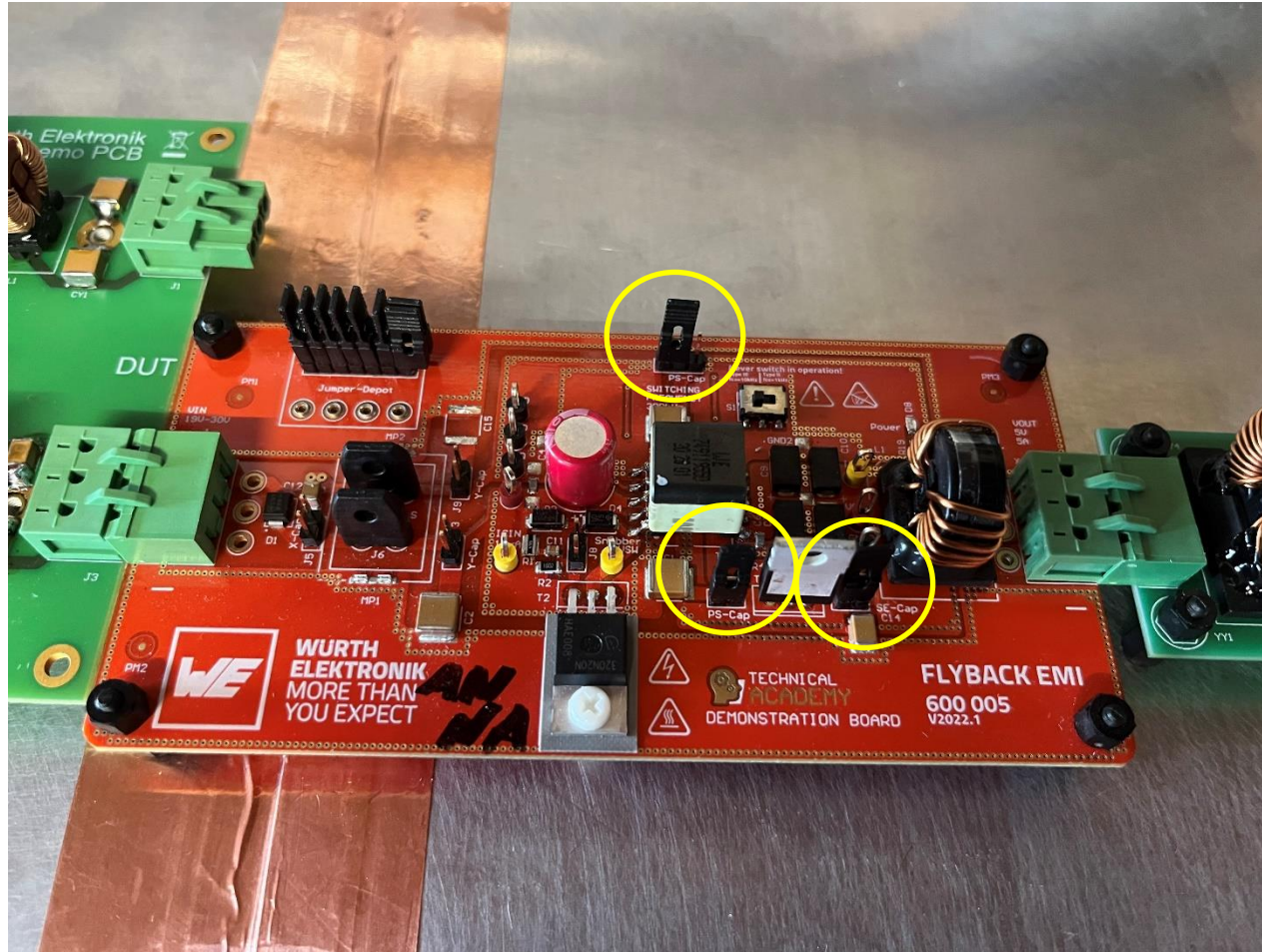
Schematic Noise Source

Isolated flyback converter 300kHz fsw



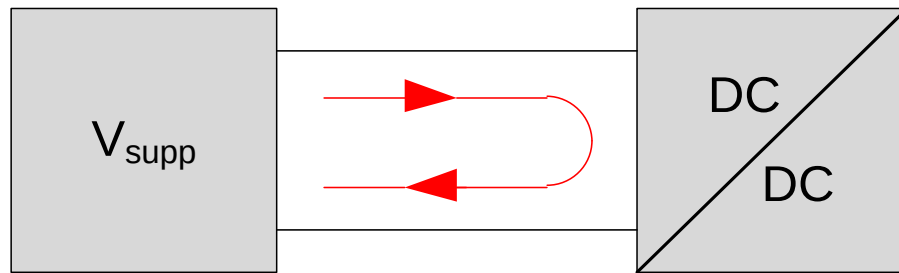
Setup

Jumper setting flyback noise source → only activate the PRI to SEC caps & SEC to PE

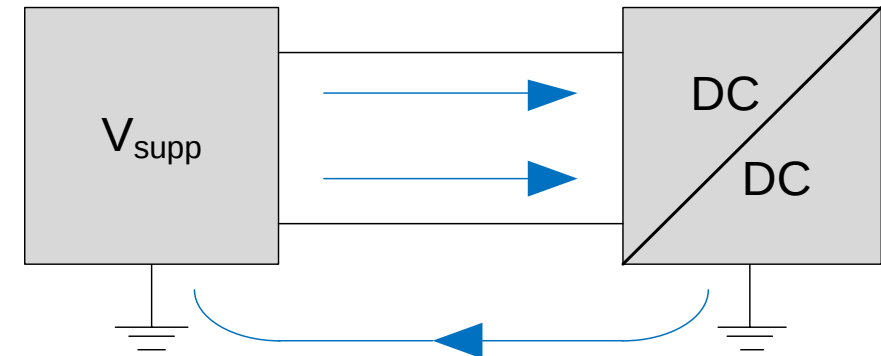


Differential Mode & Common Mode

Differential Mode



Common Mode



CM & DM comparisson

- **Differential mode currents**

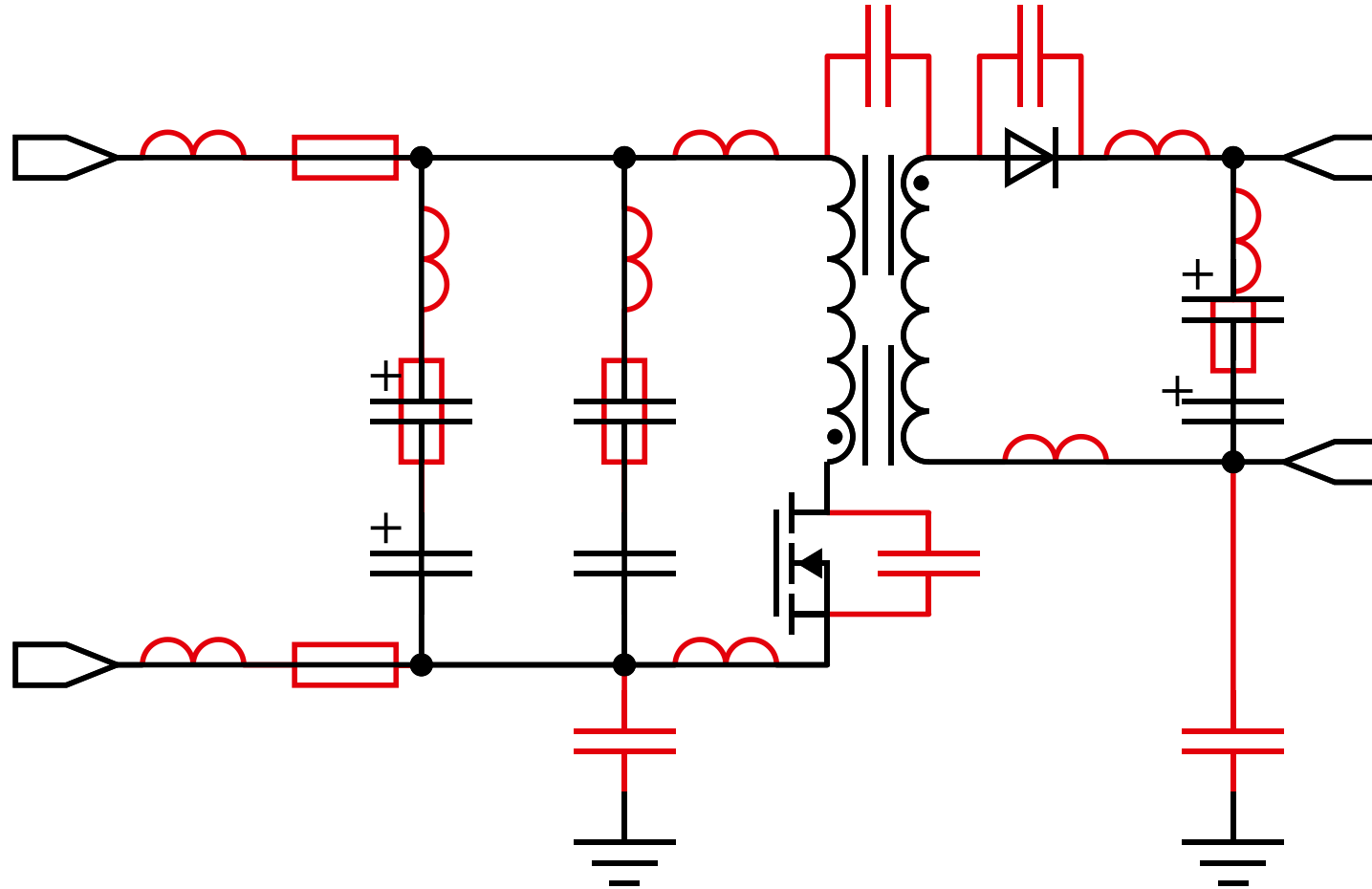
- Current path as in circuit diagram
- Easy to follow paths
- Return current path very close
- Relatively large currents
- Filtering with LC , π , T topologies
- dI / dt is dominant cause
- Conducted EMI problem

- **Common-mode currents**

- Unexpected current path
- Current flows via parasitic paths
- Return current path very large
- Relatively small currents (μA)
- Filtering with CMC and Y-caps
- dU / dt is dominant cause
- Radiated & conducted EMI problem

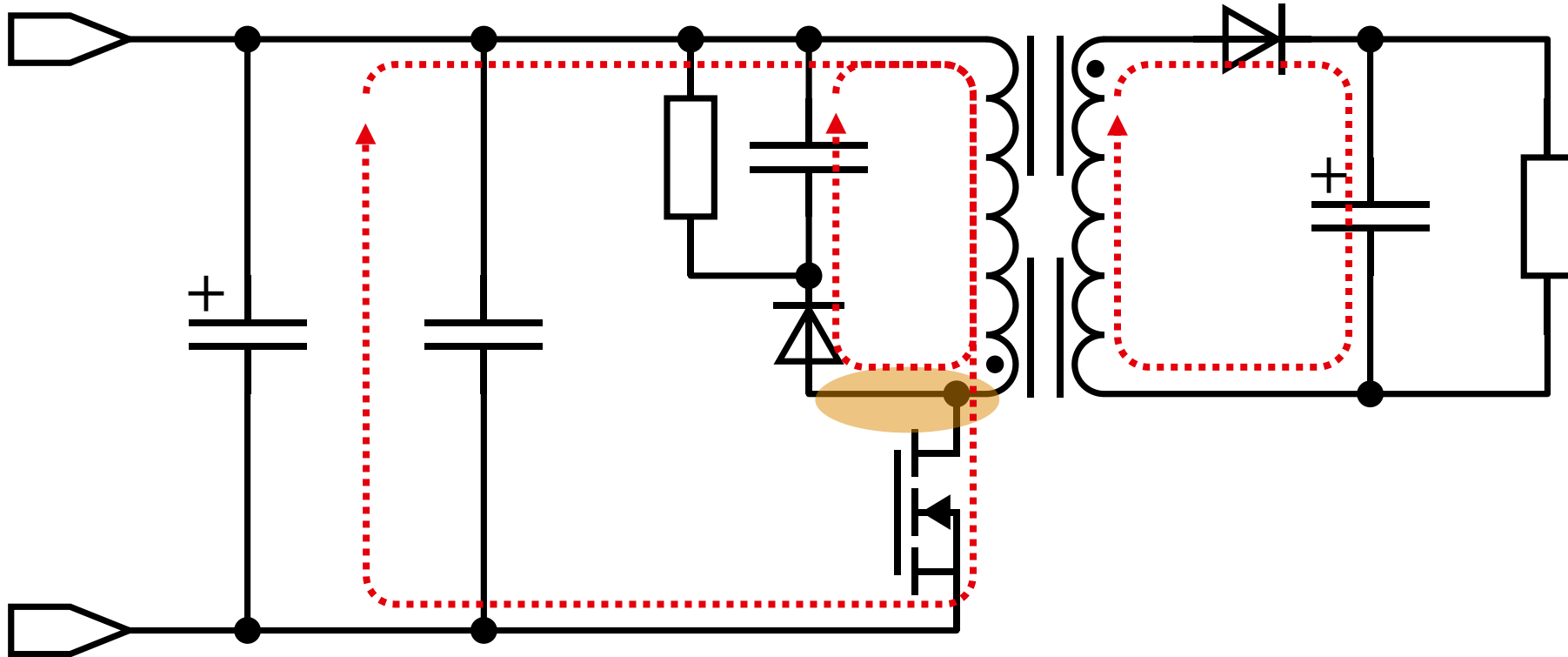
Introduction - Flyback Converter

Parasitic Model



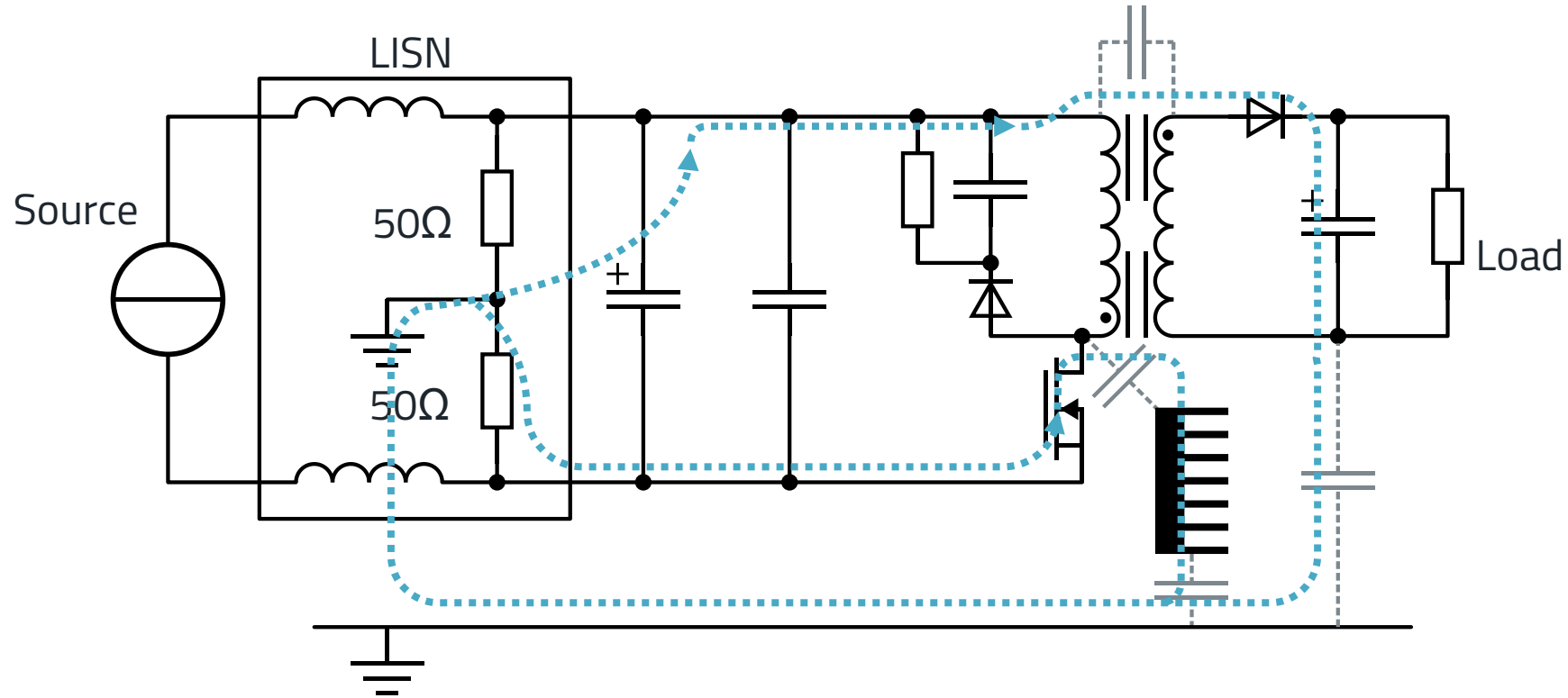
Introduction - Flyback Converter

Hot Node and Critical Loops



Introduction - Capacitive Coupling in a Flyback Converter

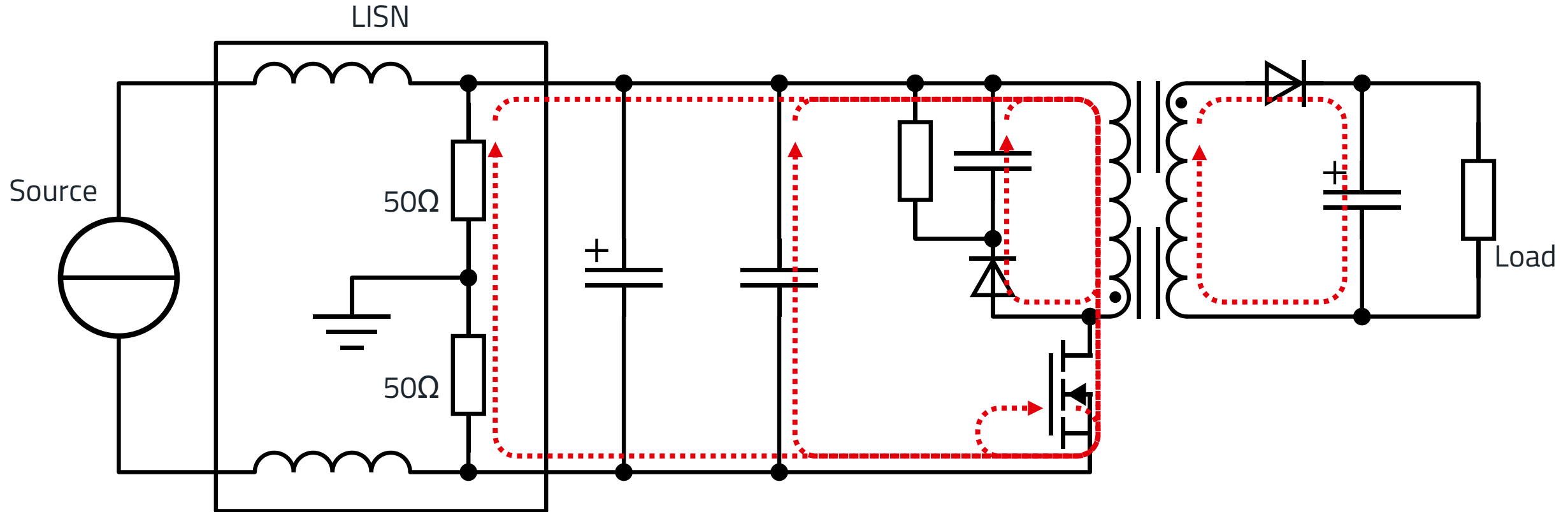
Common mode current paths



High du/dt common mode currents through parasitic capacitances
(electric dipole and monopole antennas)

Introduction - Inductive Coupling in a Flyback Converter

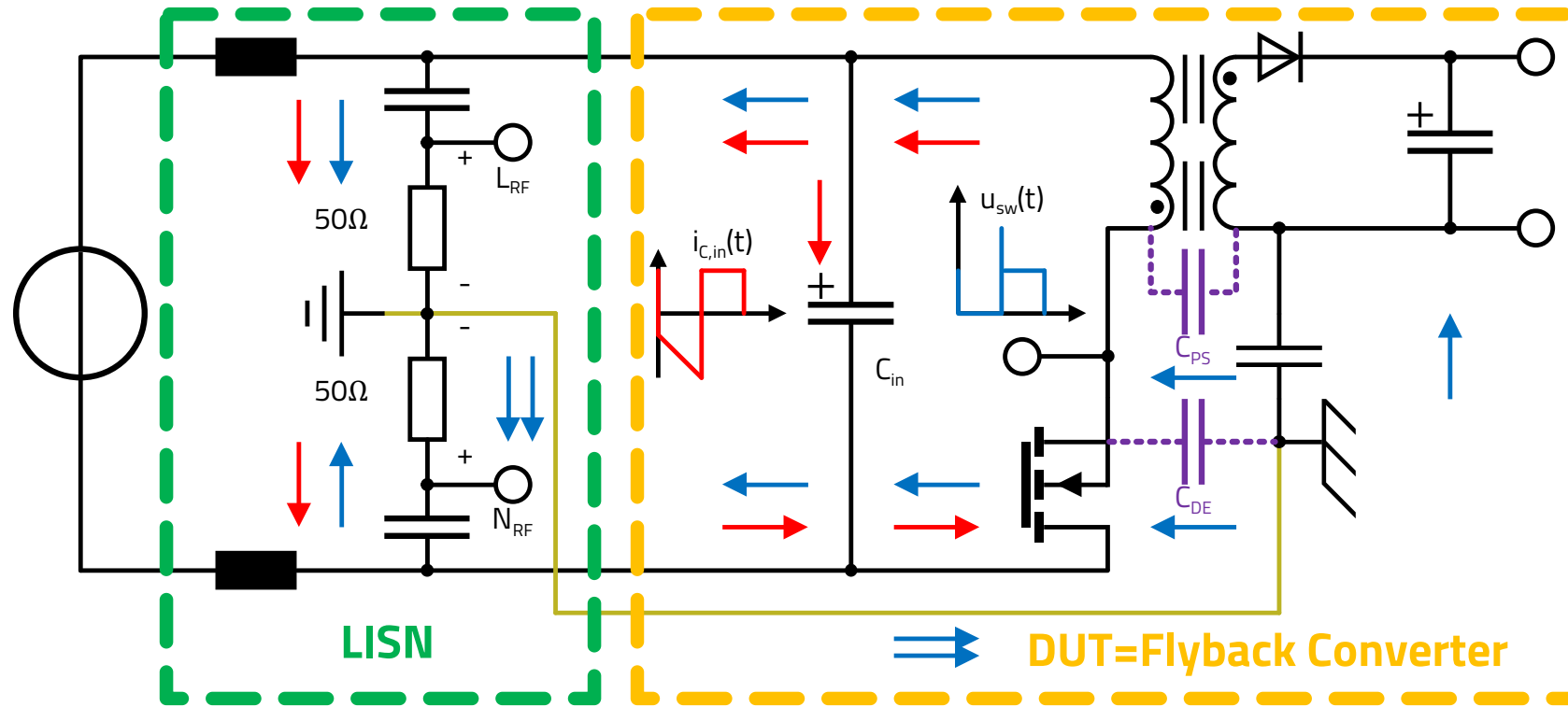
Differential mode current paths



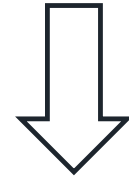
Inductive coupling caused by high di/dt differential mode currents
(magnetic loop antennas)

Test#3: Background - Noise categories

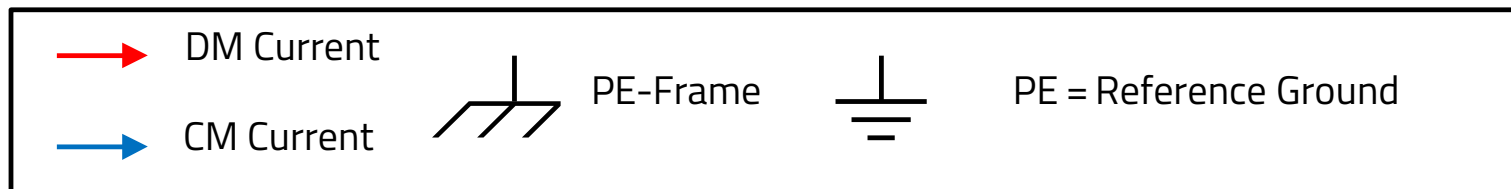
Theory: DM and CM noise path in a flyback converter



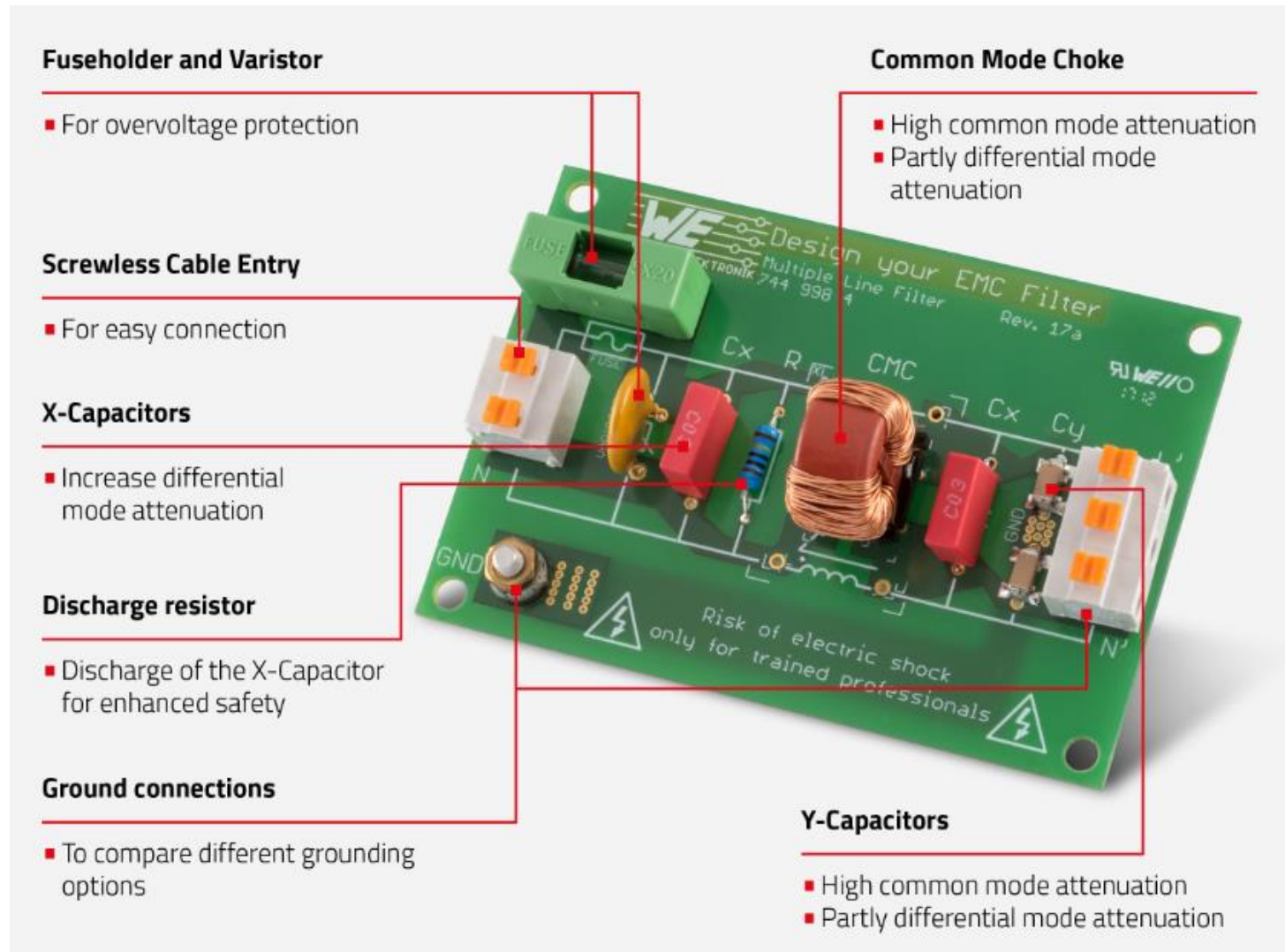
$$U_{L,RF} = U_{CM} + U_{DM}$$
$$U_{N,RF} = U_{CM} - U_{DM}$$



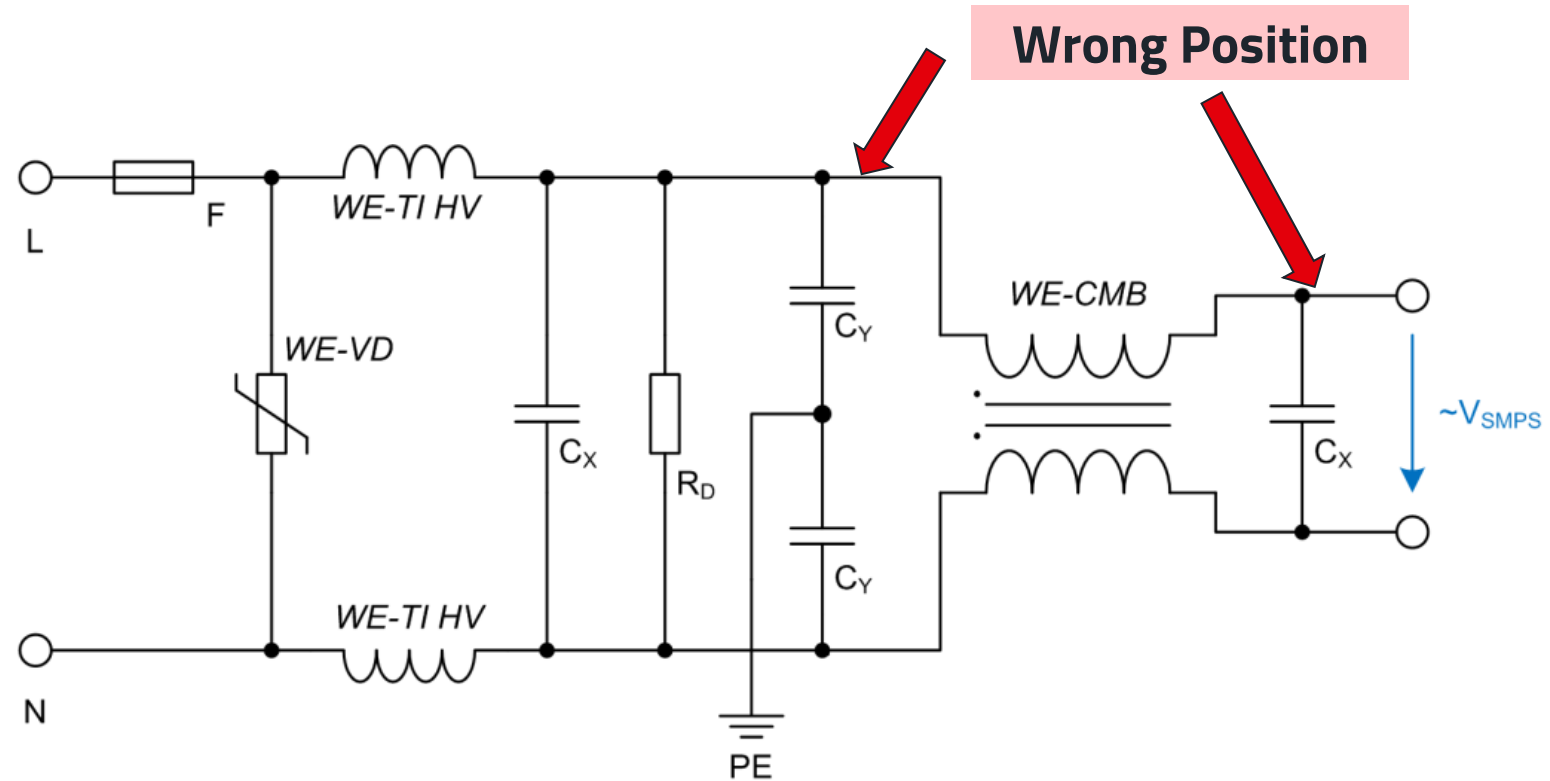
$$U_{DM} = \frac{U_{L,RF} - U_{N,RF}}{2}$$
$$U_{CM} = \frac{U_{L,RF} + U_{N,RF}}{2}$$



Filter PCB Example

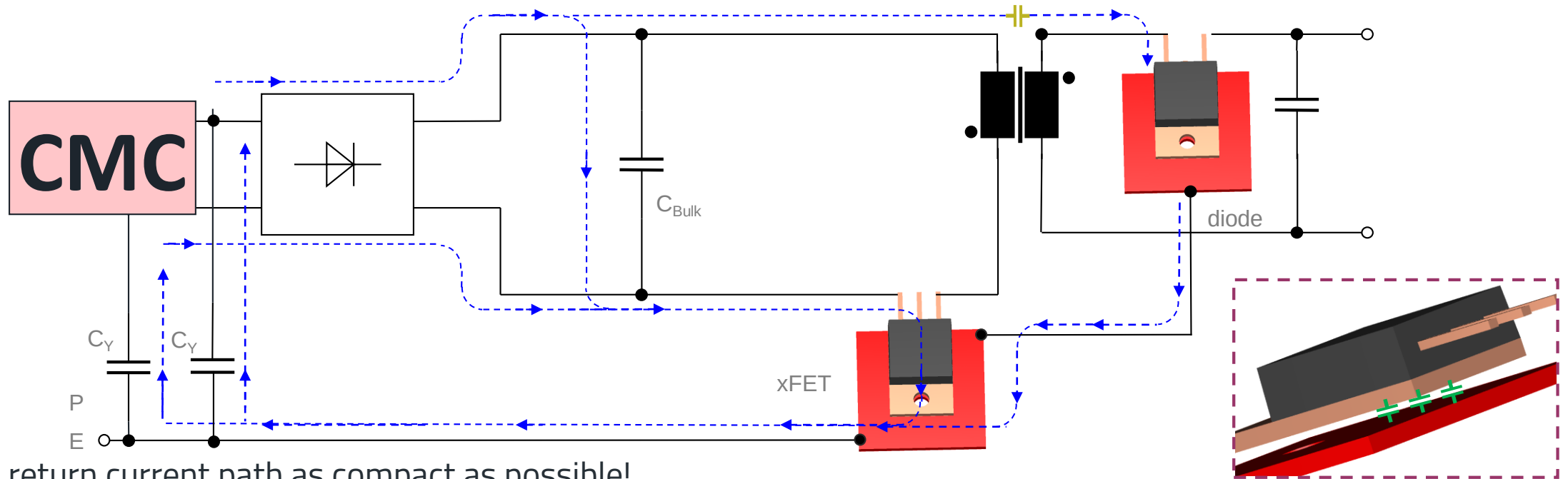


Position of the Y-caps



Differential Mode: $C_X + L_{Leak/CMB} / C_X + L_{WE-TI-HV}$
Common Mode: $C_Y + L_{WE-CMB} (C_{Y_Sum})$

Richtige Position der Y-Caps

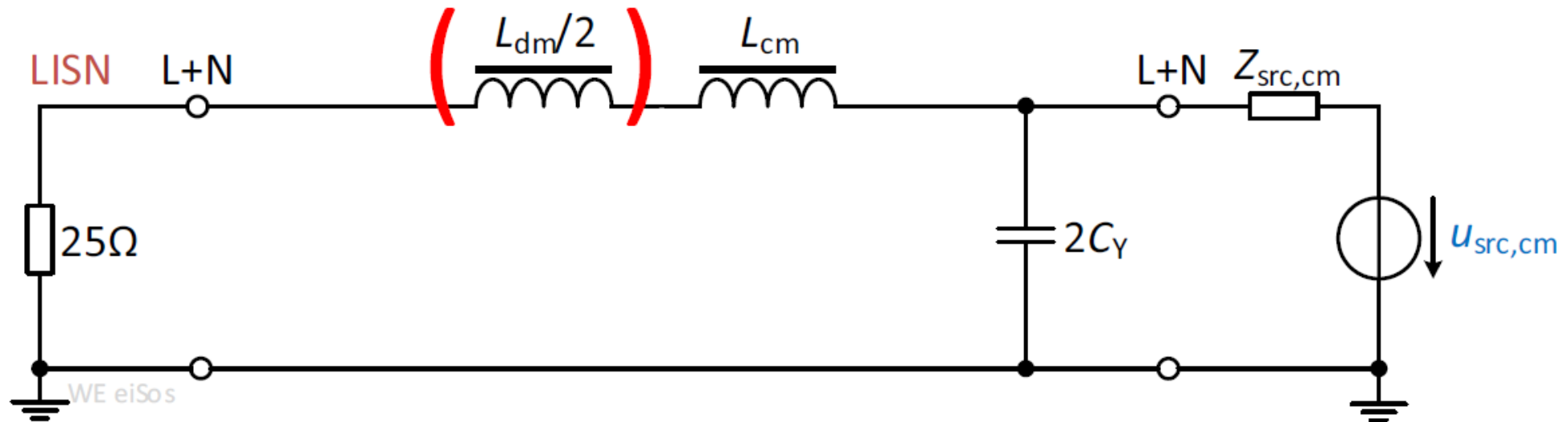


- Keep return current path as compact as possible!
- Because: DUT behaves like a high impedance current source and the LISN is a (25ohm) low impedance sink(valid for conducted emission).

Filter

CM Consideration

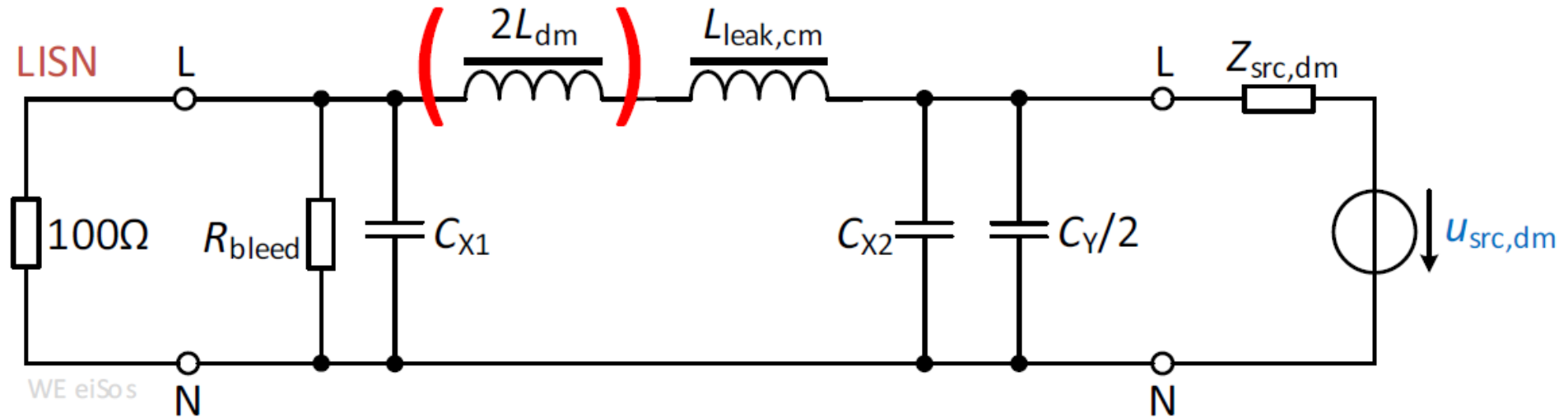
- Equivalent circuit diagram of the common mode filter, the CM source and the load
- The CM filter is considered WITHOUT the leakage inductance



Filter

DM-Consideration

- Equivalent circuit diagram of the differential mode filter, the DM source and the load



Common mode chokes

Core materials

- If the CM insertion loss of the calculated 1-stage line filter with MnZn CMC core is not sufficient for the higher frequency range (from approx. 10MHz - 100MHz), a second CMC with NiZn or nanocrystalline core must also be used



- MnZn cores
- 10kHz to 10MHz



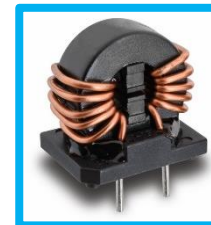
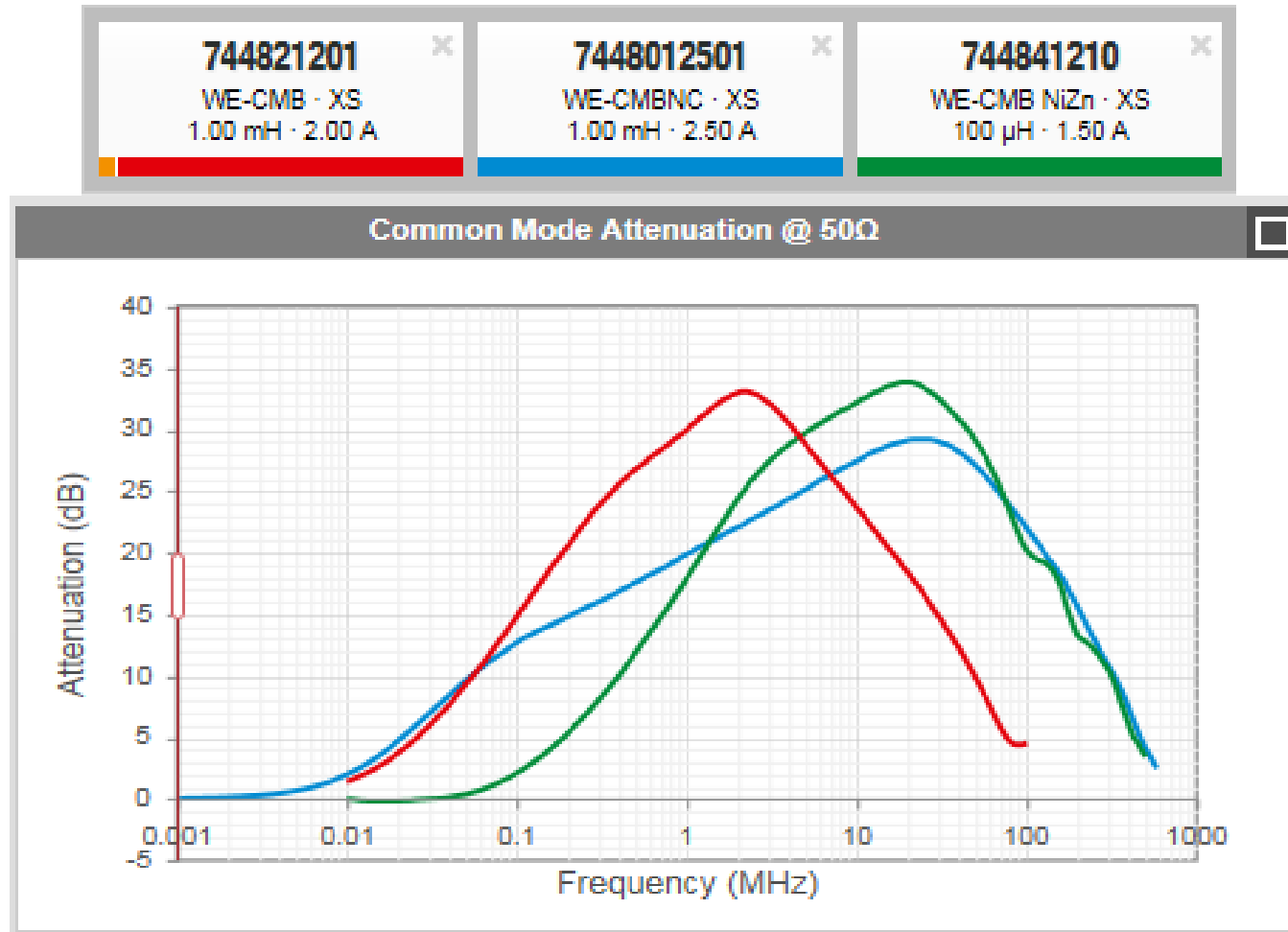
- NiZn cores
- 5MHz to 200MHz



- Nanocrystalline cores
- 1kHz to 300MHz

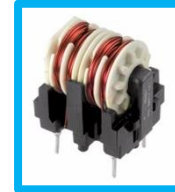
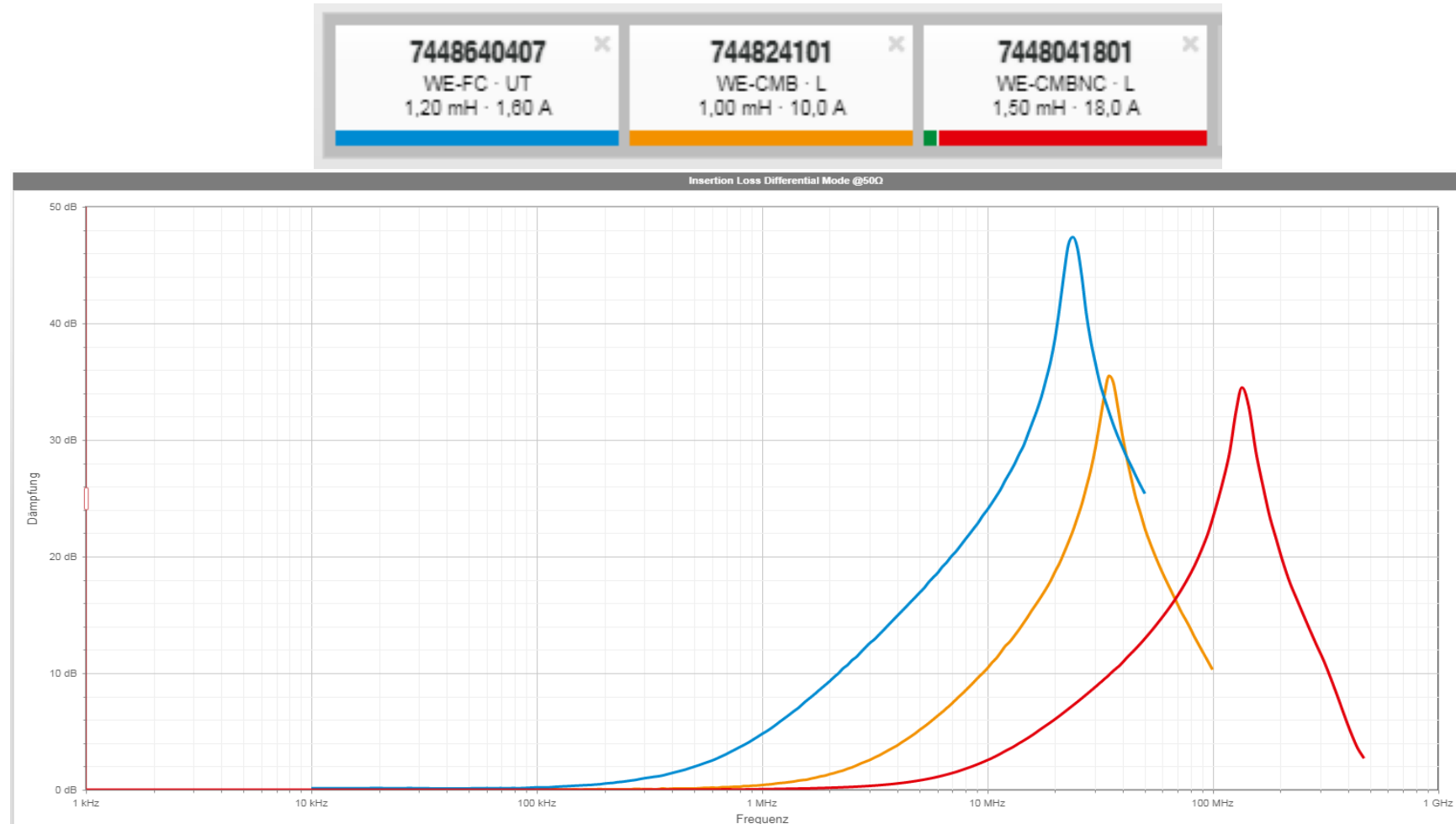
Core materials comparisson

MnZn, NC, NiZn



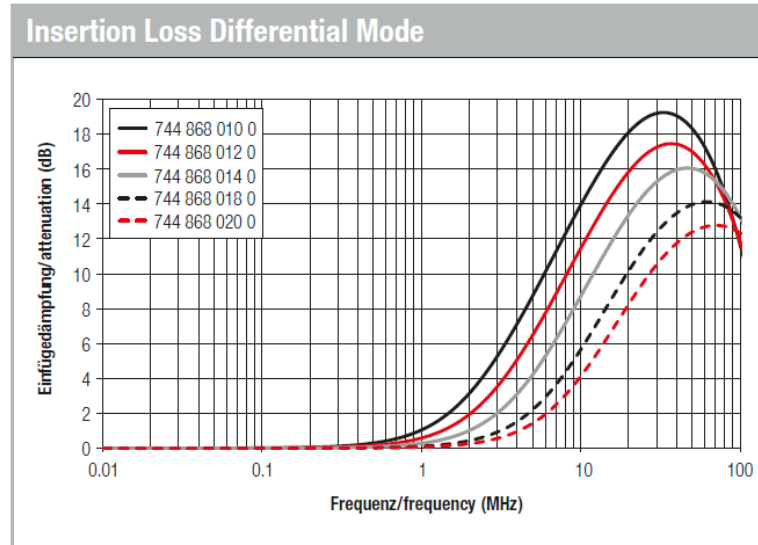
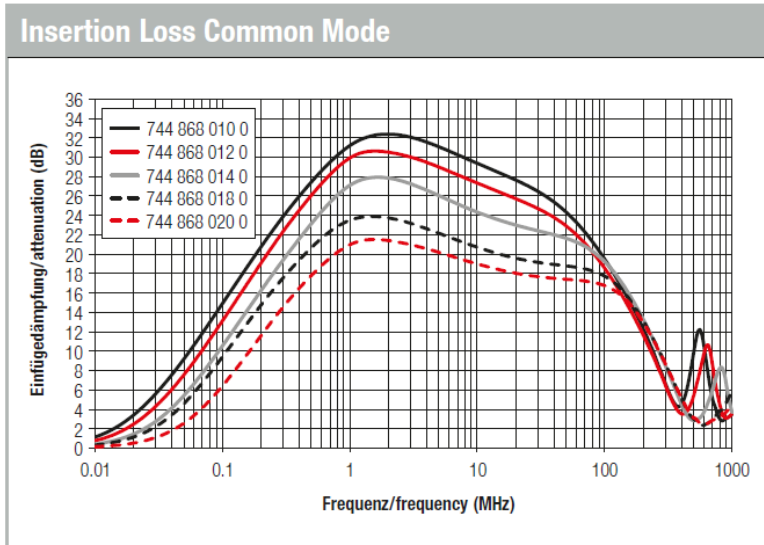
Leakage inductance

Is mainly determined by the geometry of the winding body

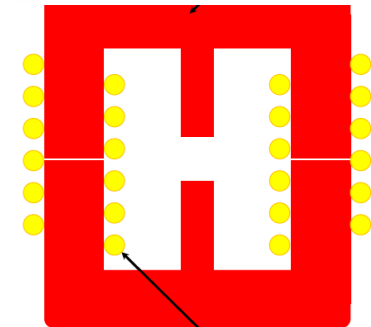


CMC with airgap

- Flat CMC with defined air gap WE-LPCC (up to 24A)
- Broadband attenuation from 100kHz to 300MHz
- Less maximum attenuation compared to cores without air gap



WE-LPCC

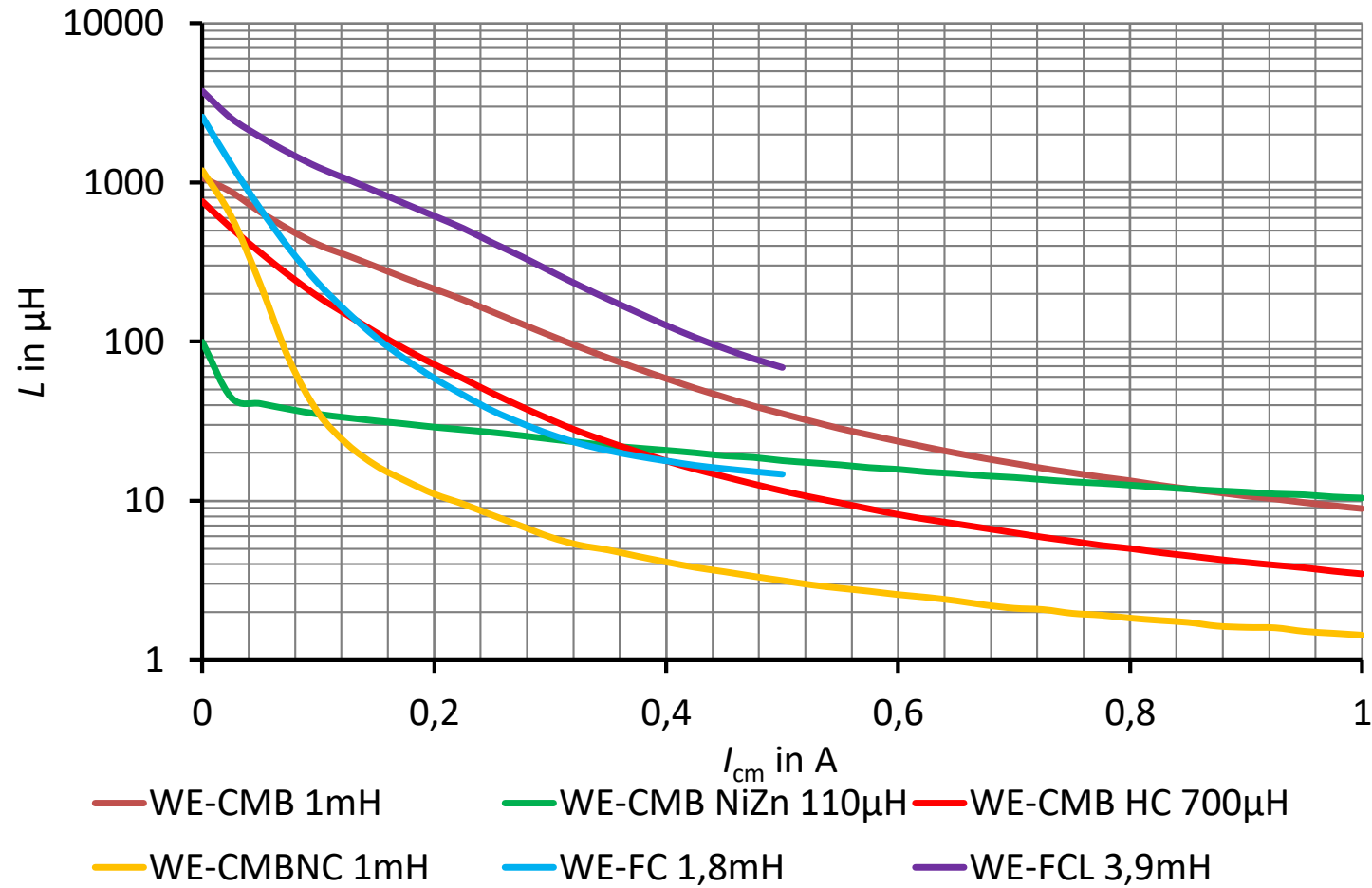


Enamelled Copper Wire

WE-LPCC

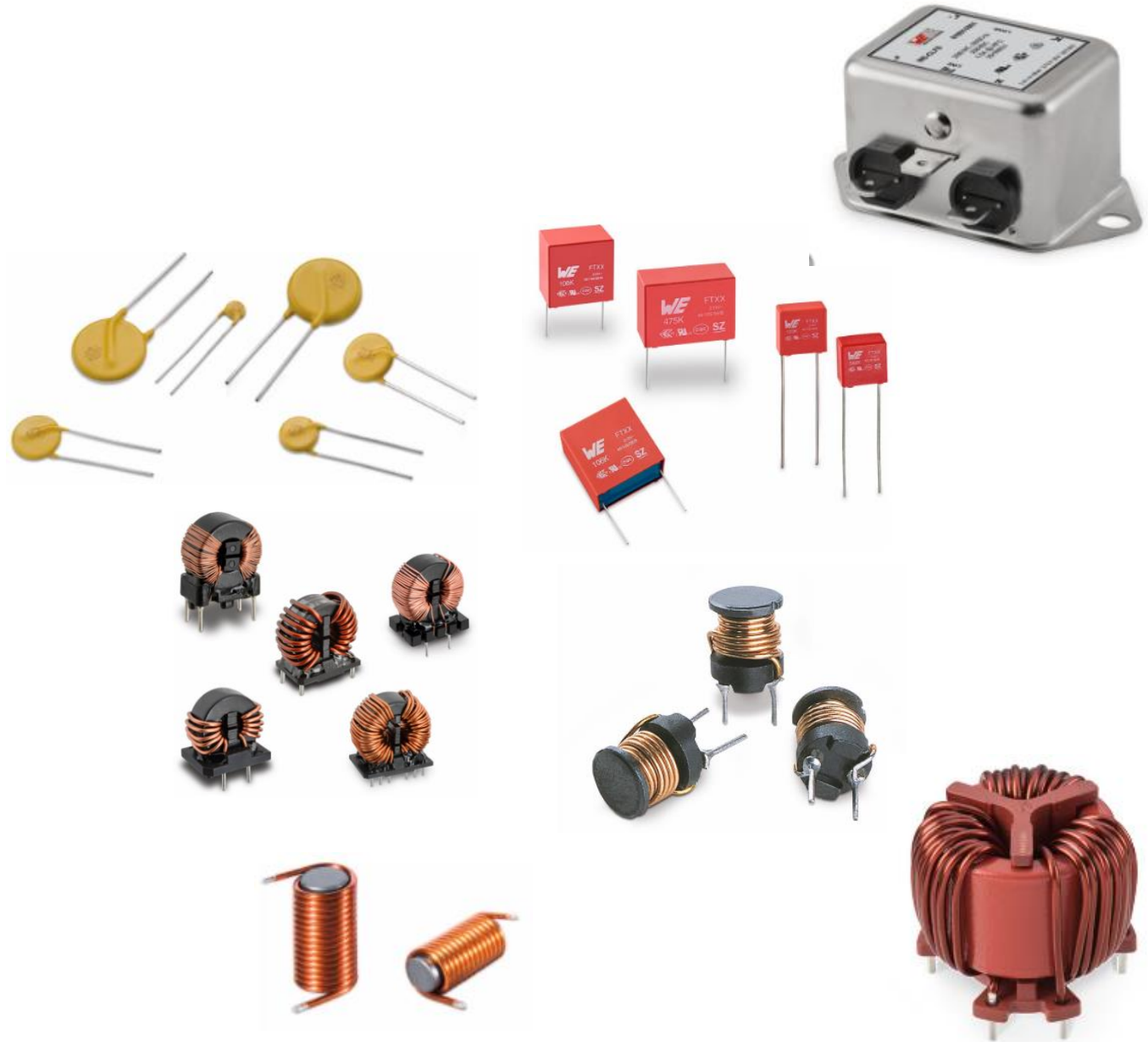
CM saturation effect

Rapid drop due to excessive CM current in the mA range



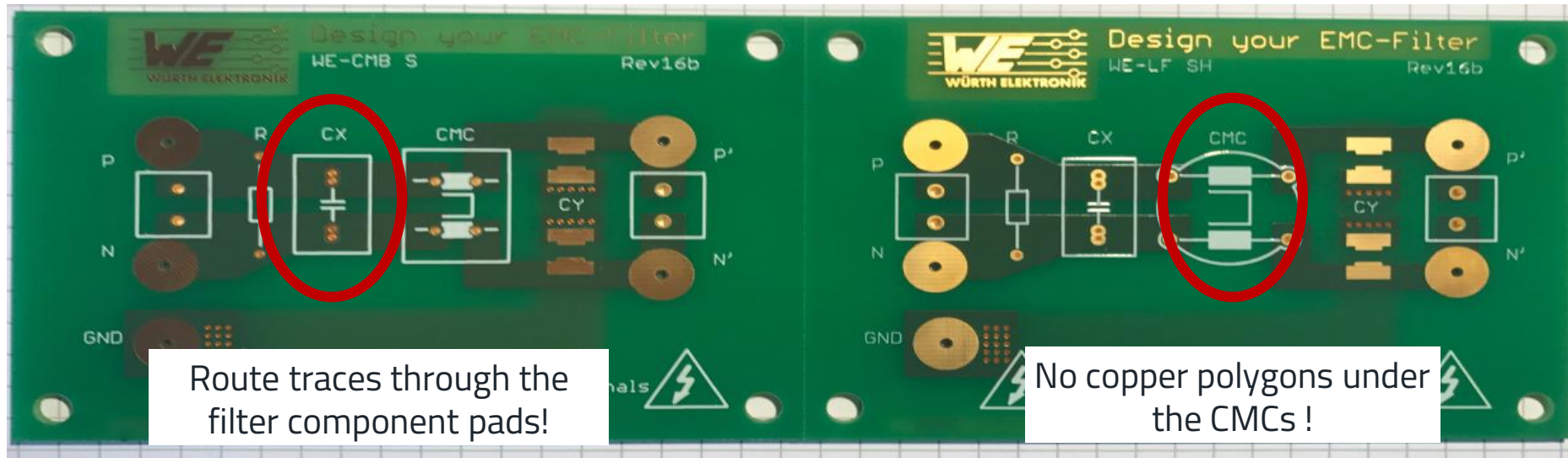
WE Portfolio

- Varistors min. 14mm! (WE-VD surge protection)
- X capacitors (WCAP-FTXX)
- Y capacitors (WCAP-CSSA & FTY2)
- Current-compensated chokes (WE-CMBx)
- Longitudinal filter chokes (WE-TIHV / WE-SD)
- Complete line filters (WE-CLFS)



Layout AC Filter

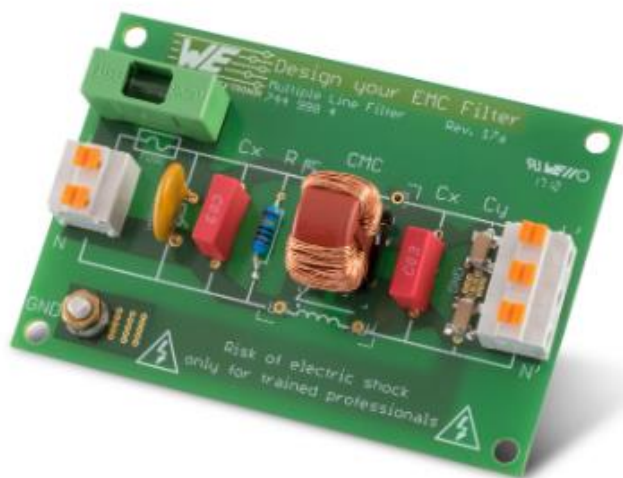
- Maintain at least **3mm** air and creepage distance between the outer conductors L & N up to 250VAC and at least **8mm** to PE (observe degree of contamination min. 2)
- The capacitors must be at least safety class X2 and Y2.
- Dimension all components according to the maximum (incl. tolerance) occurring mains voltage
- Arrange the interference suppression components and conductor tracks in such a way that their effect is not impaired by inductive and capacitive coupling (e.g. if two STROKOS are connected in series, either increase the distance or rotate one by 90°).
- Connect the Y-cap connection with low inductance



Ready to use Filte PCBs

AC & DC versions

Order Code 7449984



Order Code 744998



Filter Evaluation Boards



Evaluation Board Line Filter **NEW**



Evaluation Board Differential Filter **NEW**



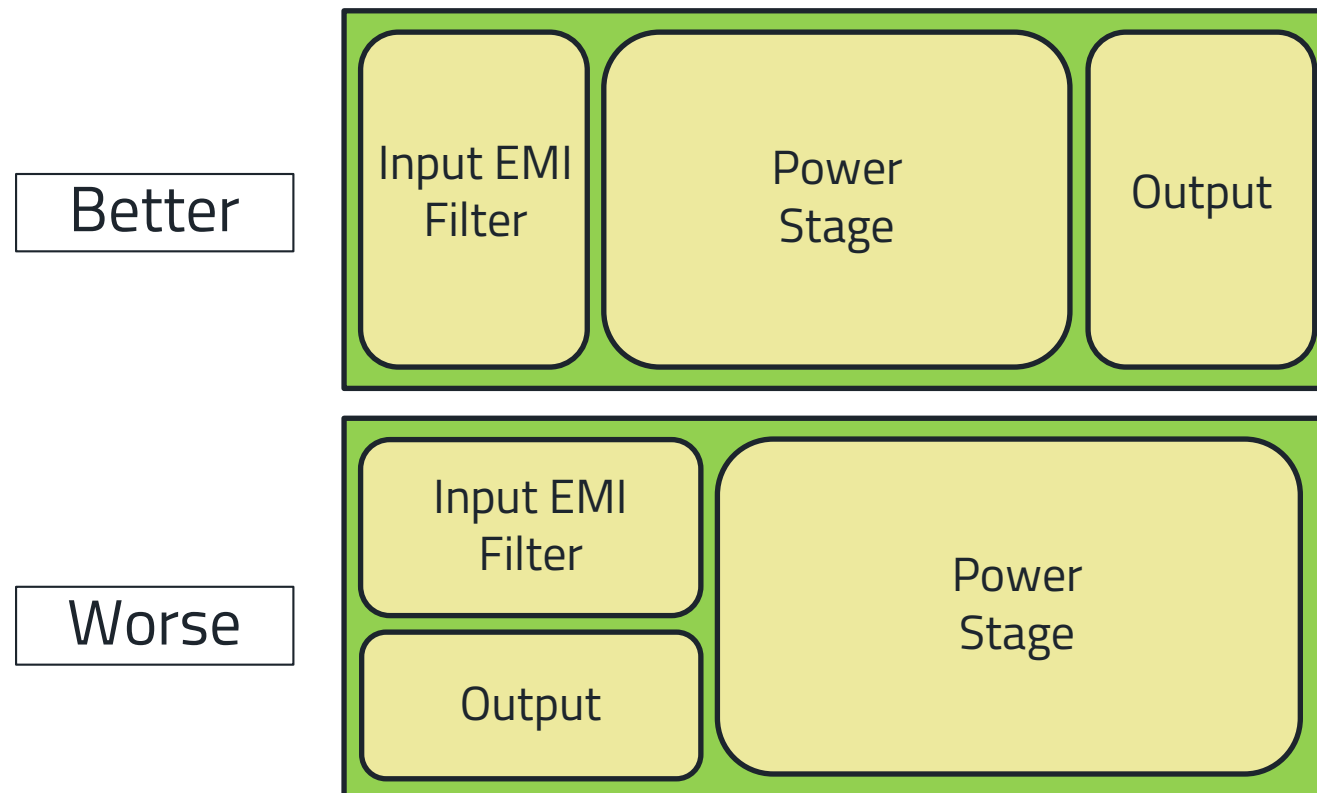
Evaluation Board DC 2stepCMC Filter **NEW**



Evaluation Board Multiple Line Filter **NEW**

Geometric arrangement of filters

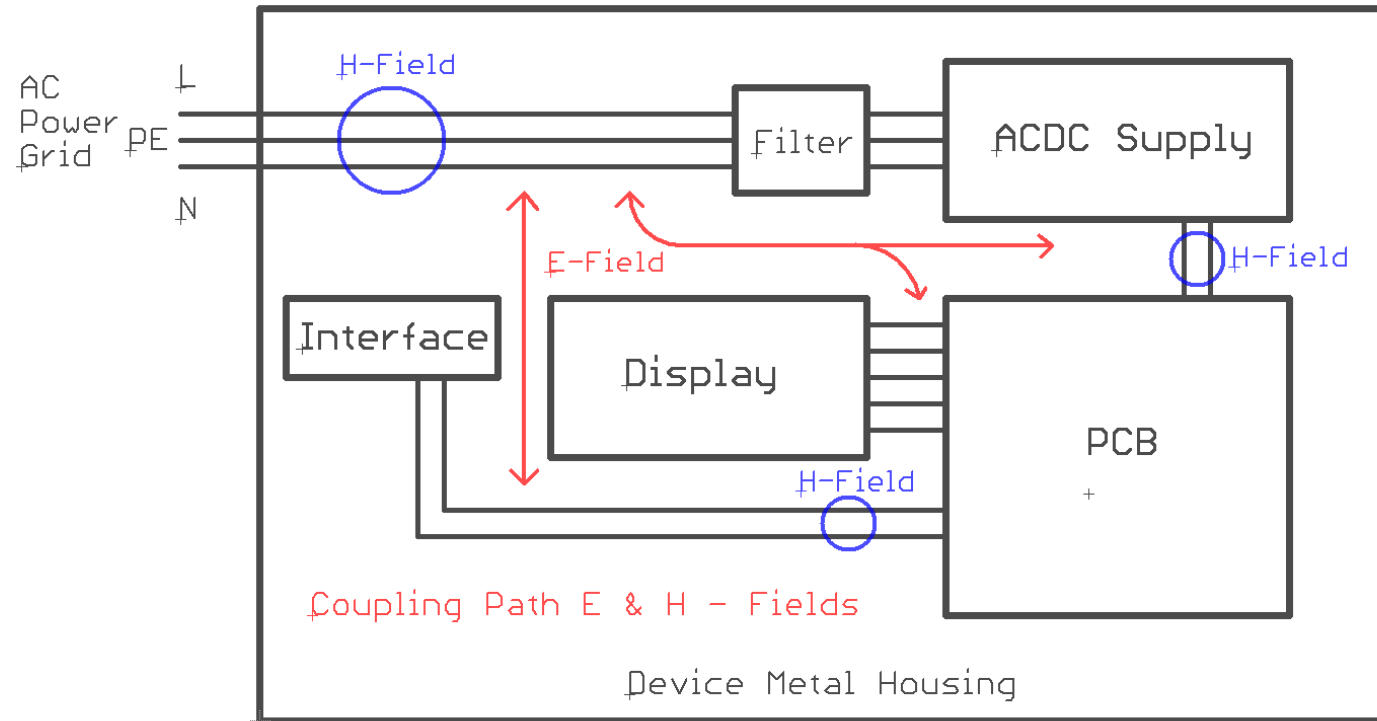
- Correct placement of the filters for maximum interference suppression effect is crucial
- Avoidance of parallel structures/coupling of I/Os



Device cabling & filter position

Bad example

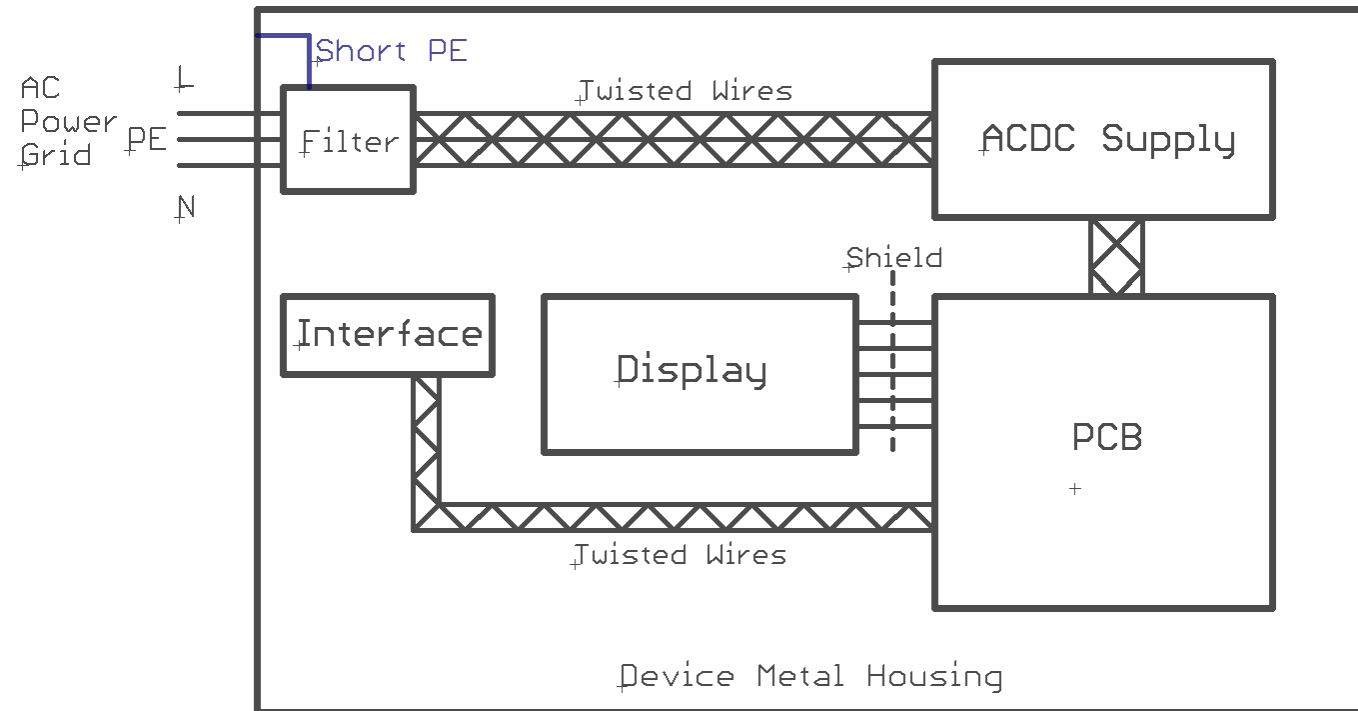
- Cable between filter and device plug is too long
- Susceptible to interference emission AND interference immunity
- Capacitive coupling (E-field) and inductive coupling (H-field)



Device cabling & filter position

Better example

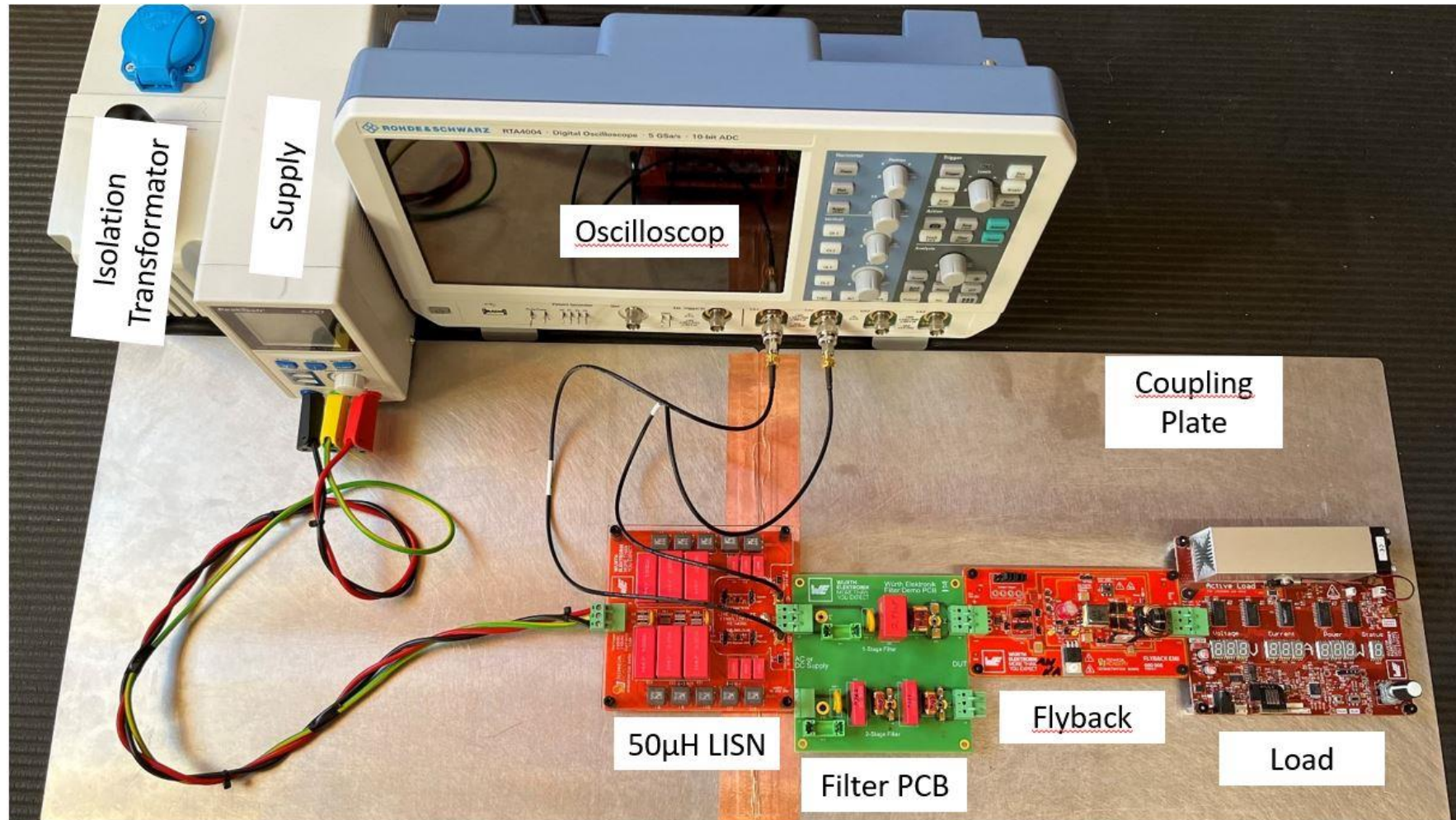
- Low-inductance and flat connection of the filter to the housing/PE
- Twisted cables (not only differential signals)
- Plan cable routing precisely
- Star-shaped connection



MATH

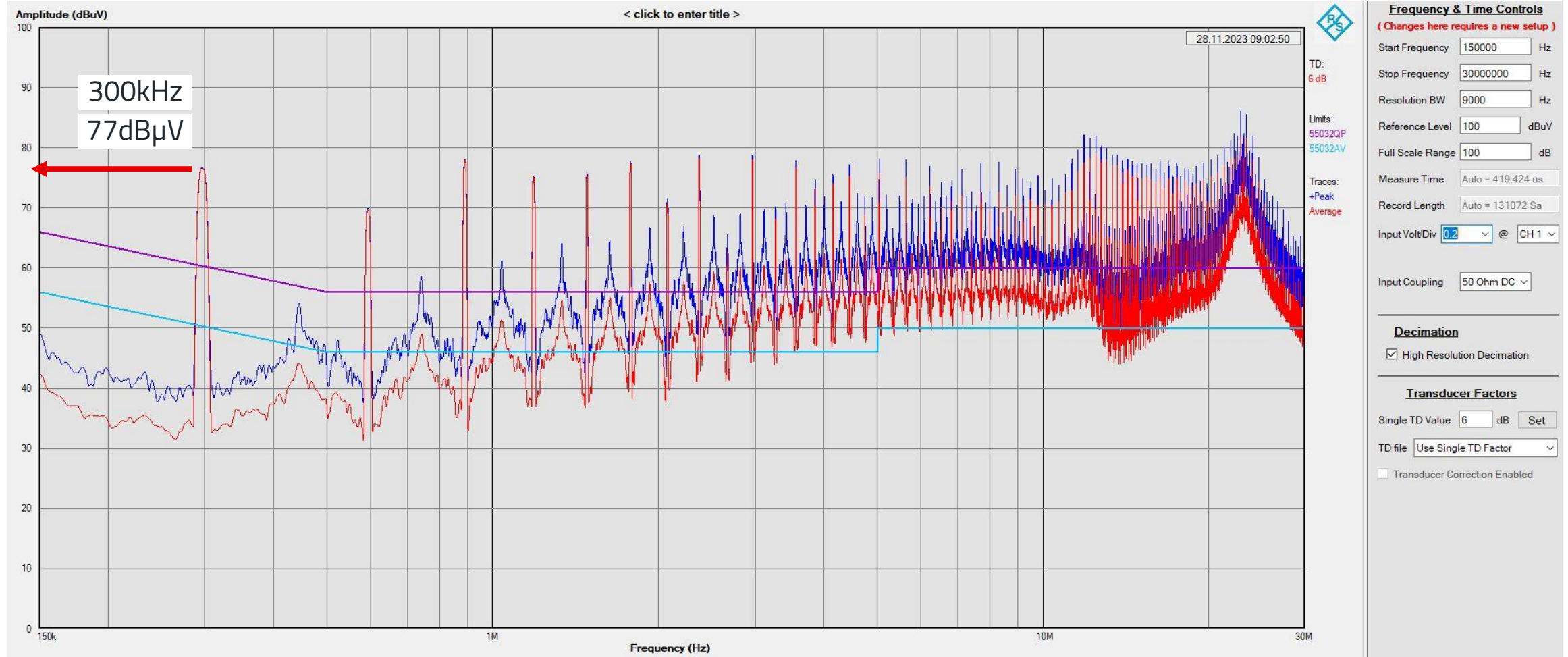
First: Measure DM and CM separate without filter

Mandatory: LISN with two RF outputs which can be used simultaneously



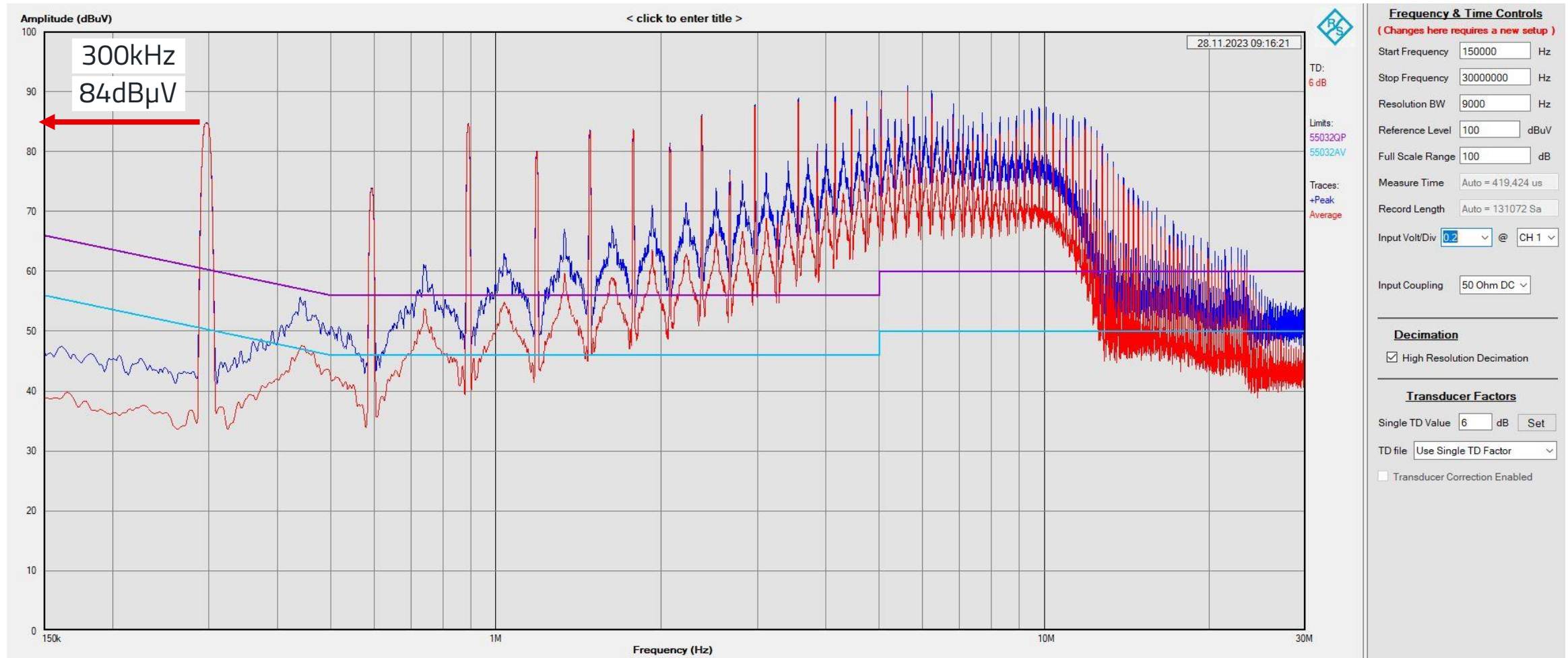
Conducted Spectrum 50uH LISN

DM no filter



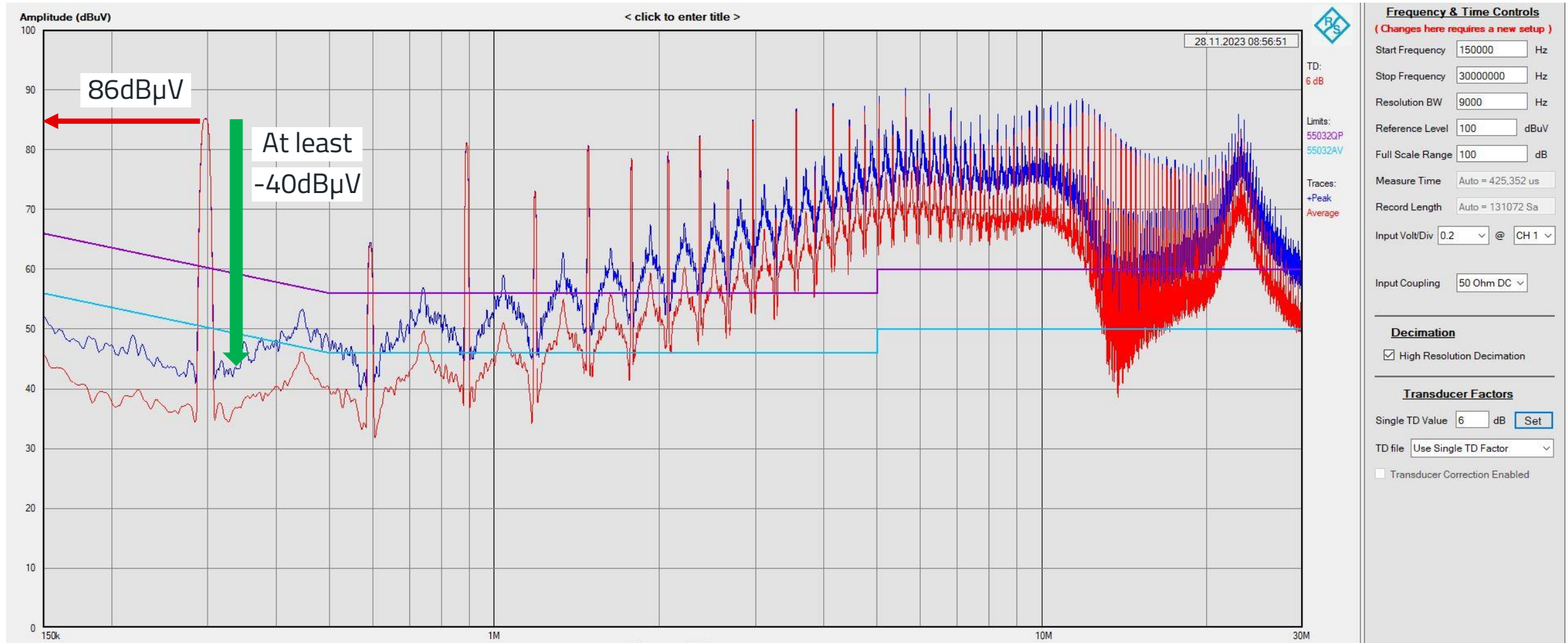
Conducted Spectrum 50uH LISN

CM no filter



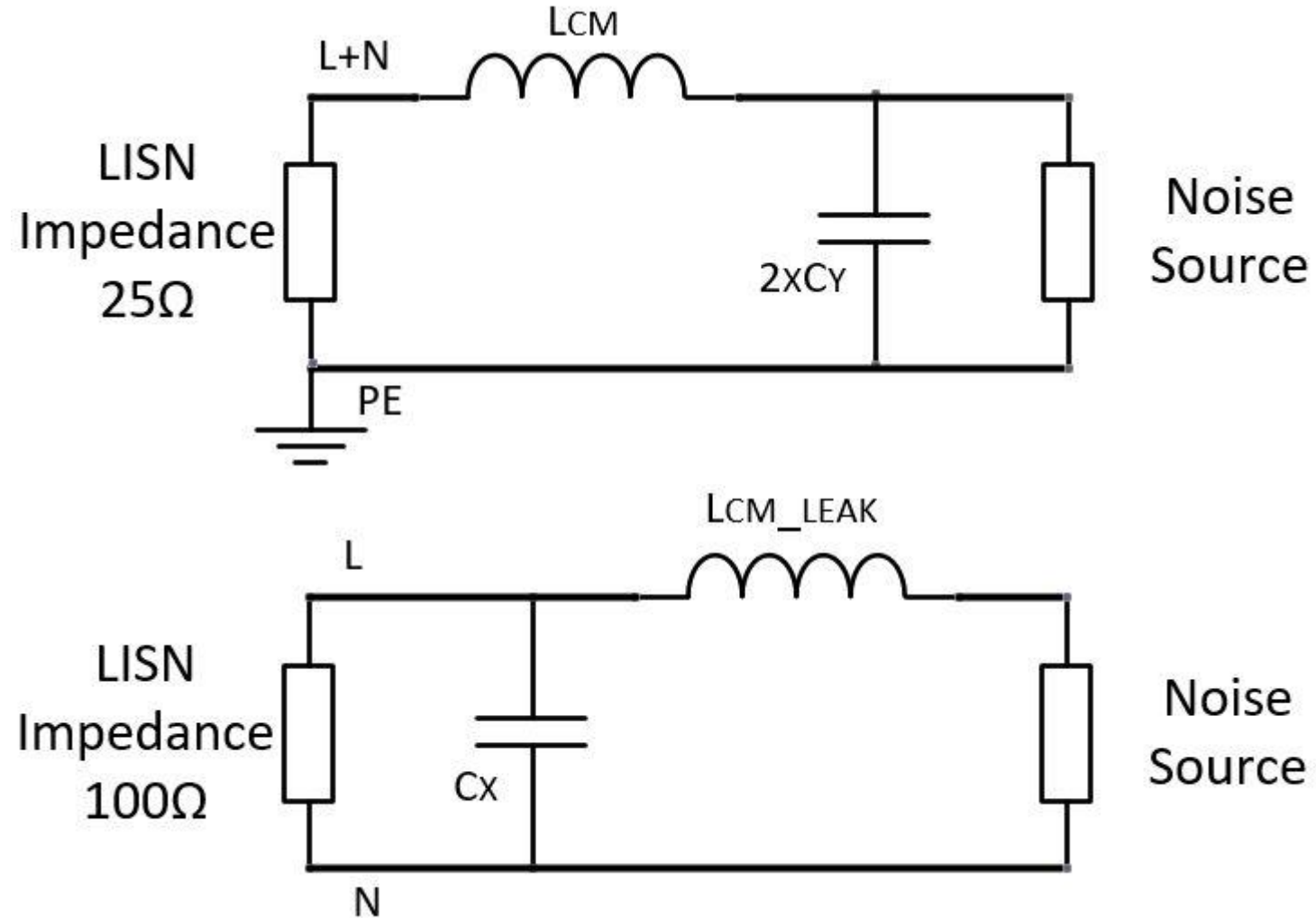
Conducted Spectrum 50uH LISN

CM + DM no Filter



CM & DM Filter Equivalent

Effective LISN Impedance and Component Arrangement

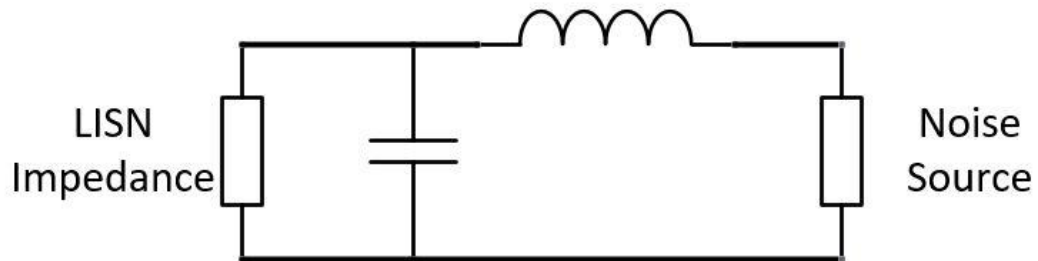


Corner Frequency for desired Attenuation at $f_{sw} = A_{fsw}$

1-Stage LC

$$A_{fsw} = \log\left(\frac{f_{sw}}{f_{co}}\right) \cdot 40dB$$

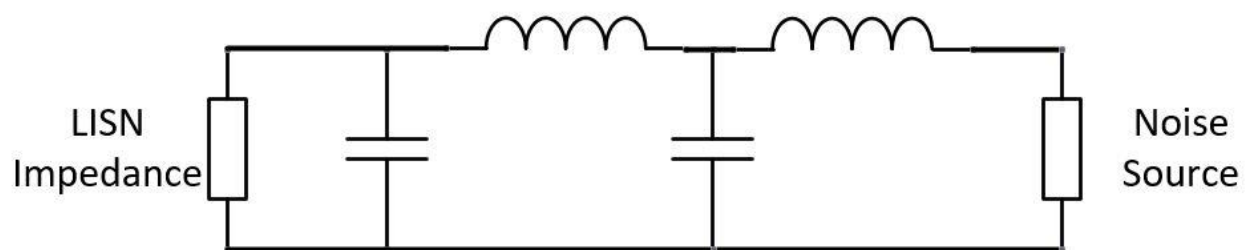
$$f_{co} = \frac{f_{sw}}{10^{\frac{A_{fsw}(dB)}{40dB}}}$$



2-Stage LC

$$A_{fsw} = \log\left(\frac{f_{sw}}{f_{co}}\right) \cdot 80dB$$

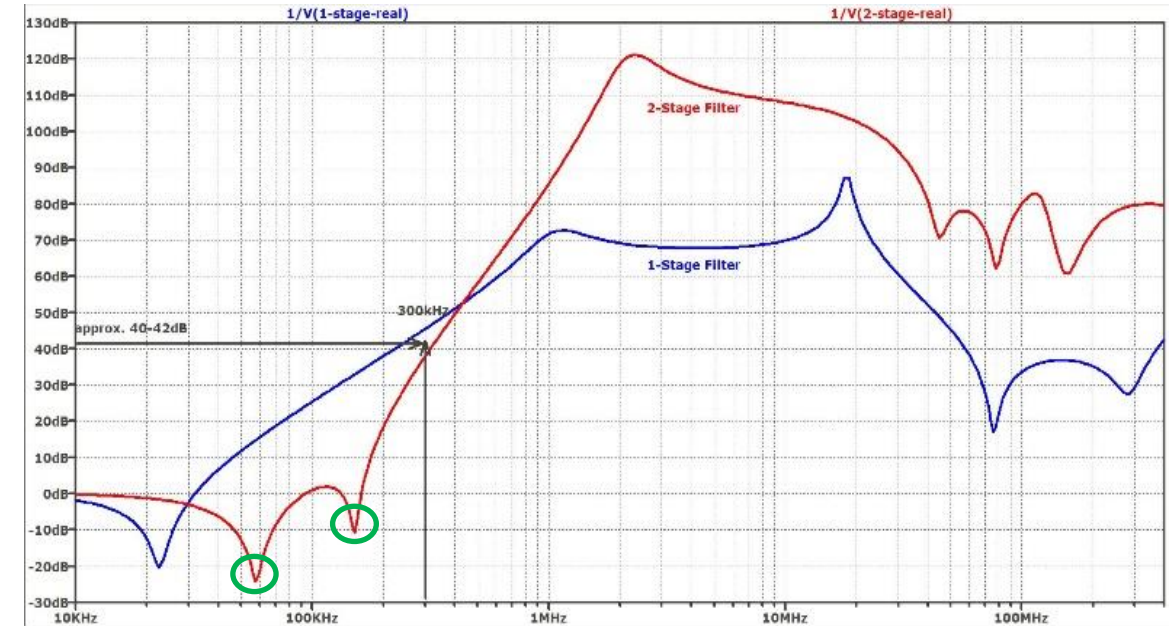
$$f_{co} = \frac{f_{sw}}{10^{\frac{A_{fsw}(dB)}{80dB}}}$$



Example: Effective Corner Frequency

If you have identical components in the 2-stage filter, you will find two resonance frequencies

- Effective $f_{co} = f_{co} = \frac{300kHz}{\frac{40dB}{10^{80dB}}} = 95 kHz$
- $f_{co1} = f_{co1} = 0.6 \cdot f_{co} = 0.6 \cdot 95kHz = 57kHz$
- $f_{co2} = f_{co2} = 1.6 \cdot f_{co} = 1.6 \cdot 95kHz = 152kHz$

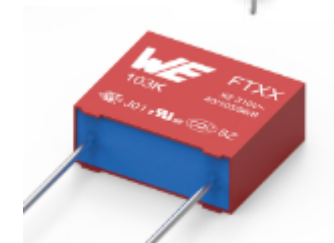
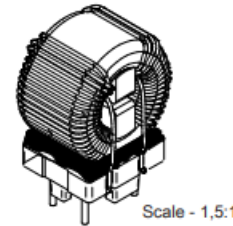
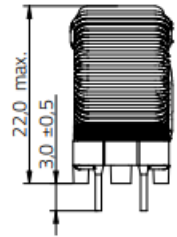
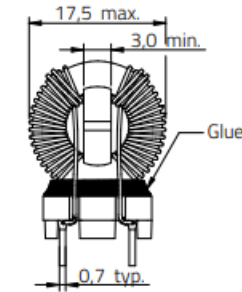


- For simplicity we will only work with the effective corner frequency (95kHz)
- This is then only valid, when the two (f_{co1} , f_{co2}) corner frequencies are at least 1 octave (2:1 frequency relationship) away from each other

1-Stage Calculation

Desired damping CM&DM = 40dB @ fsw(300kHz) → fc= 30kHz

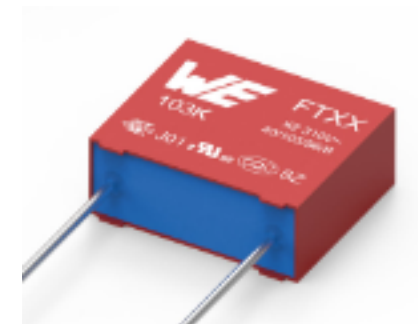
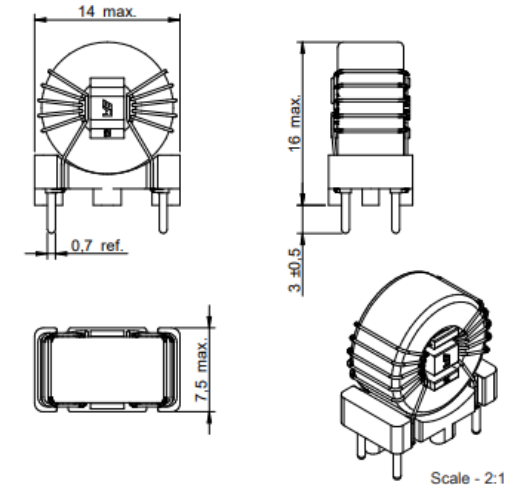
1. „LC“ = 40dB/Dec
2. Filter corner frequency: fc = 30kHz → 1/10 of fsw → 40dB @ 300kHz damping
3. Define Y-caps → 4,7nF x 2 → **9,4nF** total Cy
4. Define L_cm for fc = 30kHz → $L_{cm} = \frac{1}{(2\pi f_c)^2 C_y} = \frac{1}{(2\pi \cdot 30\text{kHz})^2 \cdot 9,4\text{nF}} = 3\text{mH} \rightarrow \mathbf{3,3\text{mH}}$ chosen (744822233)
5. $Actual\ f_{cm} = \frac{1}{2\pi\sqrt{L_{cm}C_y}} = \frac{1}{2\pi\sqrt{3,3\text{mH} \cdot 9,4\text{nF}}} = 28,5\text{kHz} \rightarrow A_{fsw_{cm}} = \log\left(\frac{300\text{kHz}}{28,5\text{kHz}}\right) \cdot 40\text{dB} = \mathbf{41\text{dB}}$
6. Calc. L_dm → XL = 92Ω@1MHz → $L_{dm} = \frac{XL}{2\pi f} = \frac{92\Omega}{2\pi \cdot 1\text{MHz}} = \mathbf{14,6\mu H}$ (use REDEXPERT for help)
7. Define Cx → $C_x = \frac{1}{(2\pi f_c)^2 L_{dm}} = \frac{1}{(2\pi \cdot 30\text{kHz})^2 \cdot 14,6\mu H} = 1,92\mu F \rightarrow \mathbf{2,2\mu F}$ chosen (890334026034CS)
8. $Actual\ f_{dm} = \frac{1}{2\pi\sqrt{L_{dm}C_x}} = \frac{1}{2\pi\sqrt{14,6\mu H \cdot 2,2\mu F}} = 28\text{kHz} \rightarrow A_{fsw_{dm}} = \log\left(\frac{300\text{kHz}}{28\text{kHz}}\right) \cdot 40\text{dB} = \mathbf{41\text{dB}}$



2-Stage Calculation

Desired damping DM&CM = 40dB @ fsw(300kHz) → fco= 95kHz

1. „2nd LC“ = 80dB/Dec
2. Filter corner frequency: fc = 95kHz → 40dB @ 300kHz damping
3. Define Y-caps → 2,2nF x 2 → **4,4nF** total Cy (8853622110151)
4. Define L_cm for fc = 95kHz → $L_{cm} = \frac{1}{(2\pi f_c)^2 C_y} = \frac{1}{(2\pi \cdot 95\text{kHz})^2 \cdot 4,4\text{nF}} = 0,64\text{mH} \rightarrow \mathbf{1,0mH}$ chosen (744821201)
5. $Actual\ f_{cm} = \frac{1}{2\pi\sqrt{L_{cm}C_y}} = \frac{1}{2\pi\sqrt{1\text{mH} \cdot 4,4\text{nF}}} = 75\text{kHz} \rightarrow A_{fsw_{cm}} = \log\left(\frac{300\text{kHz}}{75\text{kHz}}\right) \cdot 80\text{dB} = \mathbf{48dB}$
6. Calc. L_dm → XL = 41Ω@1MHz → $L_{dm} = \frac{XL}{2\pi f} = \frac{41\Omega}{2\pi \cdot 1\text{MHz}} = \mathbf{6,5\mu H}$ (use REDEXPERT for help)
7. Define Cx → $C_x = \frac{1}{(2\pi f_c)^2 L_{dm}} = \frac{1}{(2\pi \cdot 95\text{kHz})^2 \cdot 6,5\mu H} = 0,43\mu F \rightarrow \mathbf{560nF}$ chosen (890334026018CS)
8. $Actual\ f_{dm} = \frac{1}{2\pi\sqrt{L_{dm}C_x}} = \frac{1}{2\pi\sqrt{6,5\mu H \cdot 560\text{nF}}} = 83\text{kHz} \rightarrow A_{fsw_{dm}} = \log\left(\frac{300\text{kHz}}{83\text{kHz}}\right) \cdot 80\text{dB} = \mathbf{45dB}$
9. Additional DM capacitance: Varistor ~ 420pF & $(1/2 \cdot C_y) = 3.3\text{nF}$



2-Stage Calculation

If the values of the filter components differ **greatly** (DM example)

- If the size of the inductors are different in size, then there are two pairs of poles: High & Low frequency pole
- In this example has L1 the bigger inductance and C1 the bigger capacitance

- $$\Delta s = \left[1 + \frac{s}{\omega_L Q_L} + \frac{s^2}{\omega_L^2} \right] \cdot \left[1 + \frac{s}{\omega_H Q_H} + \frac{s^2}{\omega_H^2} \right]$$

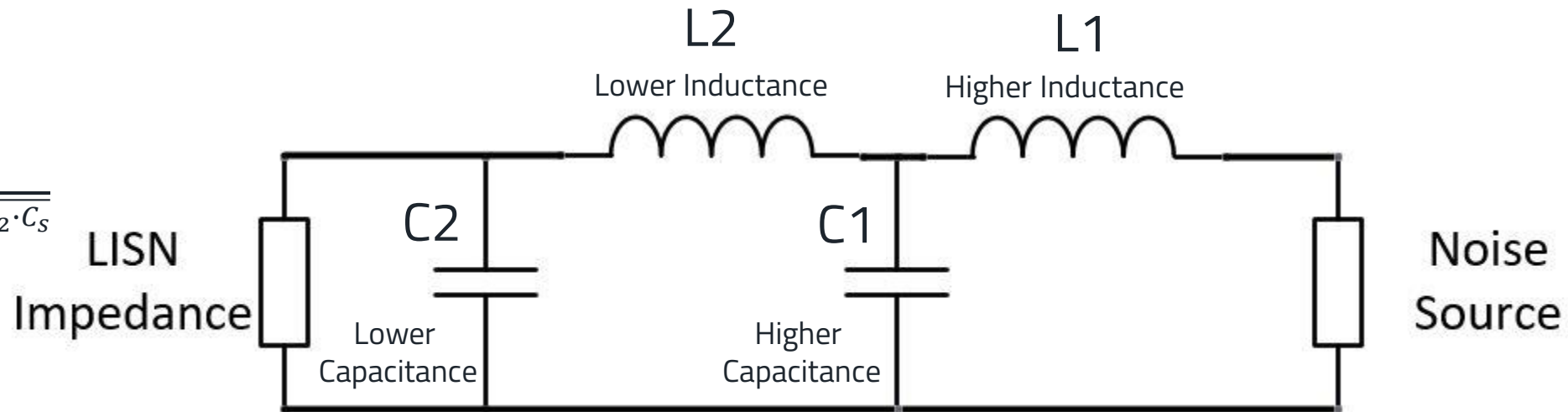
- $$\omega_L = \frac{1}{\sqrt{L_1 \cdot C_p}} \rightarrow f_{CL} = \frac{1}{2\pi \sqrt{L_1 \cdot C_p}}$$

- $$C_p = C_1 + C_2$$

- $$\omega_H = \frac{1}{\sqrt{L_2 \cdot C_s}} \rightarrow f_{CH} = \frac{1}{2\pi \sqrt{L_2 \cdot C_s}}$$

- $$C_s = \frac{1}{\left(\frac{1}{C_1} + \frac{1}{C_2} \right)}$$

$$\begin{matrix} L1 \gg L2 \\ C1 \gg C2 \end{matrix}$$



2-Stage Simulation Example

If the values of the filter components differ **greatly**

40dB/dec

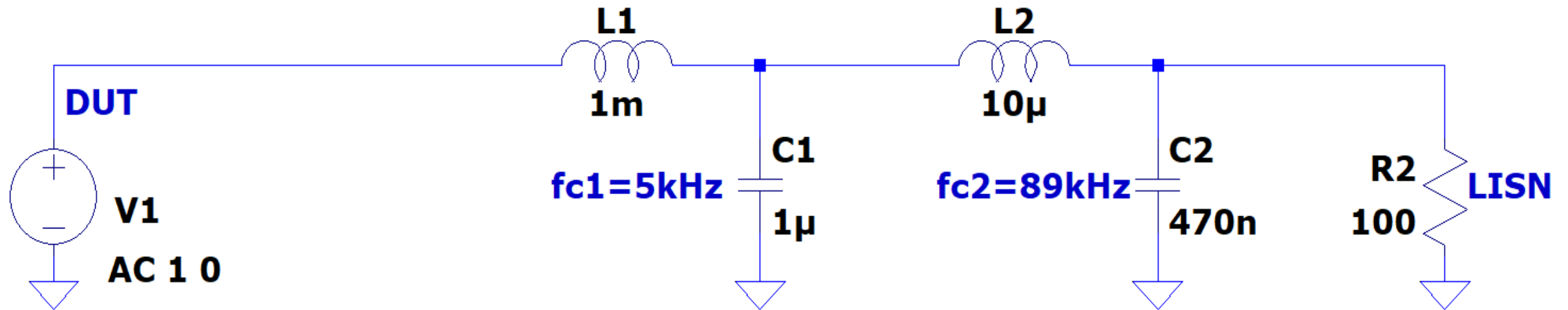
$$f_{c1} = 1/2 * \pi * \text{sqr}(L1 * C_p)$$

$$C_p = C1 + C2$$

80dB/dec

$$f_{c2} = 1/2 * \pi * \text{sqr}(L2 * C_s)$$

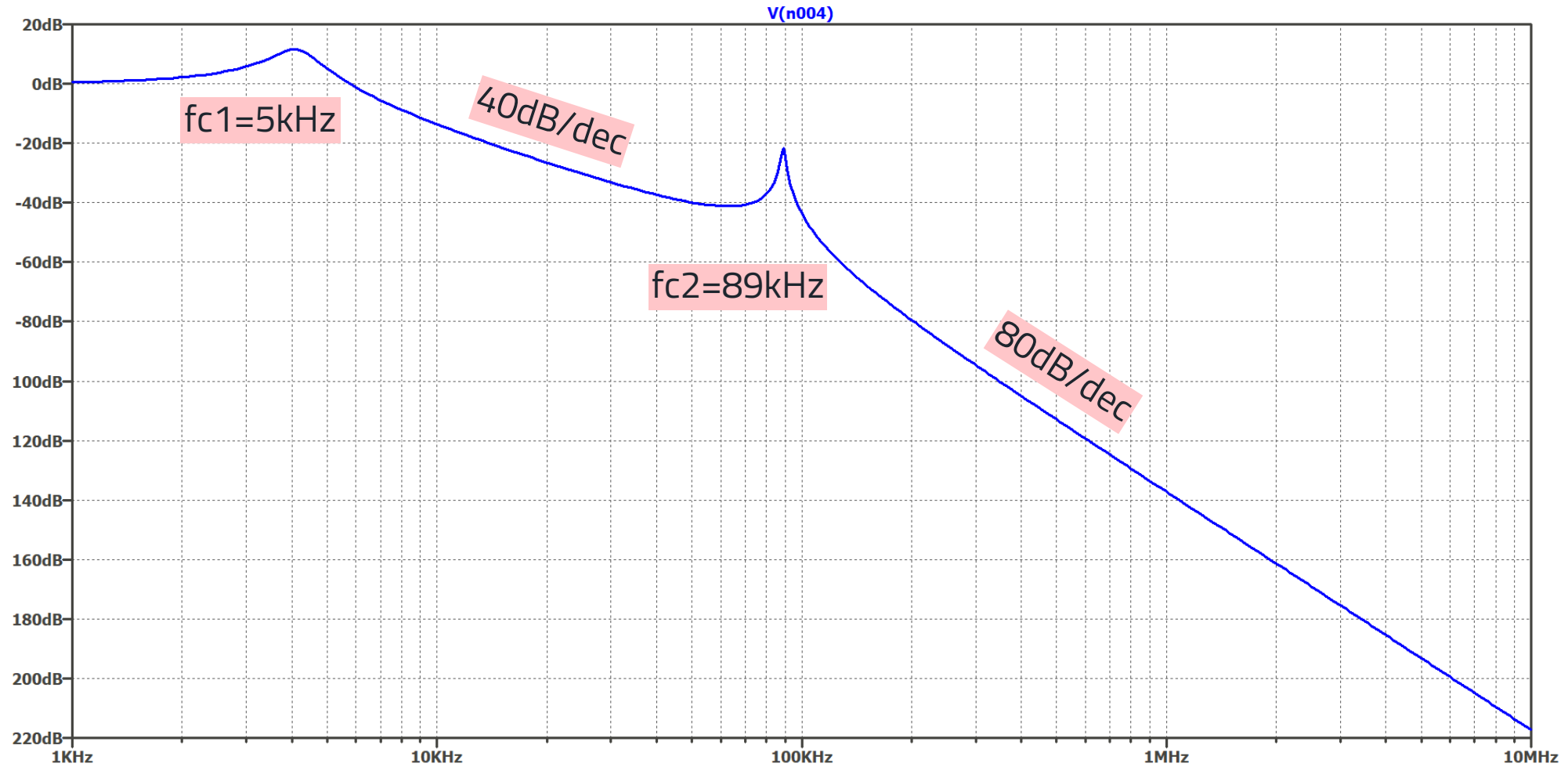
$$C_s = 1 / (1/C1 + 1/C2)$$



.ac dec 200 1000 100000000

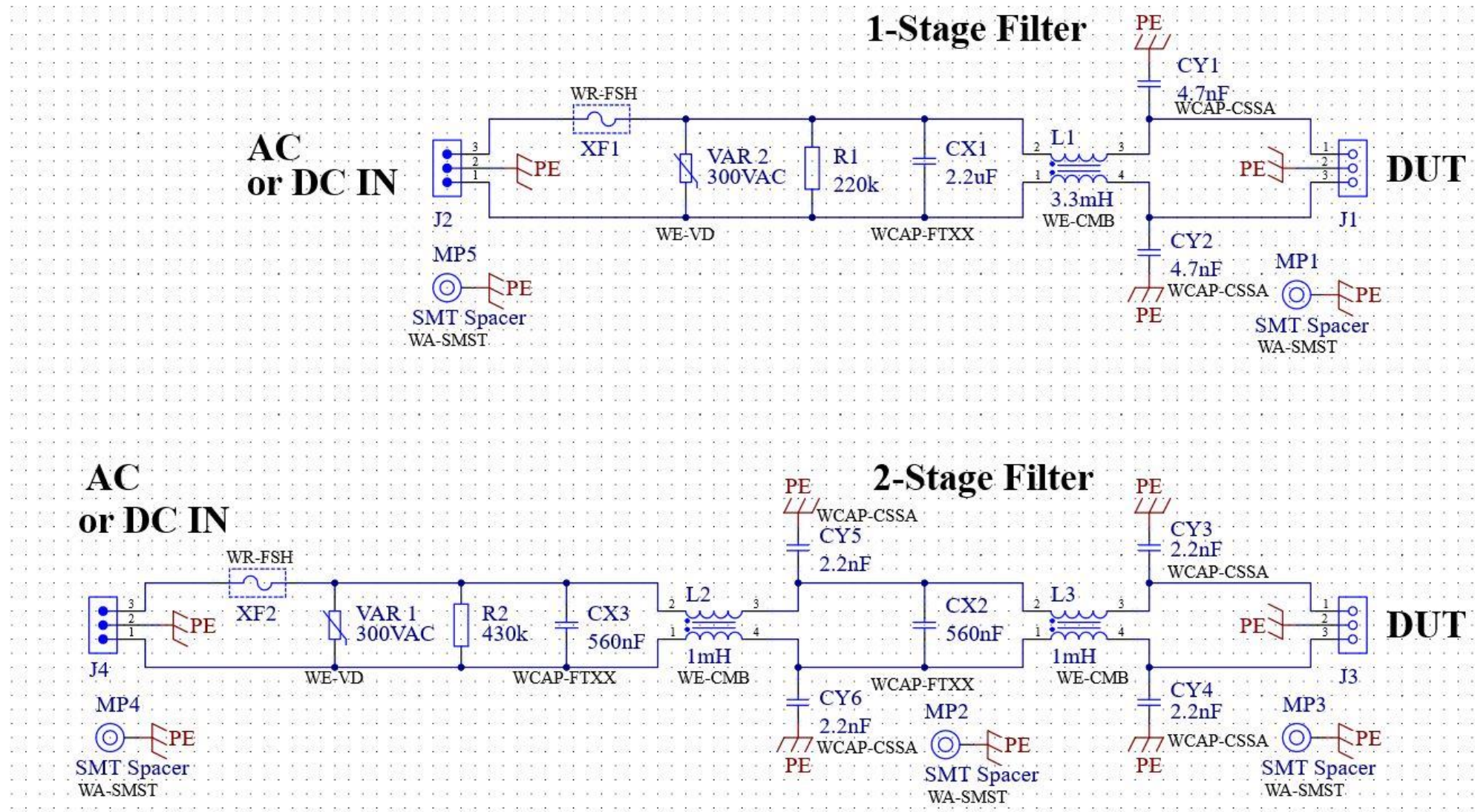
2-Stage Calculation Example

If the values of the filter components differ **greatly**

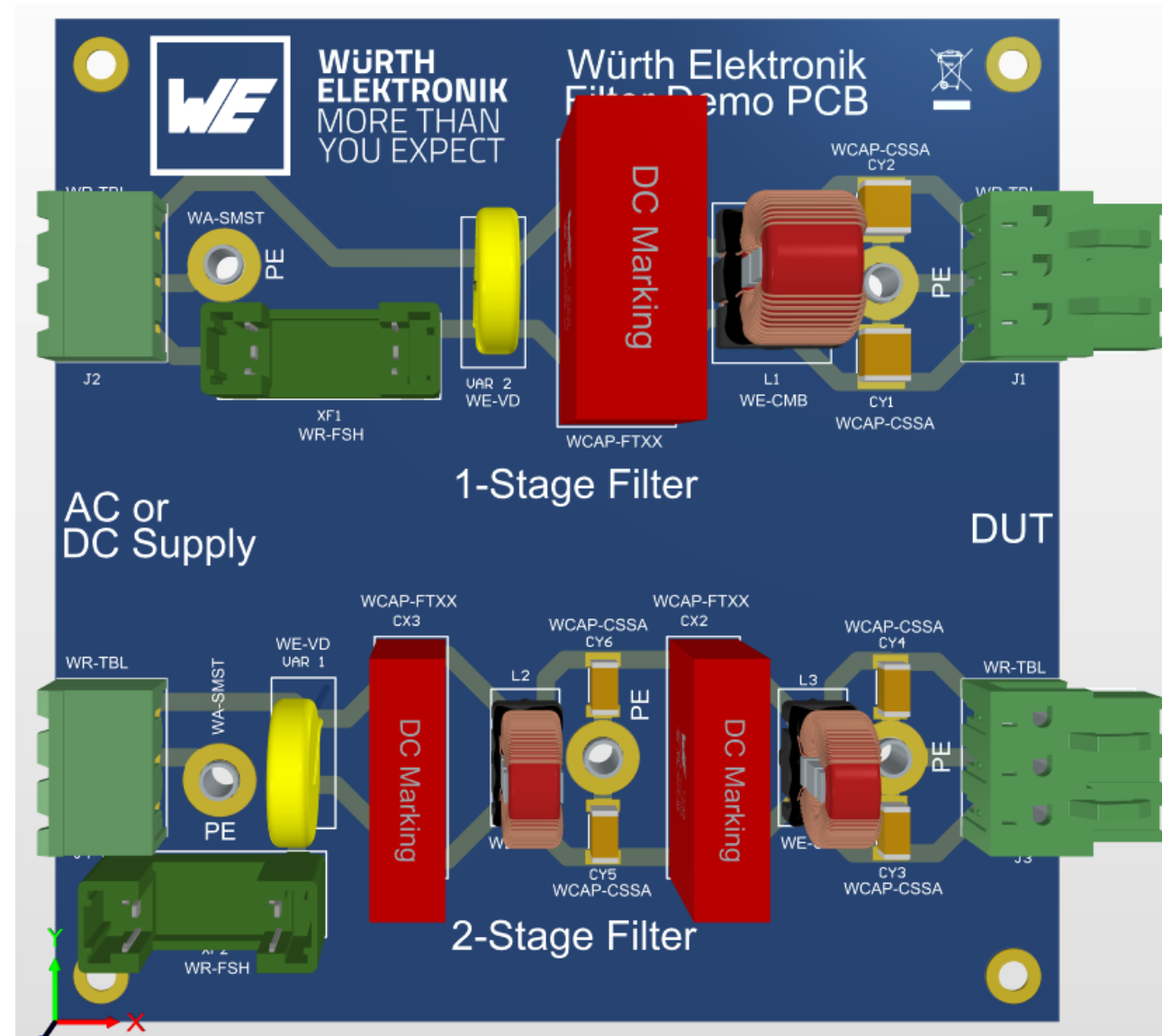


FILTER SCHEMATIC & PCB

Altium Schematic



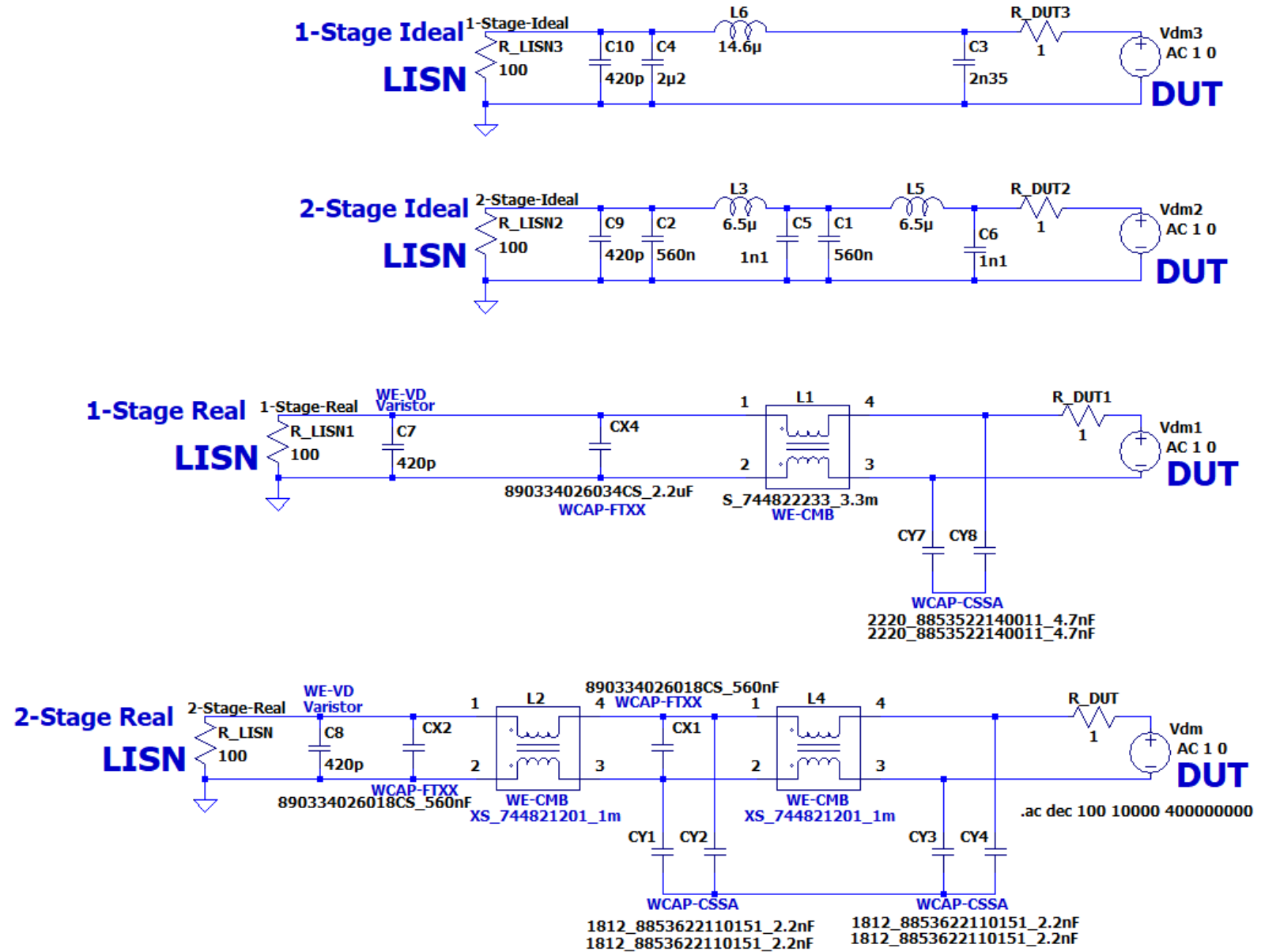
Altium PCB



SIMULATIONS

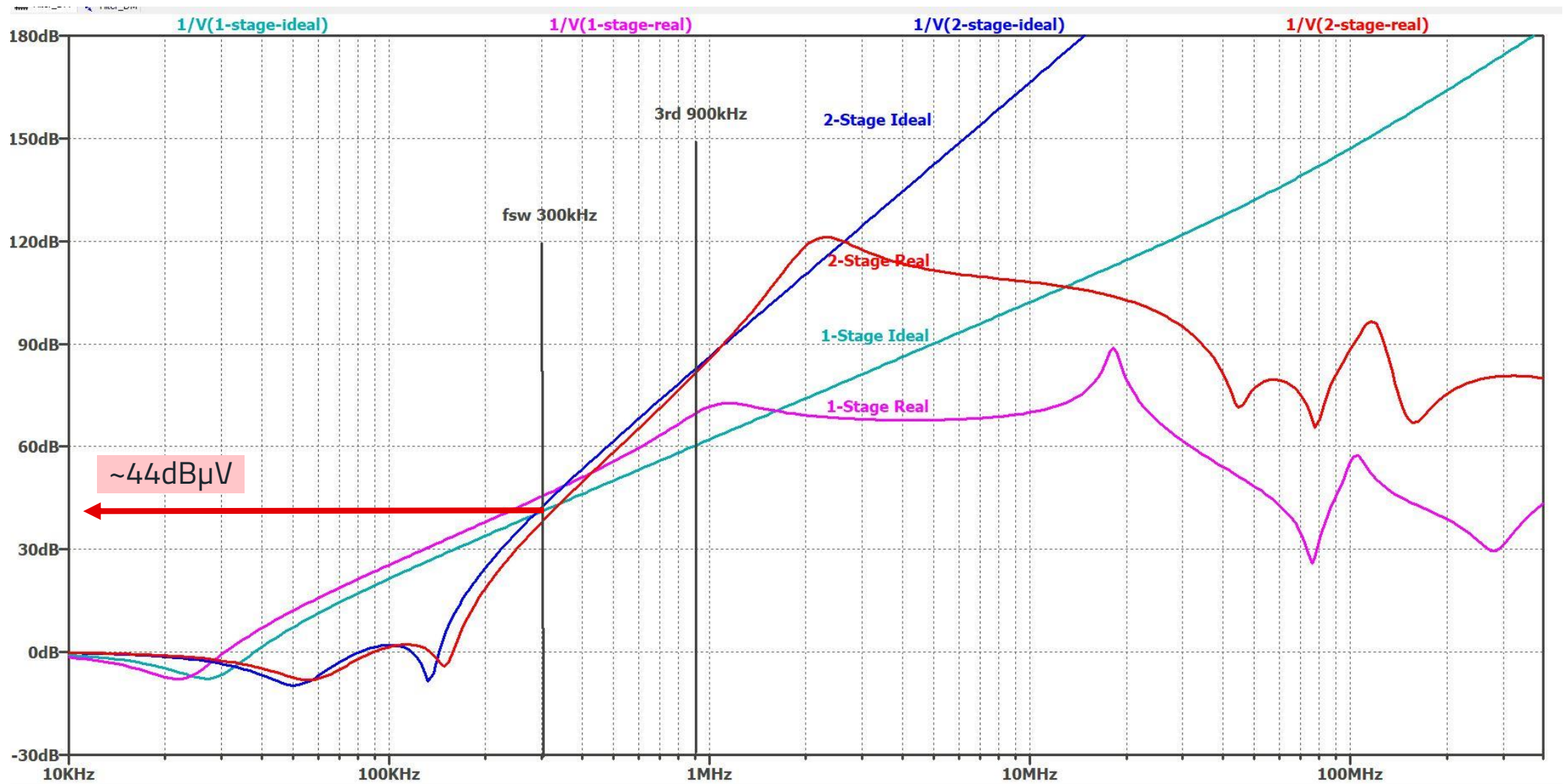
LT Spice

DM simulation



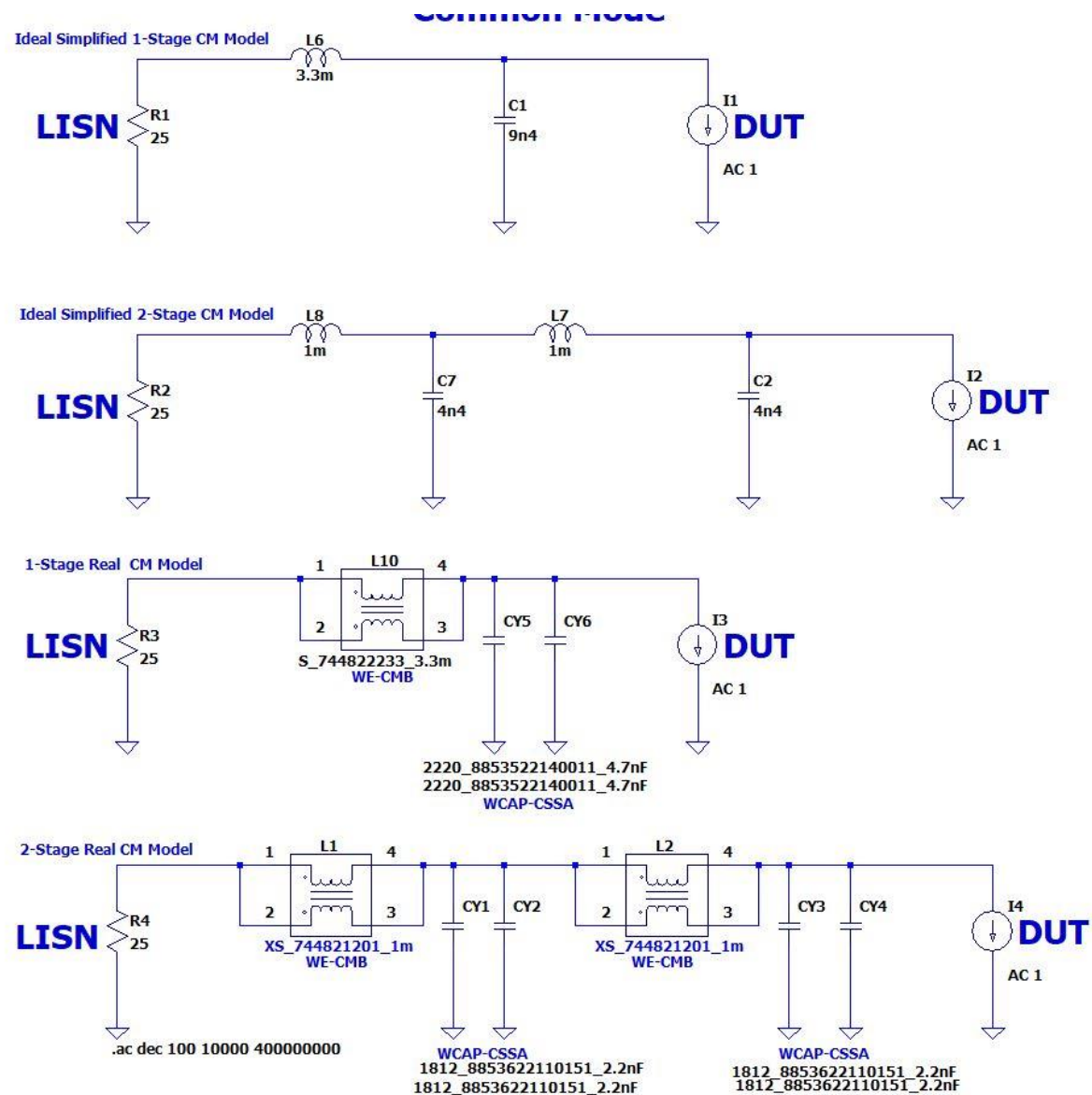
LT Spice

DM simulation



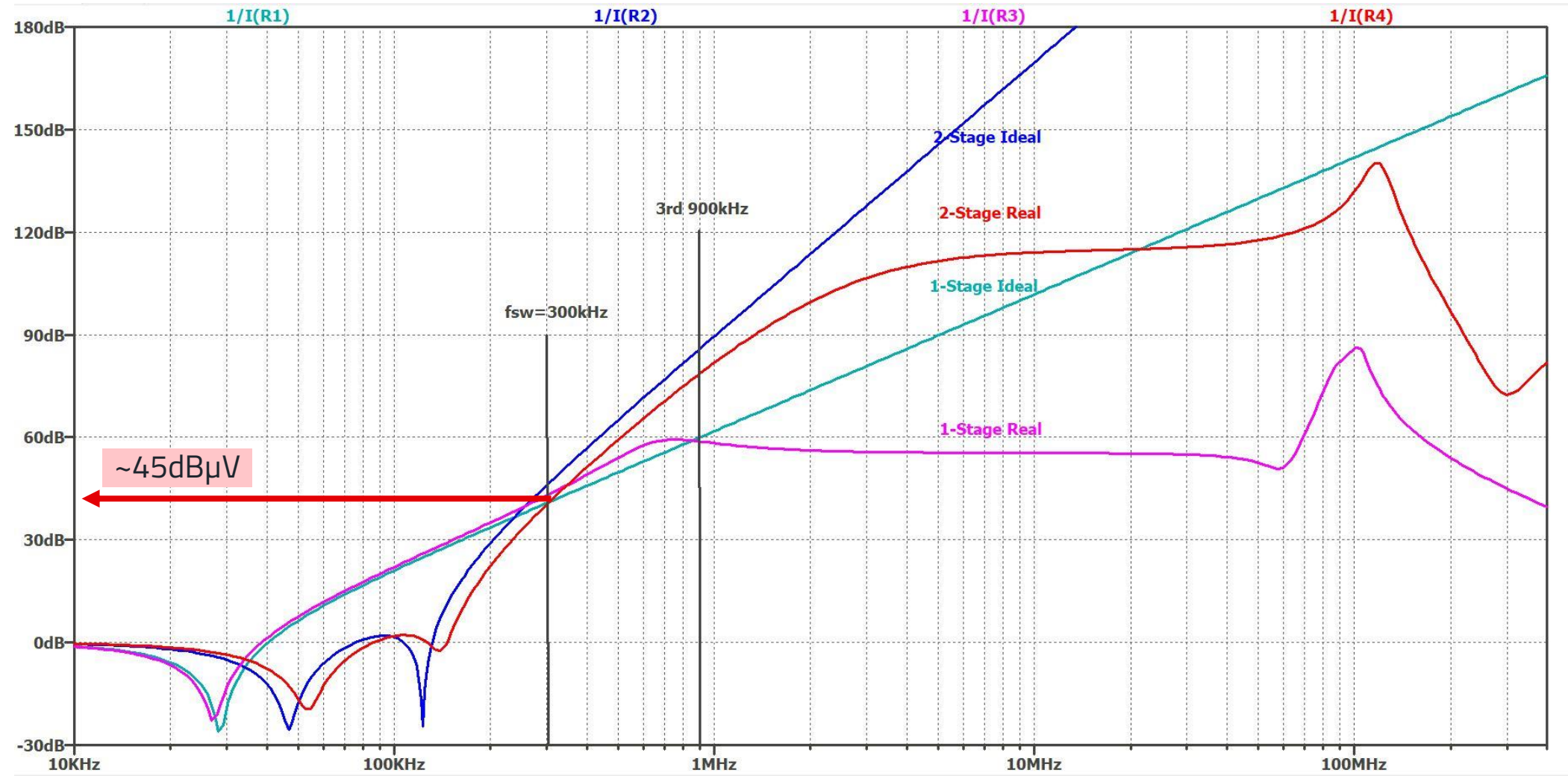
LT Spice

CM simulation



LT Spice

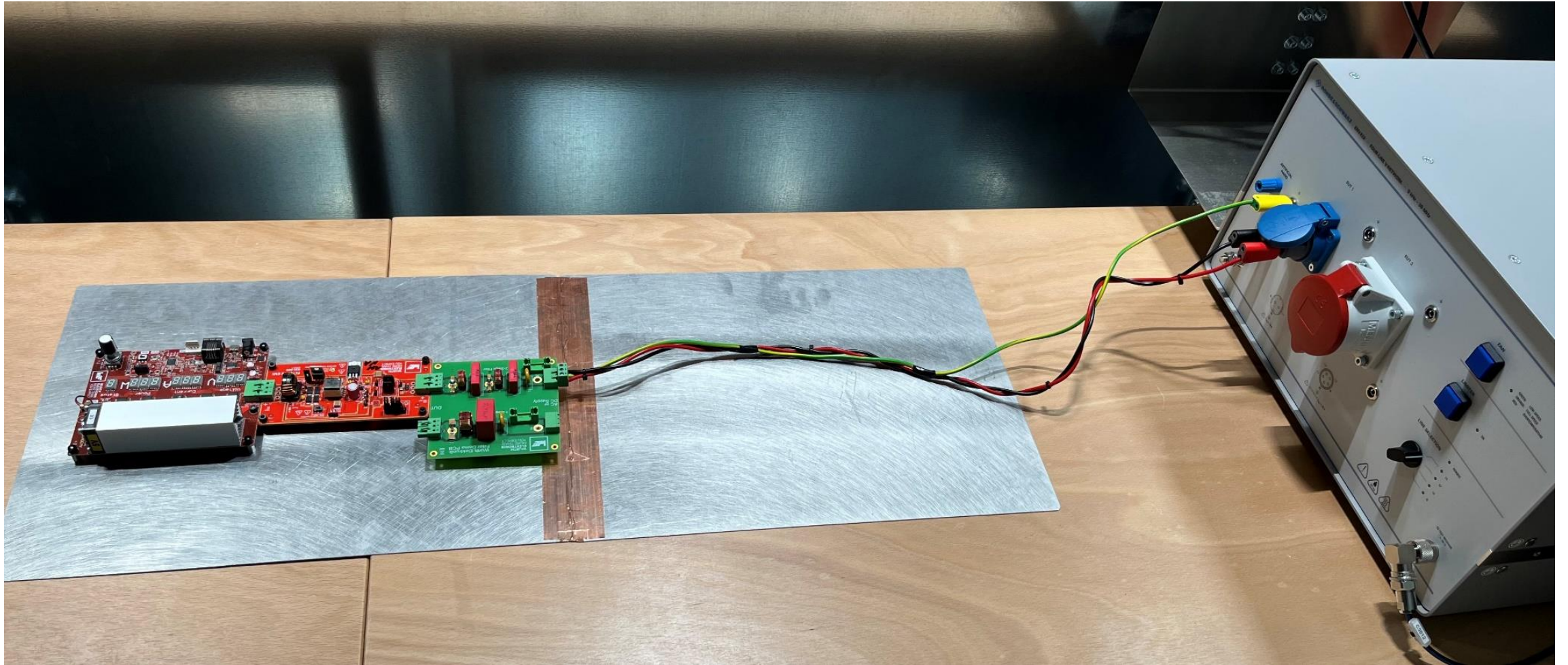
CM simulation



MEASUREMENTS FROM EMC LAB

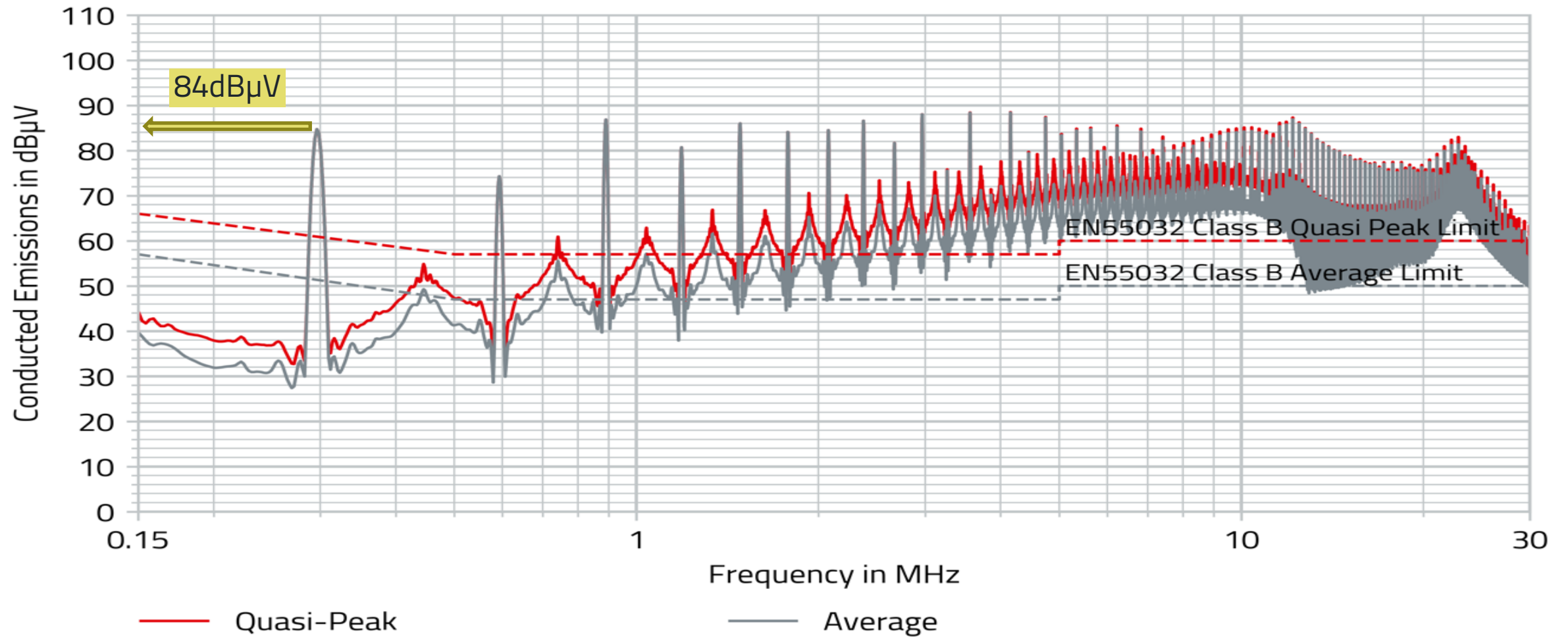
Test side: Würth Elektronik HIC Freiam

Test Setup Conducted



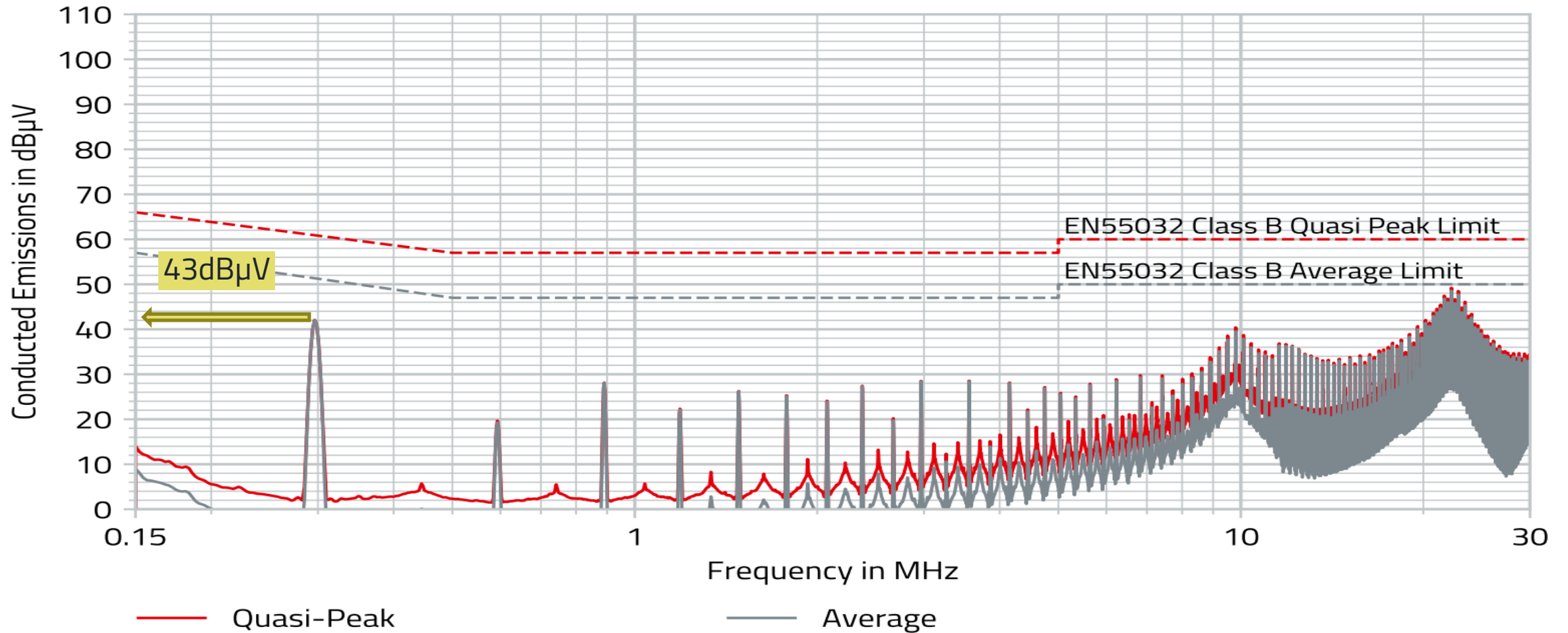
Conducted CM+DM

No Filter



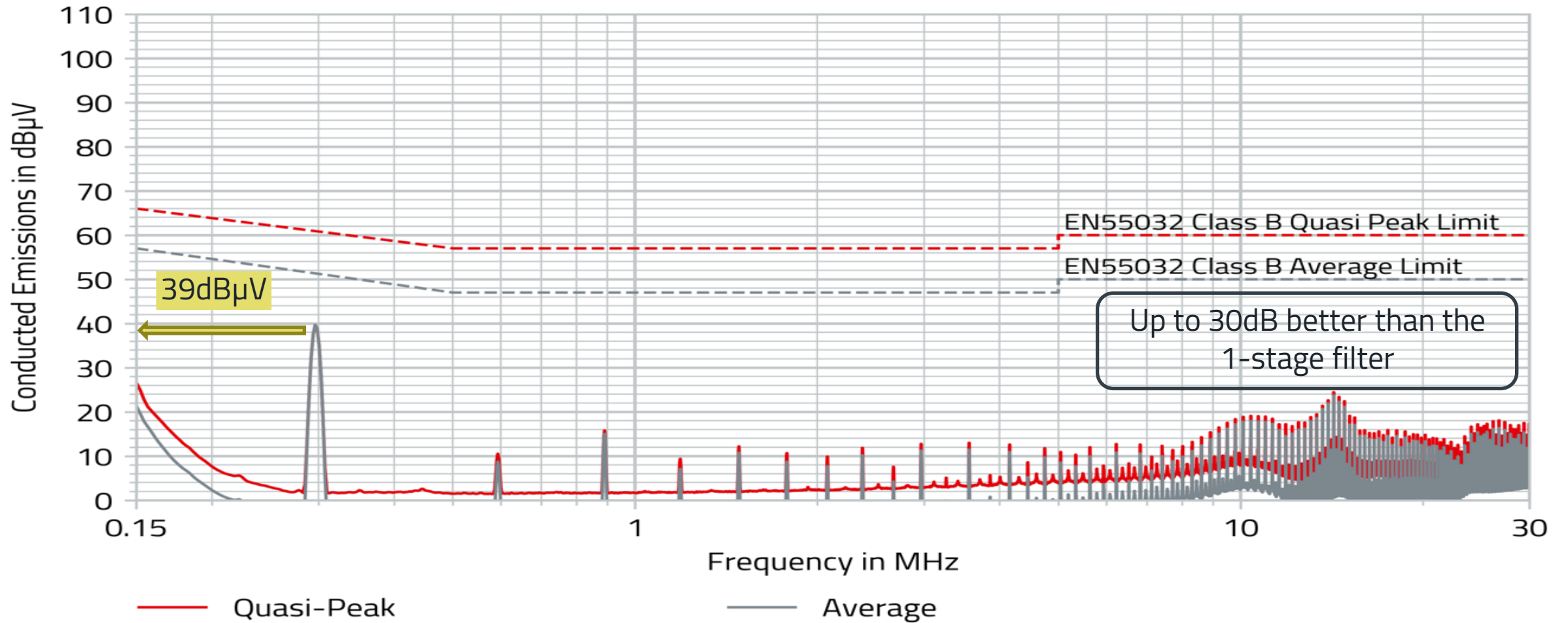
Conducted CM+DM

1-Stage Filter → approx. 41dB damping (*41dB CM & 41dB DM was calculated*)

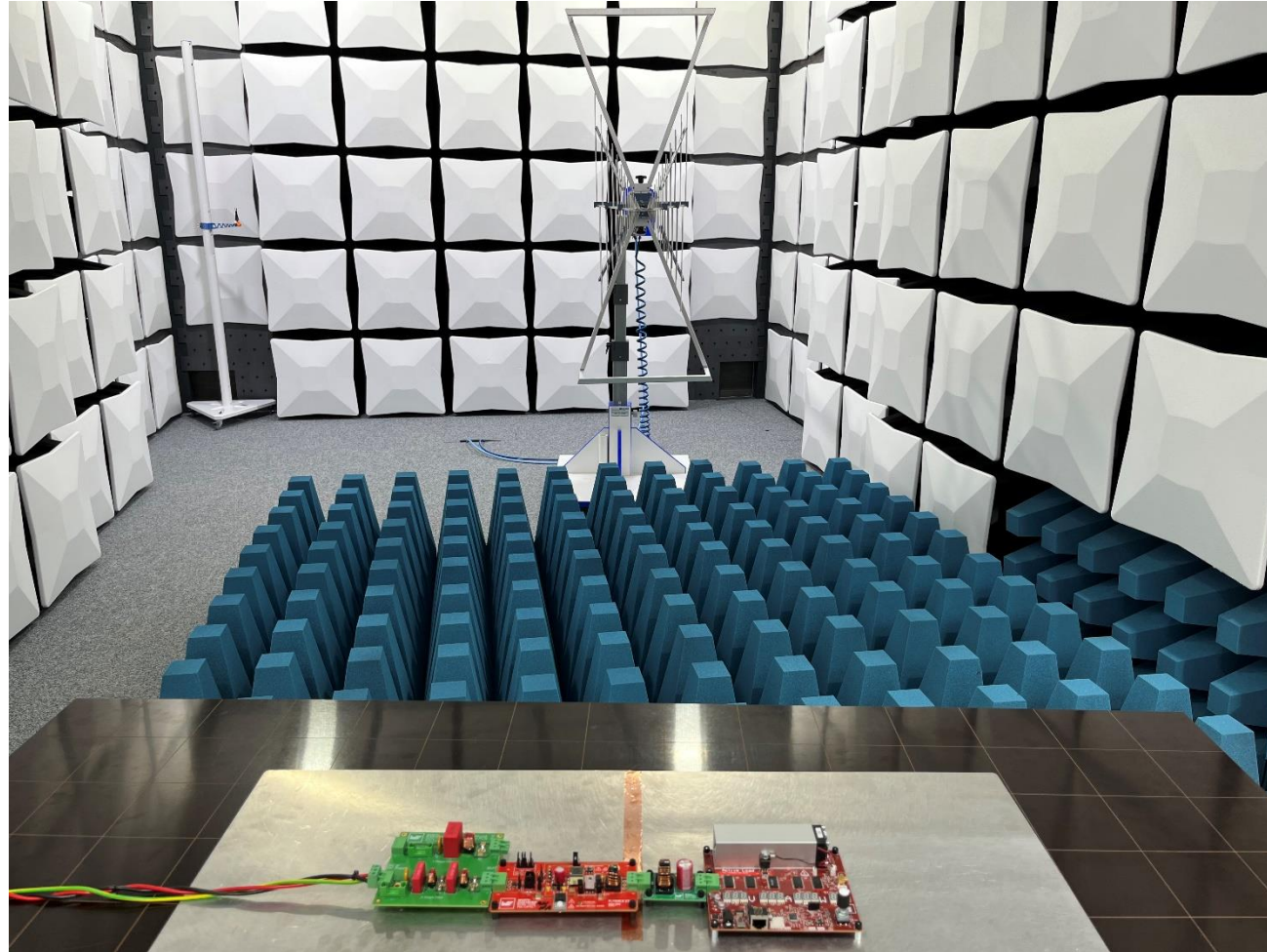


Conducted CM+DM

2-Stage Filter → approx. 45dB damping (*48dB CM & 45dB DM was calculated*)

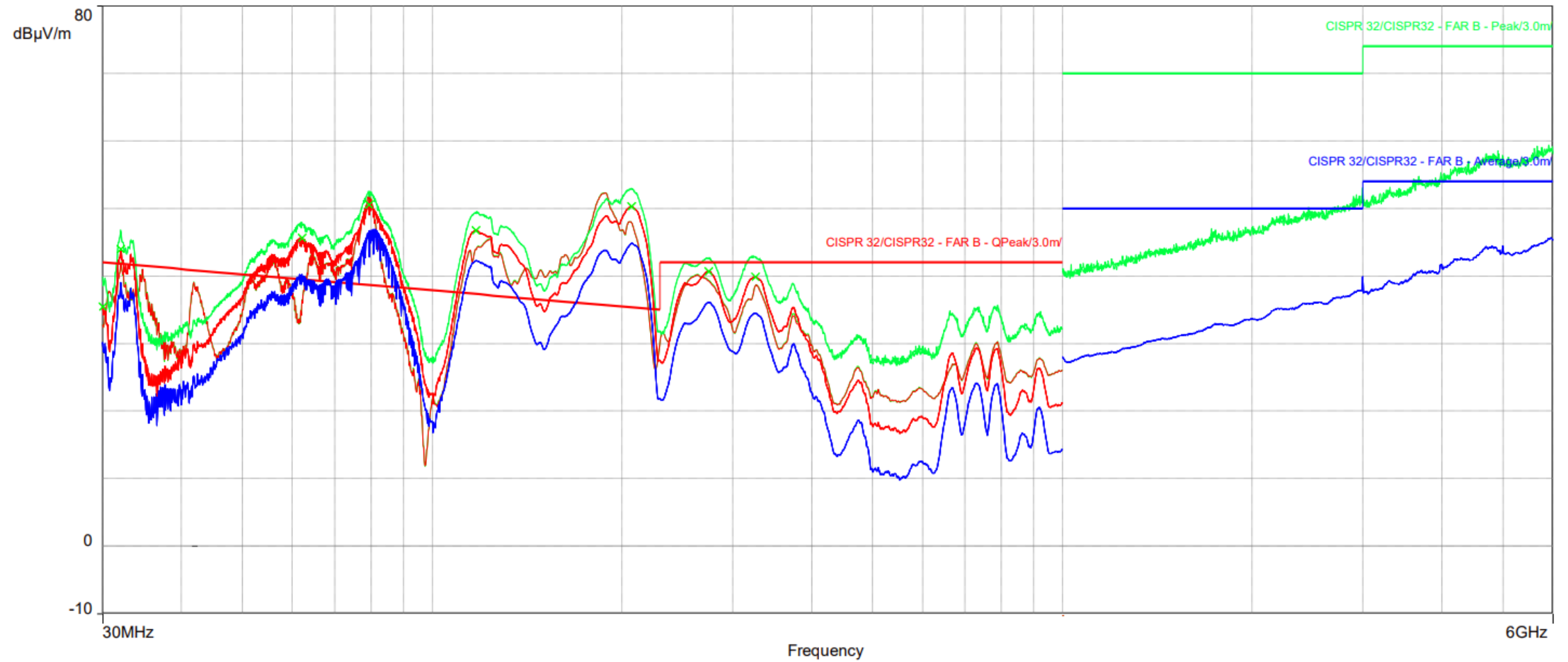


Test Setup Radiated



Radiated

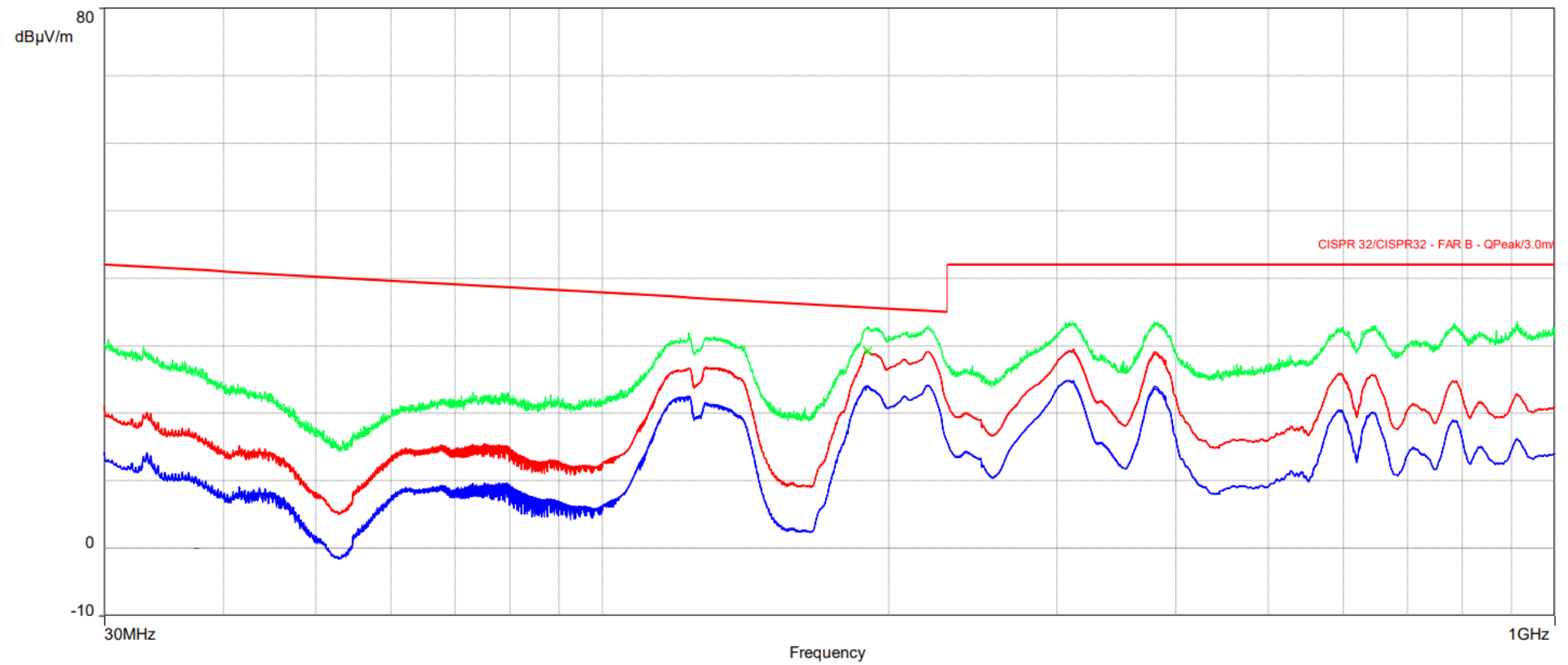
- No Filter



Radiated

1-Stage Filter

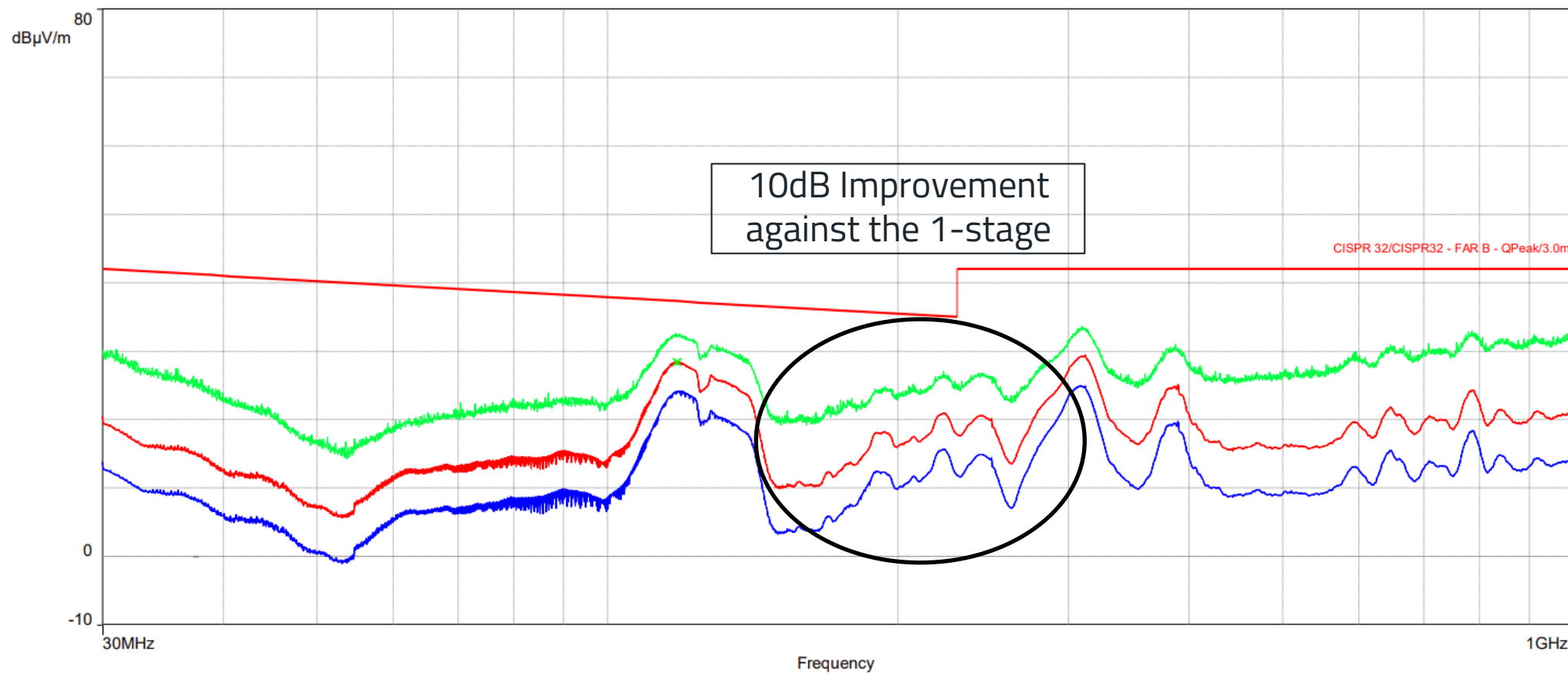
Global Graph:



Radiated

2-Stage Filter

Global Graph:



Possible Deviations

Measurement vs Math vs Simulation

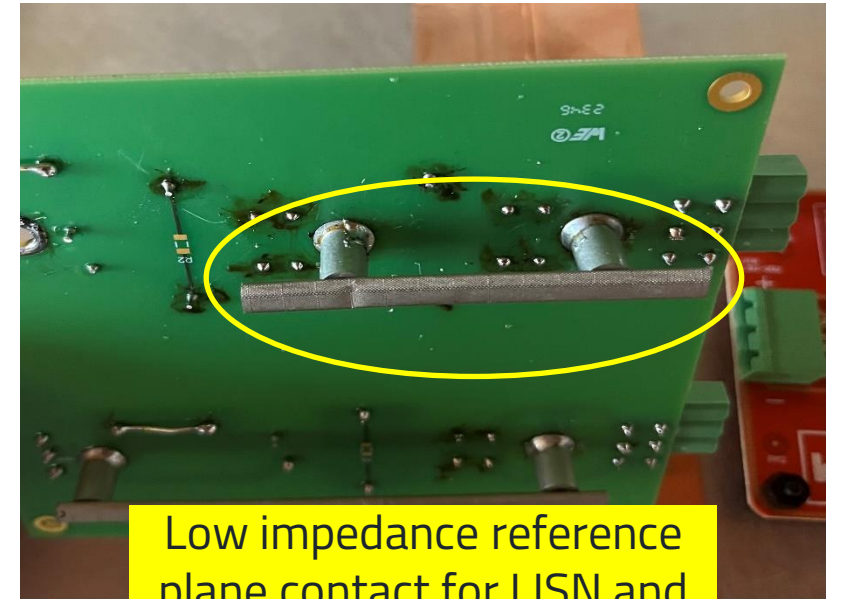
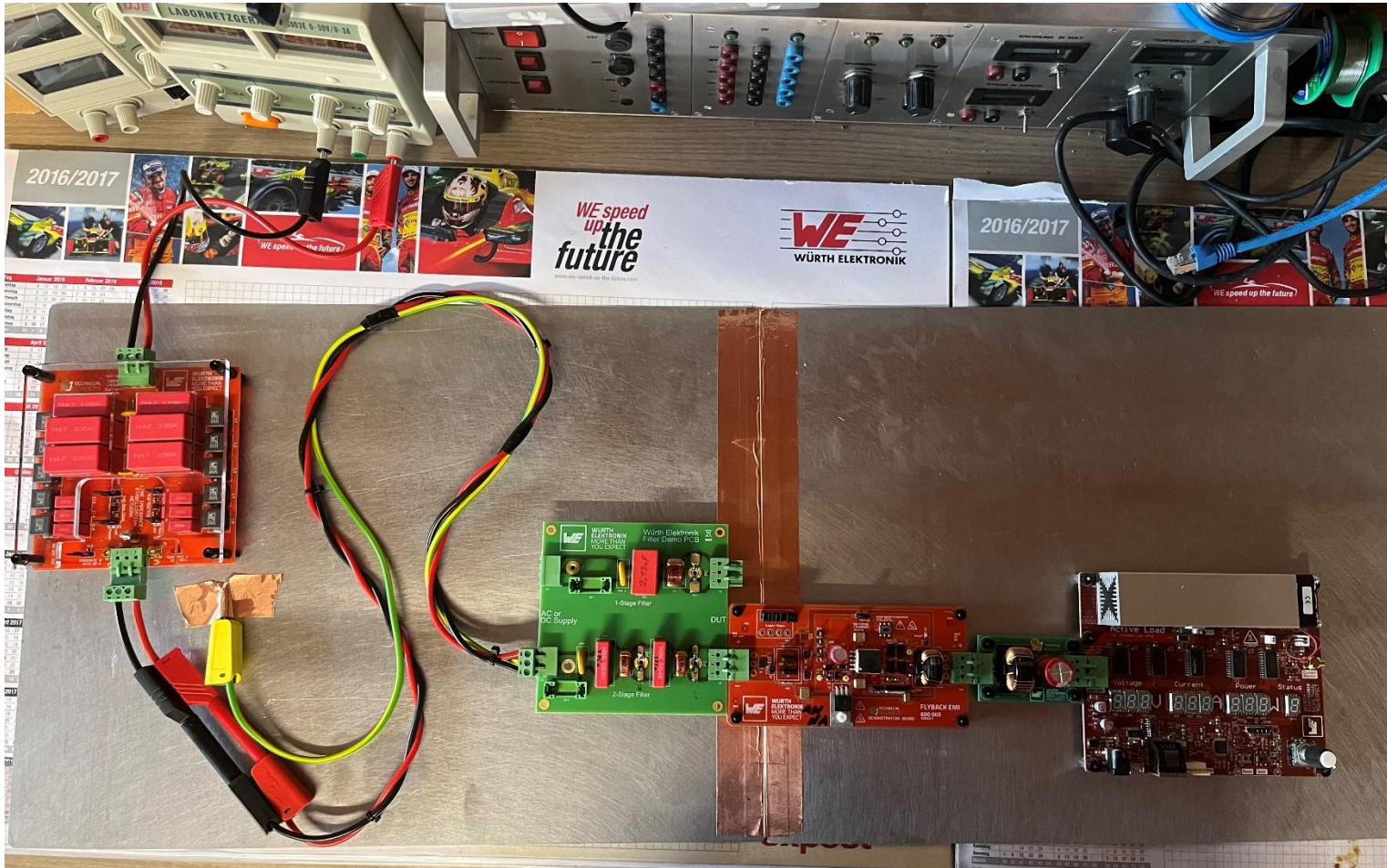
- Inductance of supply wires
- Inaccuracy of simulation models
- Tolerance of the inductive and capacitive components
- Grounding of Y-caps
- CMC saturation effects
- ESL of filter PCB
- Parasitic inductive & capacitive coupling between the filter components
- Varistor capacitance

MEASUREMENTS

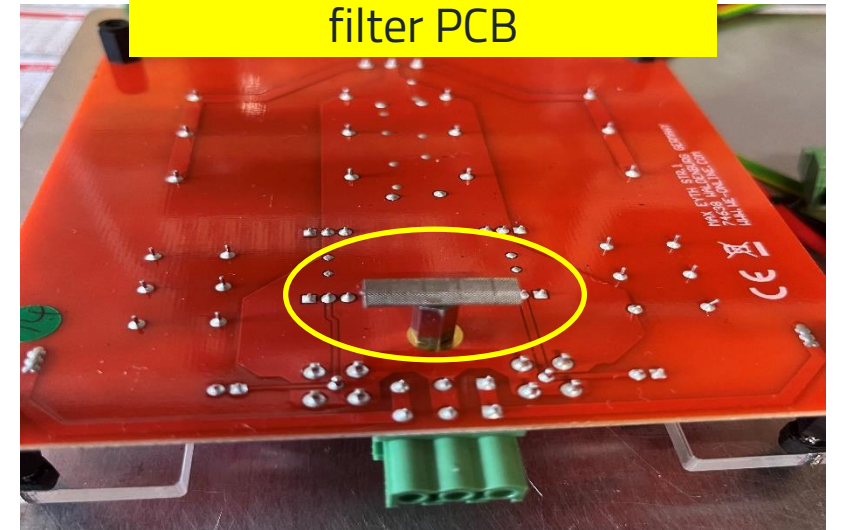
DESKLAB

Setup

External filter PCB with 1- & 2-stage CM/DM filter

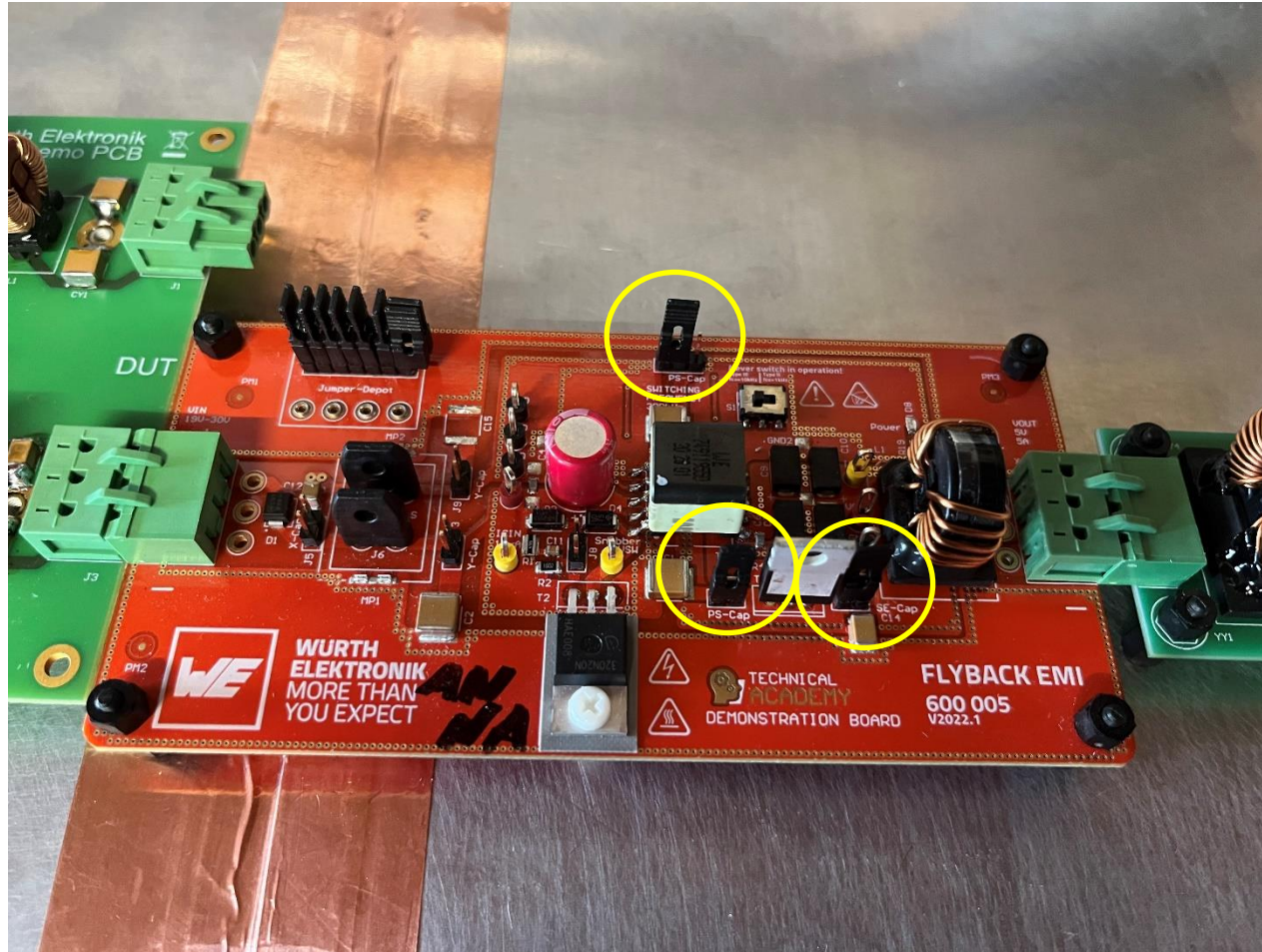


Low impedance reference plane contact for LISN and filter PCB



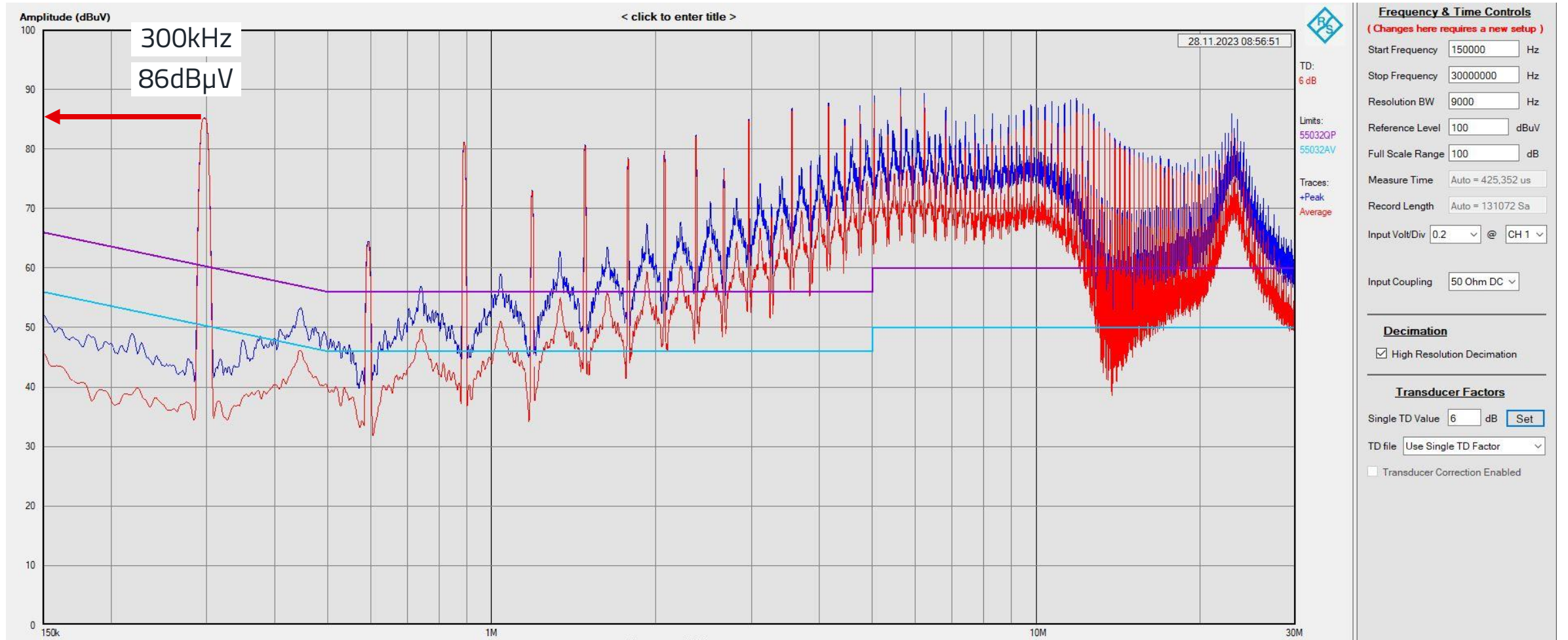
Setup

Jumper setting flyback noise source → only activate the PRI to SEC caps & SEC to PE



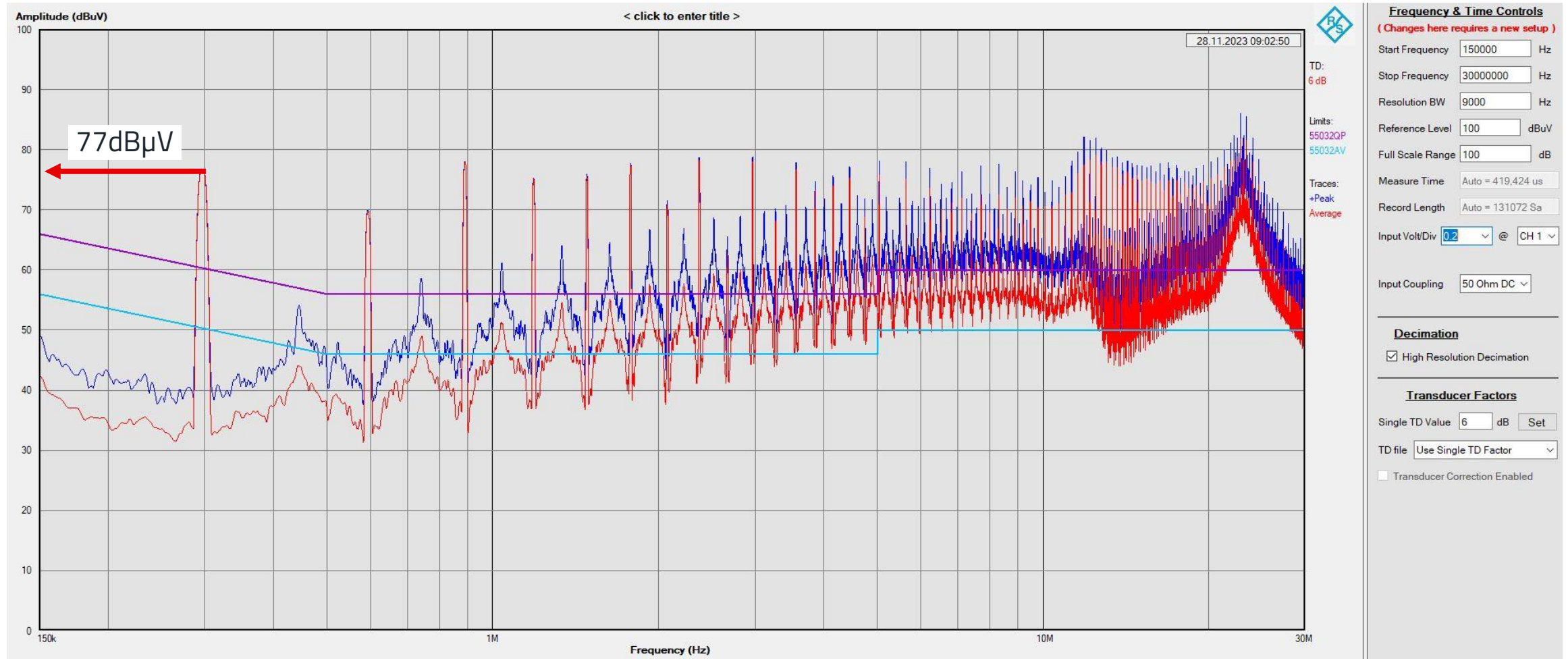
Conducted Spectrum 50uH LISN

CM + DM no filter



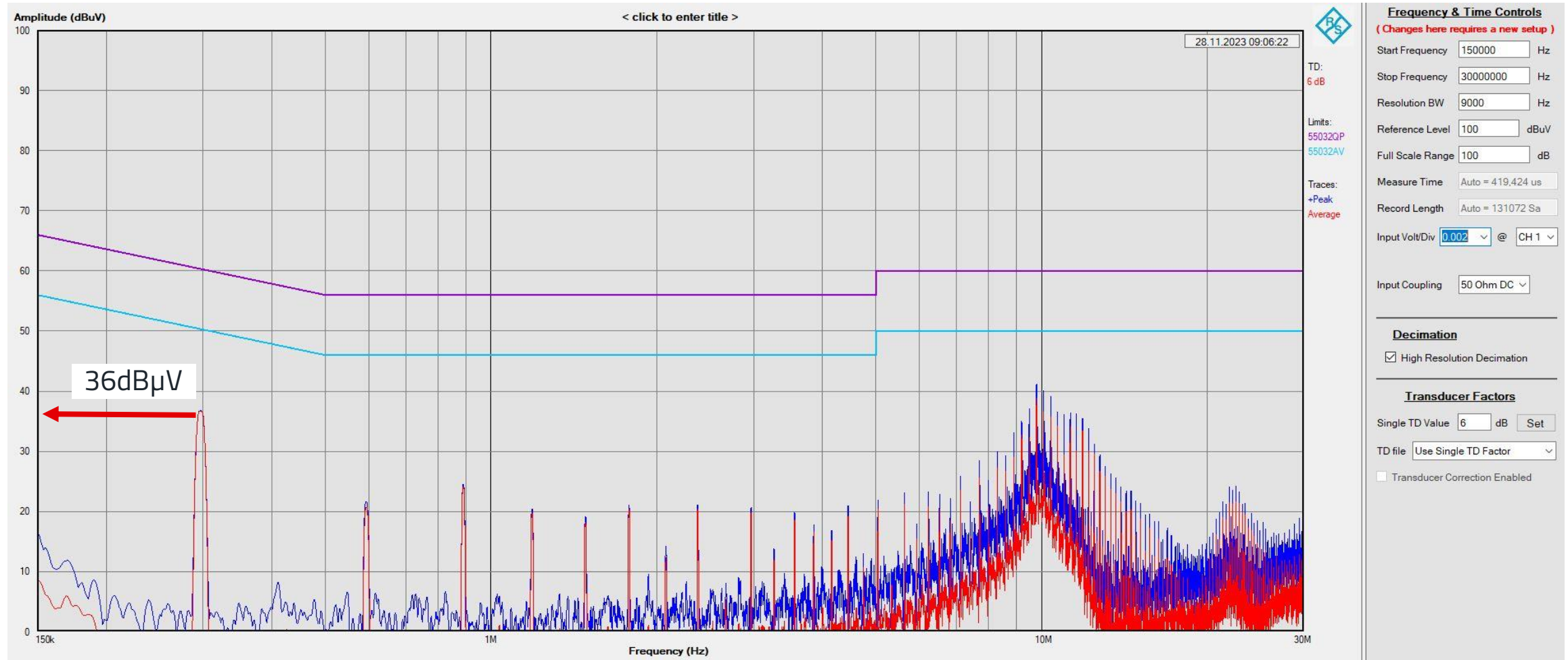
Conducted Spectrum 50uH LISN

DM no filter



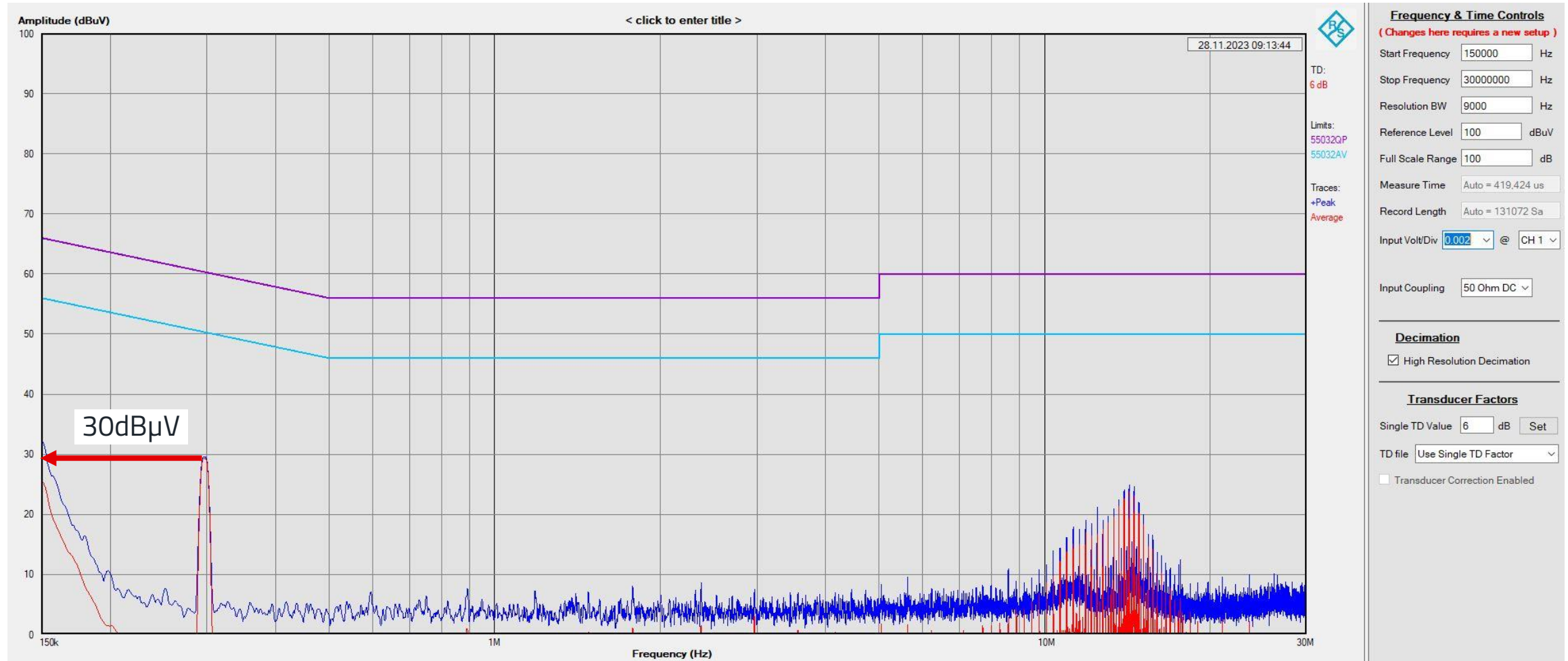
Conducted Spectrum

DM 1-Stage Filter = ~41dB att. @ fsw 300kHz



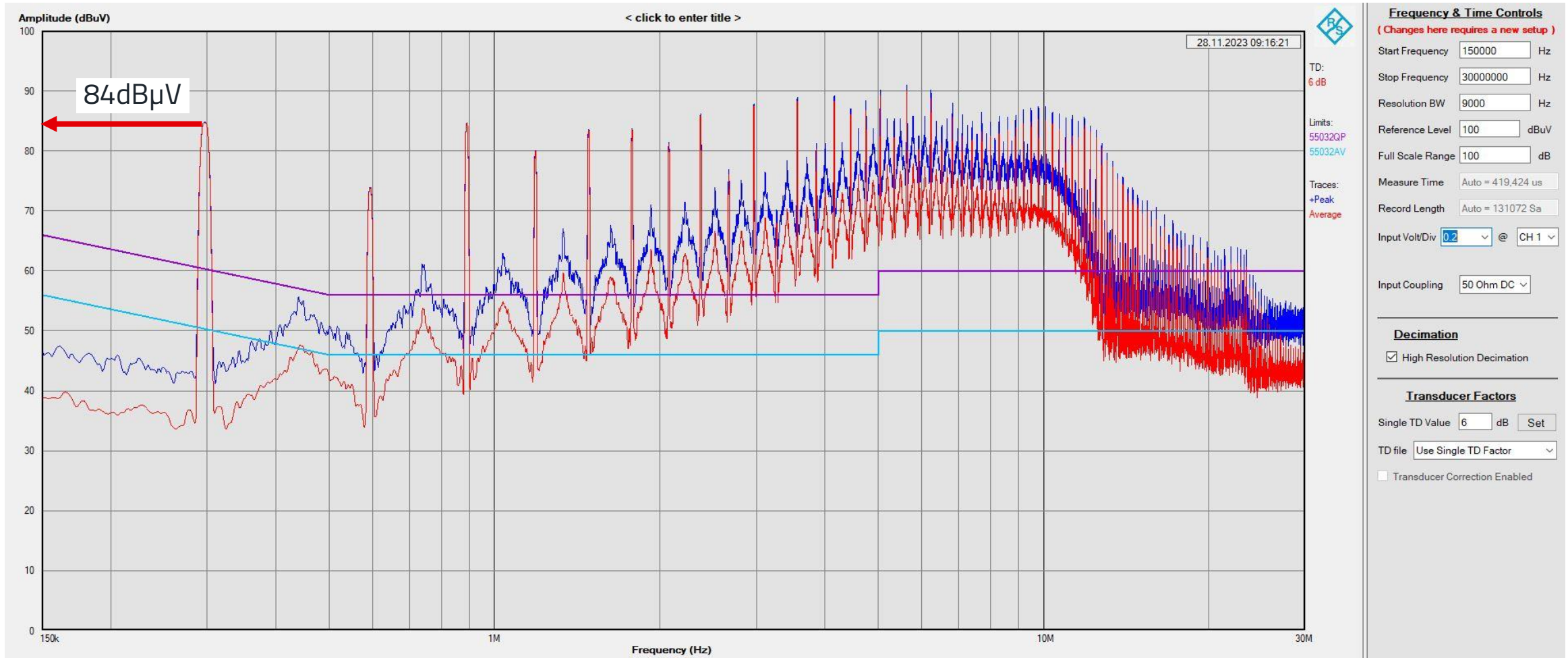
Conducted Spectrum

DM 2-Stage Filter= ~47dB att. @ fsw 300kHz



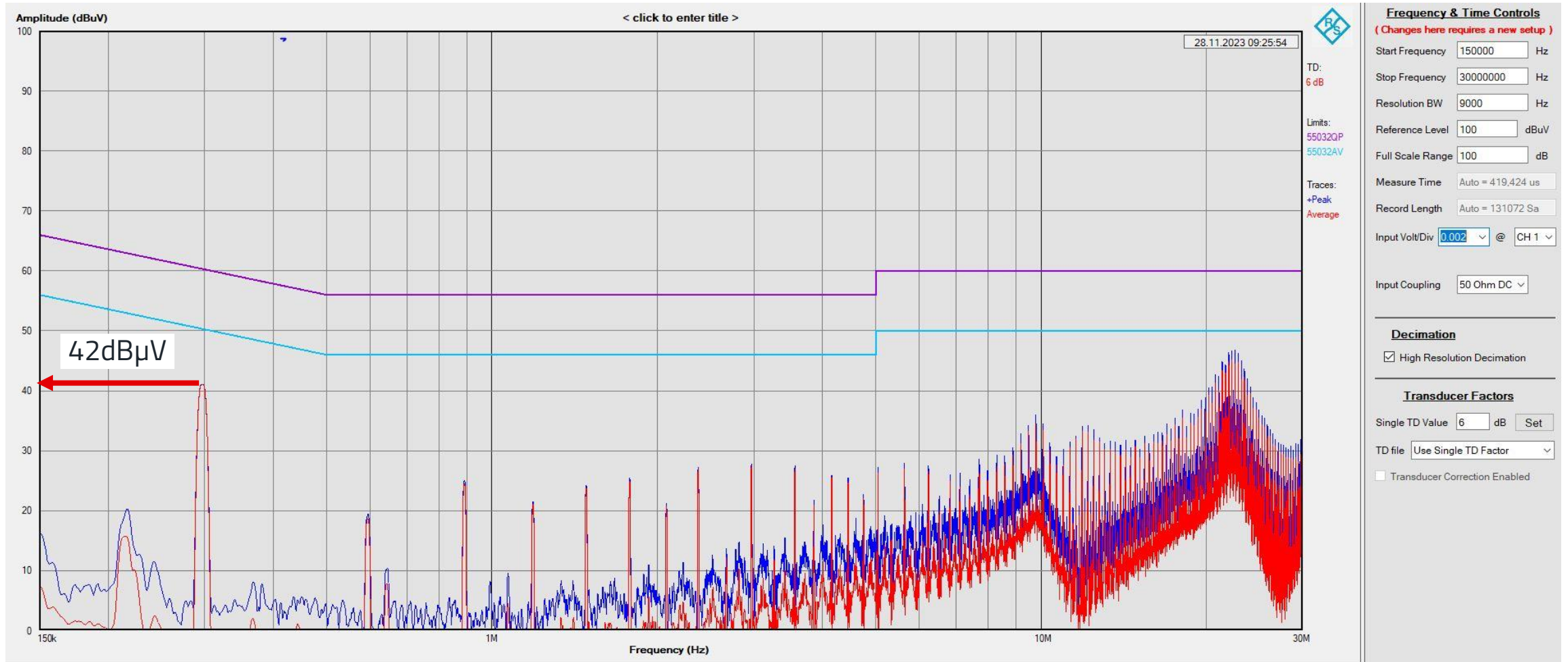
Conducted Spectrum

CM no filter



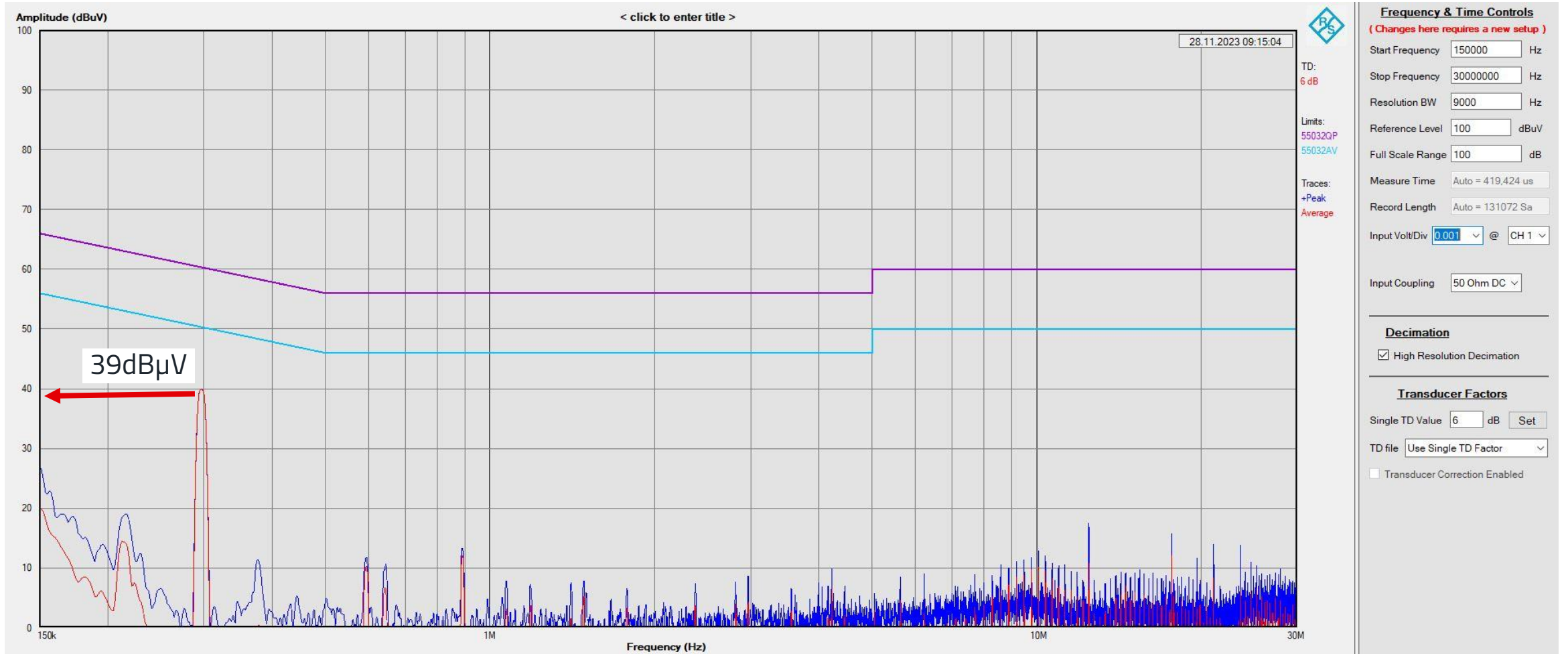
Conducted Spectrum

CM 1-Stage Filter= ~42dB att. @ fsw 300kHz



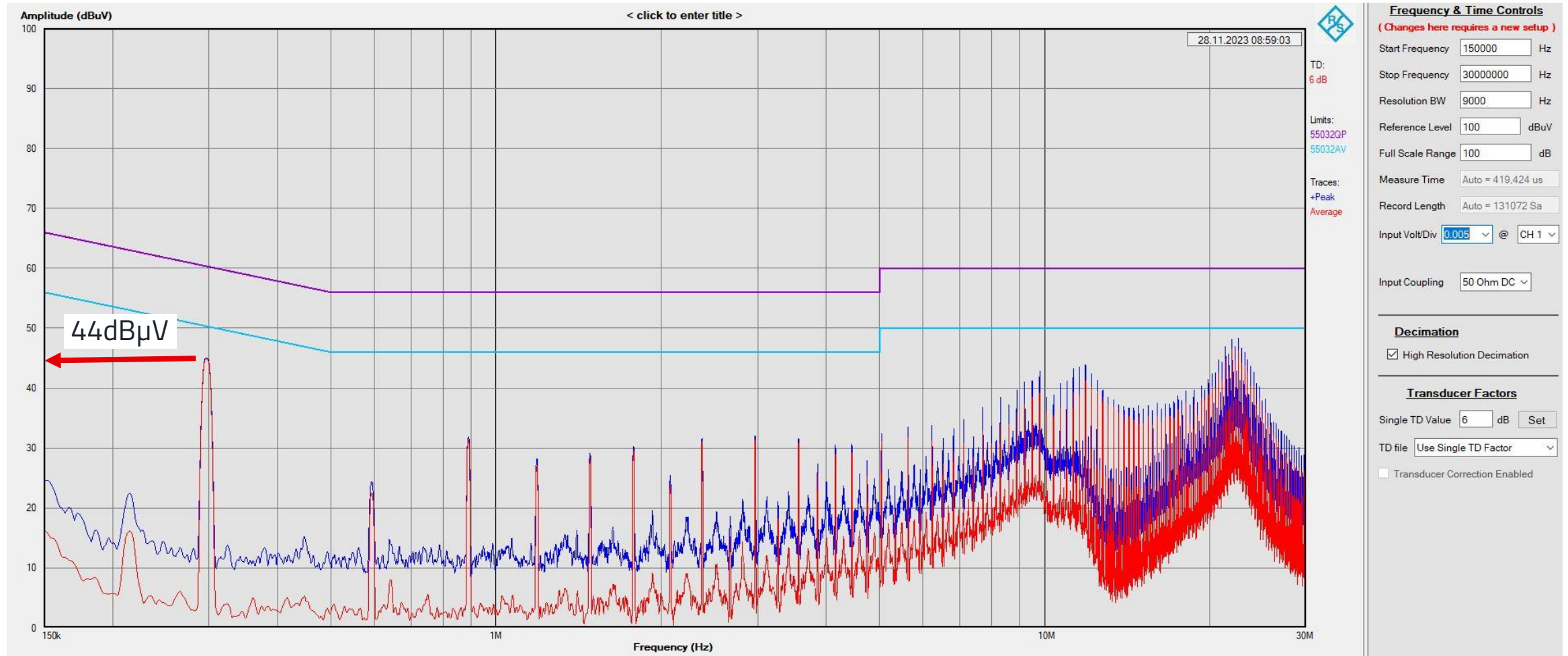
Conducted Spectrum

CM 2-Stage Filter= ~45dB att. @ fsw 300kHz



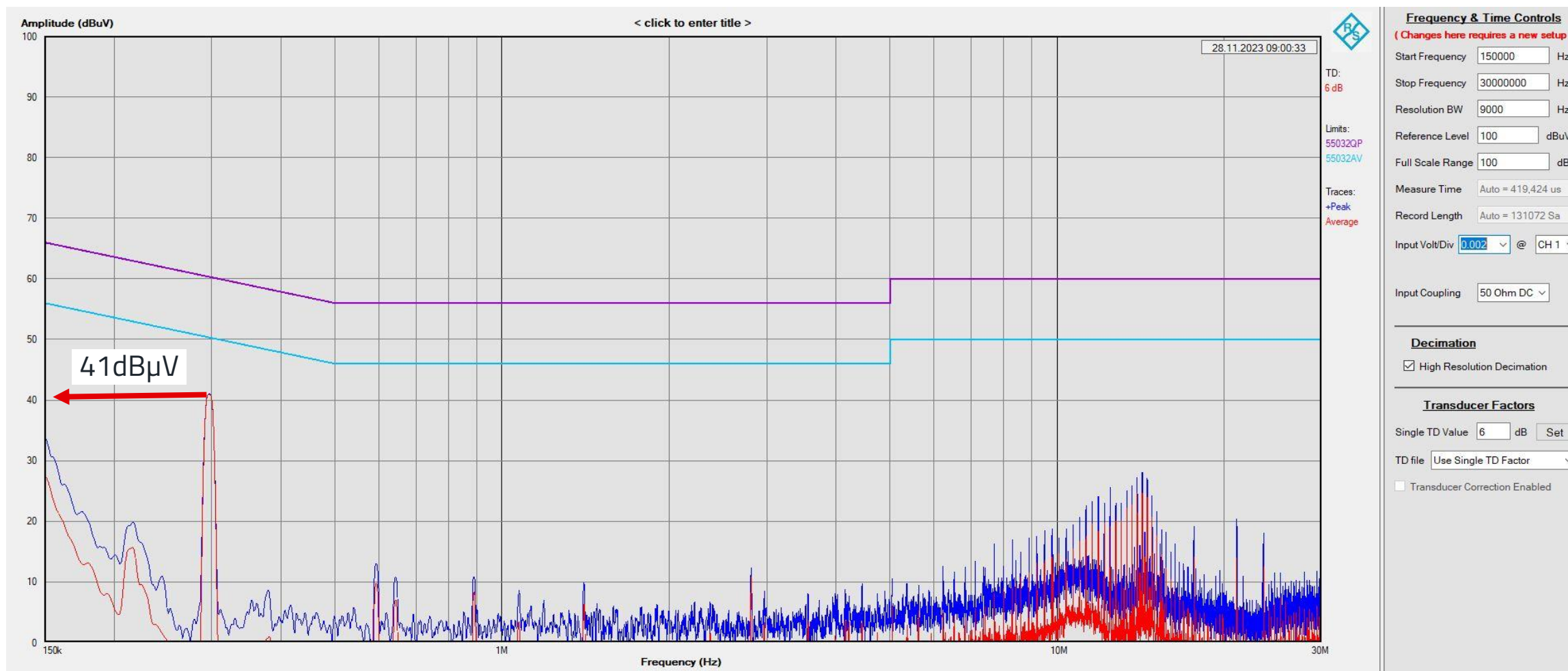
Conducted Spectrum

CM + DM 1-Stage Filter



Conducted Spectrum

CM + DM 2-Stage Filter



Y-CAP LEAKAGE

STANDARDS:

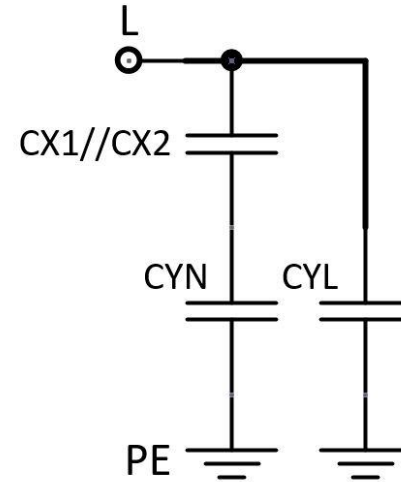
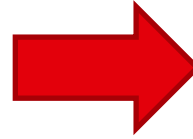
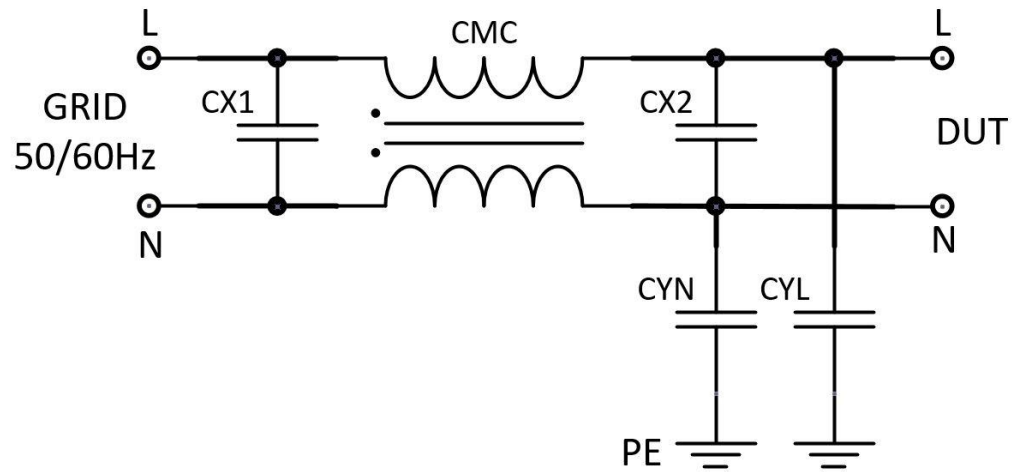
IEC 60950 office equipment

IEC 60601 medical equipment

IEC 60335 domestic equipment

Y-Cap Leakage Currents

1-phase system



Largest leakage current occur:

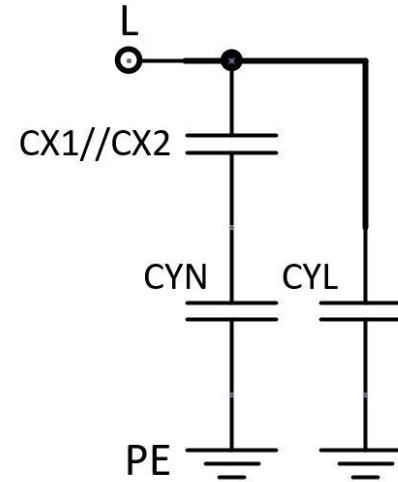
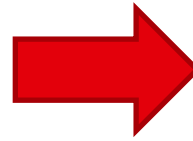
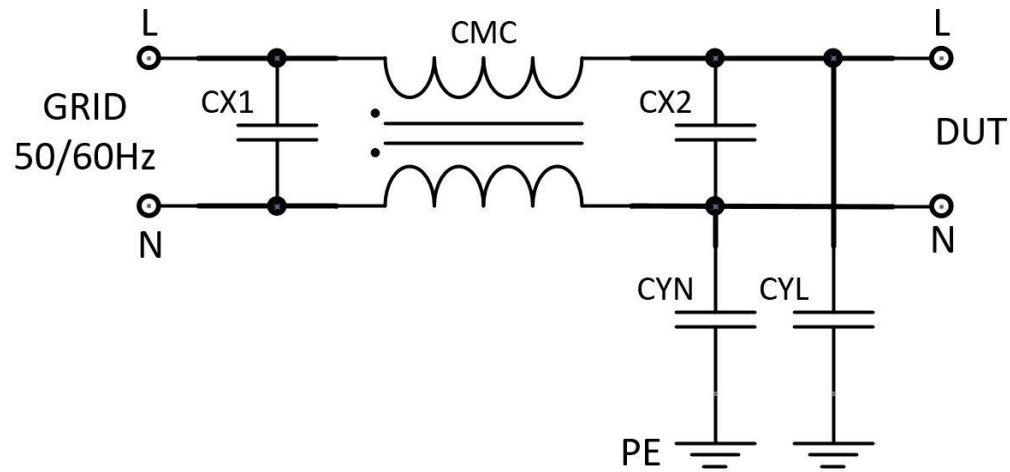
- Grid voltage tolerance +10%
- Grid frequency 60Hz
- Capacitor tolerance +20%

$$I_{leak} = U_{grid} \cdot 2\pi \cdot f_{grid} \cdot \left(C_{YL} + \frac{C_{XG} \cdot C_{YN}}{C_{XG} + C_{YN}} \right)$$

$$C_{XG} = C_{X1} + C_{X2}$$

Y-Cap Leakage Currents

1-phase system



Example: CX1 / CX2 = each 1 μF ; CY1 / CY2 = each 5.6 nF;
253 V / 50 Hz

Worst Case Tolerance: Grid Voltage +10%; Capacitance +20%

$$I_{Leak} = 253 \text{ V} \cdot 2\pi 50 \text{ Hz} \cdot \left(5.6 \text{ nF} + \frac{2.4 \mu\text{F} \cdot 5.6 \text{ nF}}{2.4 \mu\text{F} + 5.6 \text{ nF}} \right)$$
$$= \mathbf{0.89 \text{ mA}}$$

DISCHARGE RESISTOR

Caps Discharge Resistor

Voltage after power plug removed: **60V** in less than **1 second!** (IEC/EN/UL 60950 / 60335 / 62368)

- Total capacitance: $C_{tot} = C_{X1} + C_{X2} + C_{ParaMOV} + \left(\frac{1}{\frac{1}{C_{YL}} + \frac{1}{C_{YN}}} \right)$

- Cap discharge voltage decay: $V = V_{Grid} \cdot e^{-(t/RC)}$

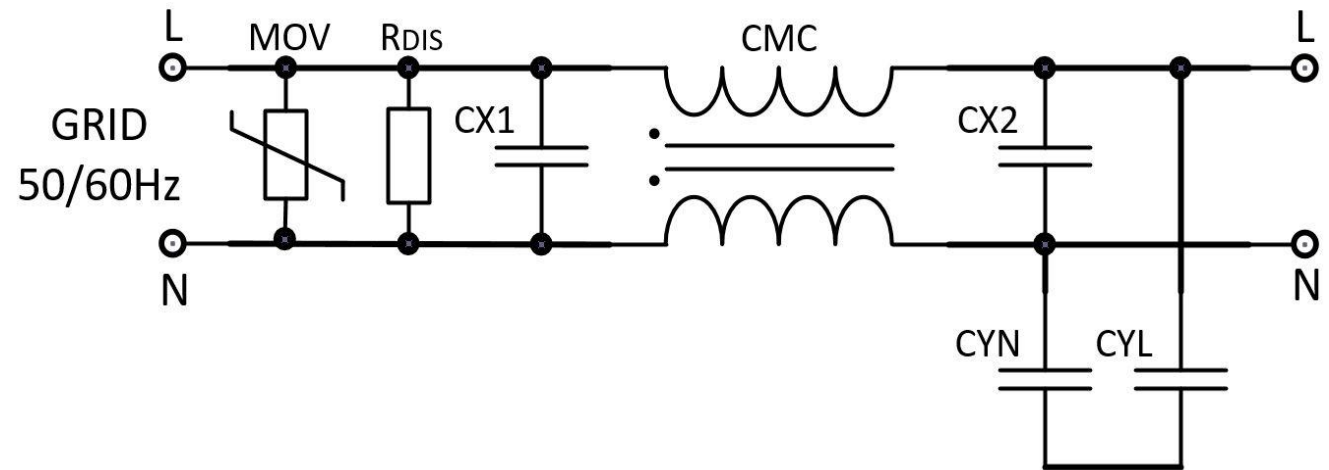
- $V_{Grid} = 325V (= 230V\sqrt{2})$

- $\Rightarrow R_{discharge} = \frac{\left(\frac{-\ln\left(\frac{V_{discharge}}{V_{Grid}}\right)}{t} \right)^{-1}}{C_{tot}}$

- Power rating resistor: $P_v = \frac{V_{Line}^2}{R_{Discharge}}$

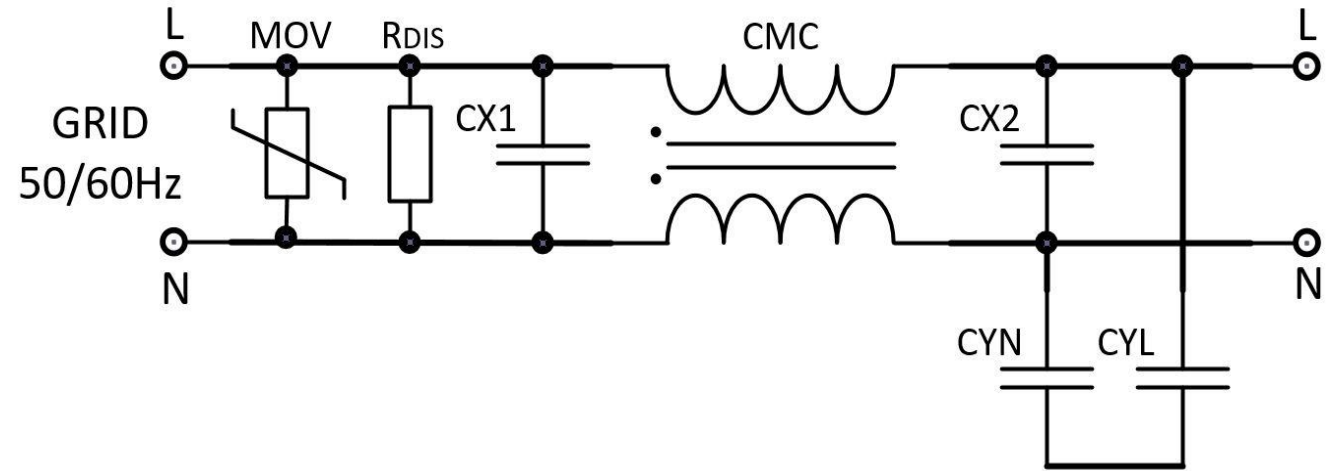
- [Capacitor Safety Discharge Conversion Calculator | DigiKey](#)

- Take care of: creepage, clearance and voltage rating when choosing the discharge resistor!



Caps Discharge Resistor

Voltage after power plug removed: **60V** in less than **1 second!** (IEC/EN/UL 60950 / 60335 / 62368)



Example: $C_{\text{tot}} = 2.1 \mu\text{F} + 20\%$ / $V_{\text{Grid}} = 230 \text{ V} + 10\%$

- $R_{\text{discharge}} < \frac{\left(\frac{-\ln\left(\frac{60\text{V}}{358\text{V}}\right)}{1\text{s}}\right)^{-1}}{2.52 \mu\text{F}} = 222 \text{ k}\Omega \rightarrow \text{Chosen } 220 \text{ k}\Omega$

- $P_v (\text{Discharge Resistor}) = \frac{(253 \text{ V})^2}{240 \text{ k}\Omega} = 267 \text{ mW}$

VARISTOR CALCULATION

Varistor

1. Operating voltage:

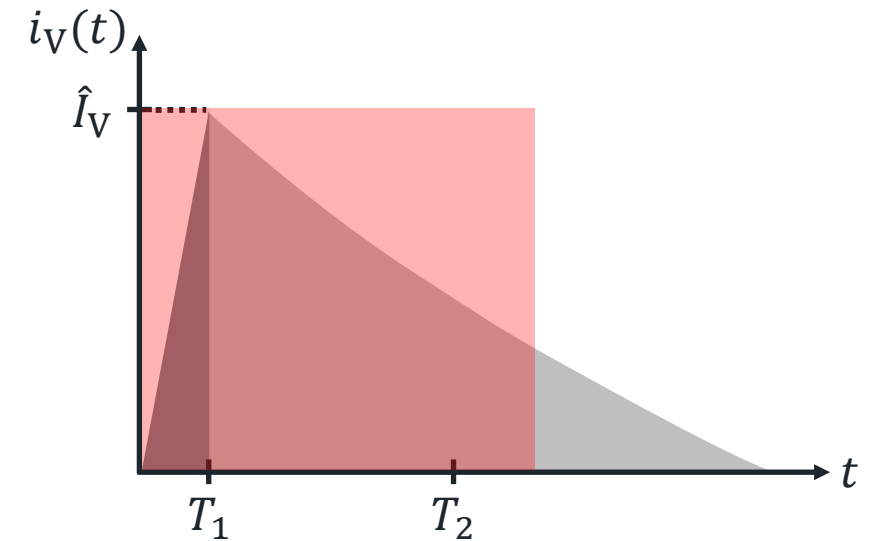
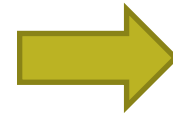
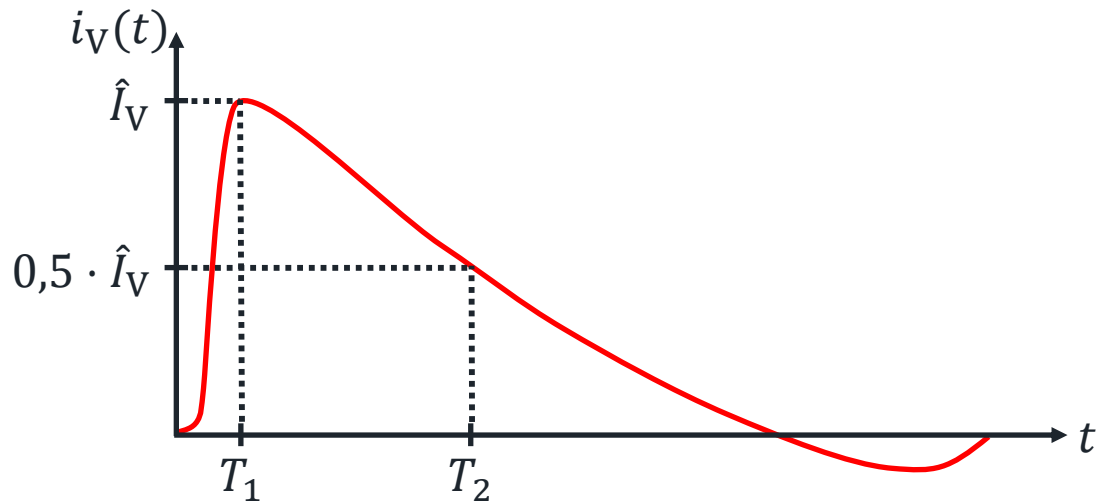
- The operating voltage of the varistor must be greater than the maximum possible supply voltage for the application.
- Consideration of the supply voltage and varistor tolerance
- Example AC voltage supply:
 - $U_{(Q,eff)} = 230V + 10\% + 10\% = 230V \cdot 1,21 = 278,3V$
 - Varistor operating voltage: $U_{rms} = 300V$
 - **Alternative:** 275Vrms type 820542711
- Example DC voltage supply:
 - $U_Q = 24V + 10\% + 10\% = 24V \cdot 1,21 = 29,04V$
 - Varistor operating voltage: $U_{dc} = 31V$

Electrical Properties:

Properties		Test conditions	Value	Unit	Tol.
AC Operating Voltage	V_{RMS}		300	V	max.
DC Operating Voltage	V_{DC}		385	V	max.
Clamping Voltage	V_{Clamp}	50 A @ 8/20 μ s	775	V	max.
(Reverse) Peak Pulse Current	I_{Peak}	8/20 μ s	4500	A	max.
Power Dissipation	P_{Diss}		0.6	W	max.
Energy Absorption	W_{max}	10/1000 μ s	140	J	max.
Nominal Discharge Current	I_n		3	kA	max.
Measured Limiting Voltage	V_{ML}		1200	V	max.
(Reverse) Breakdown Voltage	V_{BR}	1 mA	470	V	$\pm 10\%$
(Channel) Input Capacitance	C_{Ch}	1 kHz	420	pF	typ.

Varistor

2. Pulse energy: Estimation of the energy of a surge pulse in the varistor:



- Equivalent duration of a rectangular pulse with amplitude $\hat{I}_V$:
 - Linearer rise: $t_{eq,1} = 0,5 \cdot T_1$
 - Exponential decrease: $t_{eq,2} = 1,43 \cdot (T_2 - T_1)$
 - Combination: $t_{eq} = t_{eq,1} + t_{eq,2}$

$$W_V = \int_0^{\infty} i_V(t) \cdot u_V(t) dt \sim \hat{I}_V \cdot \hat{U}_V \cdot t_{eq}$$

Varistor

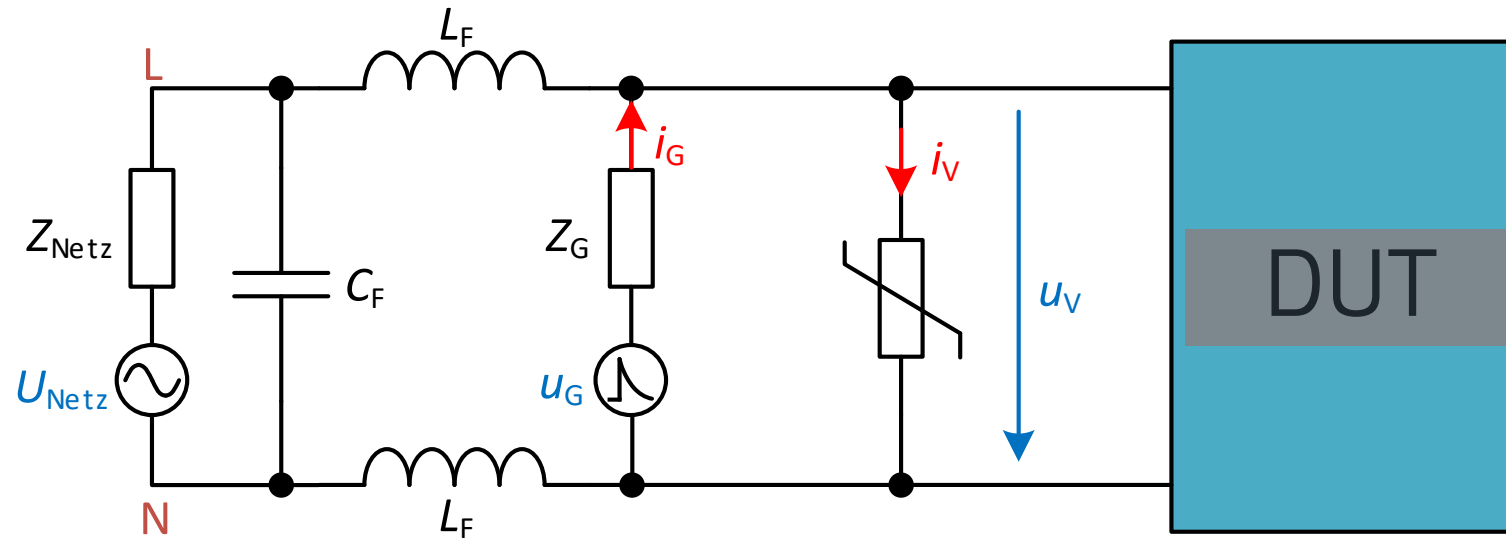
4. Clamping Voltage

- The resulting maximum clamping voltage $U_{(c,max)}$ must be lower than the dielectric strength of the circuit to be protected.
- The data sheet specification is only a comparative value within a series and deviates from the actual clamping voltage.
- The actual clamping voltage can only be determined with the aid of the U-I characteristic curve.
- 14 mm disk varistor WE-VD 820543011

Properties		Test conditions	Value	Unit	Tol.
AC Operating Voltage	V_{RMS}		300	V	
DC Operating Voltage	V_{DC}		385	V	
Clamping Voltage	V_{Clamp}	50 A @ 8/20 μ s	775	V	
(Reverse) Peak Pulse Current	I_{Peak}	8/20 μ s	4500	A	
Power Dissipation	P_{Diss}		0.6	W	
Energy Absobtion	W_{max}	10/1000 μ s	140	J	
Nominal Discharge Current	I_n		3	kA	
Measured Limiting Voltage	V_{ML}		1200	V	
(Reverse) Breakdown Voltage	V_{BR}	1 mA	470	V	$\pm 10\%$
(Channel) Input Capacitance	C_{Ch}	1 kHz	420	pF	

Varistor

- Circuit diagram with test generator between L and N:



Varistor

Typical 1-Phase , **2kV** Surge Test

- Dimensioning example 1:
- Industrial PC with 230 V AC power supply; max. allowed Voltage 1kV
- Surge test according to DIN EN 55024
- Coupling type: symmetrical (phase to neutral conductor)
- Test voltage: $U^*_G=2\text{kV}$
- Coupling impedance (= internal resistance of the test generator): $Z_G=2\Omega$ 1000A max.
- Selection of the varistor:
- Operating voltage taking into account the tolerance: $U_{\text{rms}}=300\text{V}$
- 14 mm disk varistor WE-VD 820543011



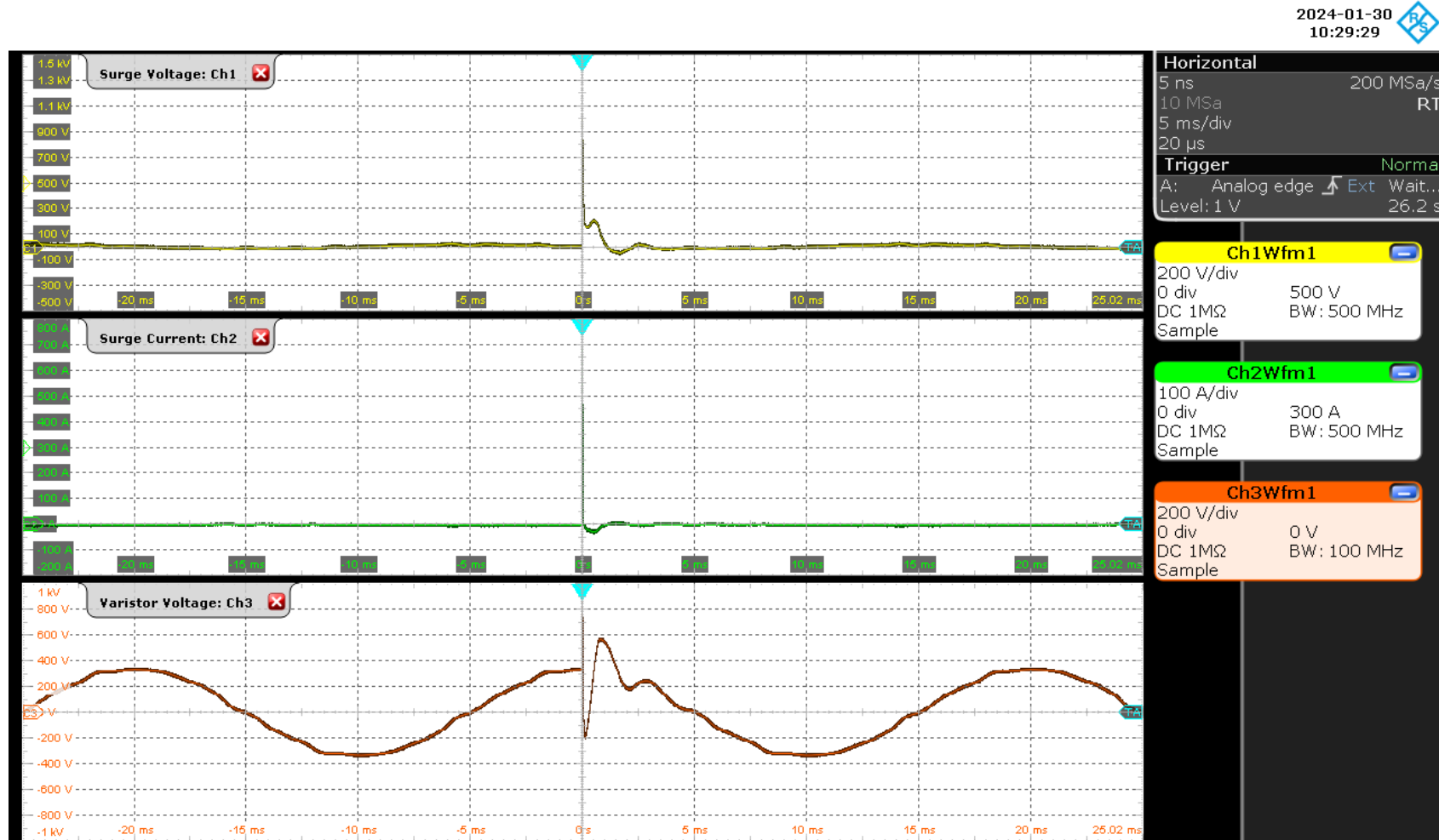
Varistor 1-Phase / 2kV Surge

Determination of operating point Alternative iterative calculation (use Redexpert V/I max. curve for reading)

- **Step 1:** Determination of the maximum clamping voltage for the short-circuit current of the test generator
($\hat{V}_G=2\text{kV} / Z_G = 2\Omega / 1\text{kA} / 230\text{V} \times 1.41$) $\rightarrow I_G = \frac{V_G + V_{Grid}}{Z_G} = \frac{2\text{kV} + 325\text{V}}{2\Omega} = 1163\text{A}$
V/I characteristic curve +10% tolerance: $V_{CALMP_{max}@1163\text{A}} \rightarrow 1.02\text{kV}$
- **Step 2:** Calculation of the new current value through the varistor using the known coupling impedance Z_G and $V_{\text{clamp_max}}$:
 $I_{C1} = \frac{V_G + V_{Grid} - V_{CLAMP_{max}}}{Z_G} = \frac{2\text{kV} + 325\text{V} - 1.02\text{kV}}{2\Omega} = 653\text{A}$
- V/I characteristic curve +10% tol.: $V_{CALMP@653\text{A}} \rightarrow 971\text{V} \checkmark$
- **Step 3:** Determine max. I_{clamp} through varistor using the V/I curve with -10%
- $V_{CALMP_{min}@653\text{A}} \rightarrow 794\text{V}$
- **Step 4:** Calculate the max. current through the varistor: $I_{C_{max}} = \frac{V_G + V_{Grid} - V_{CLAMP_{min}}}{Z_G} = \frac{2\text{kV} + 325\text{V} - 794\text{V}}{2\Omega} = 766\text{A}$

Lab Measurement Overview

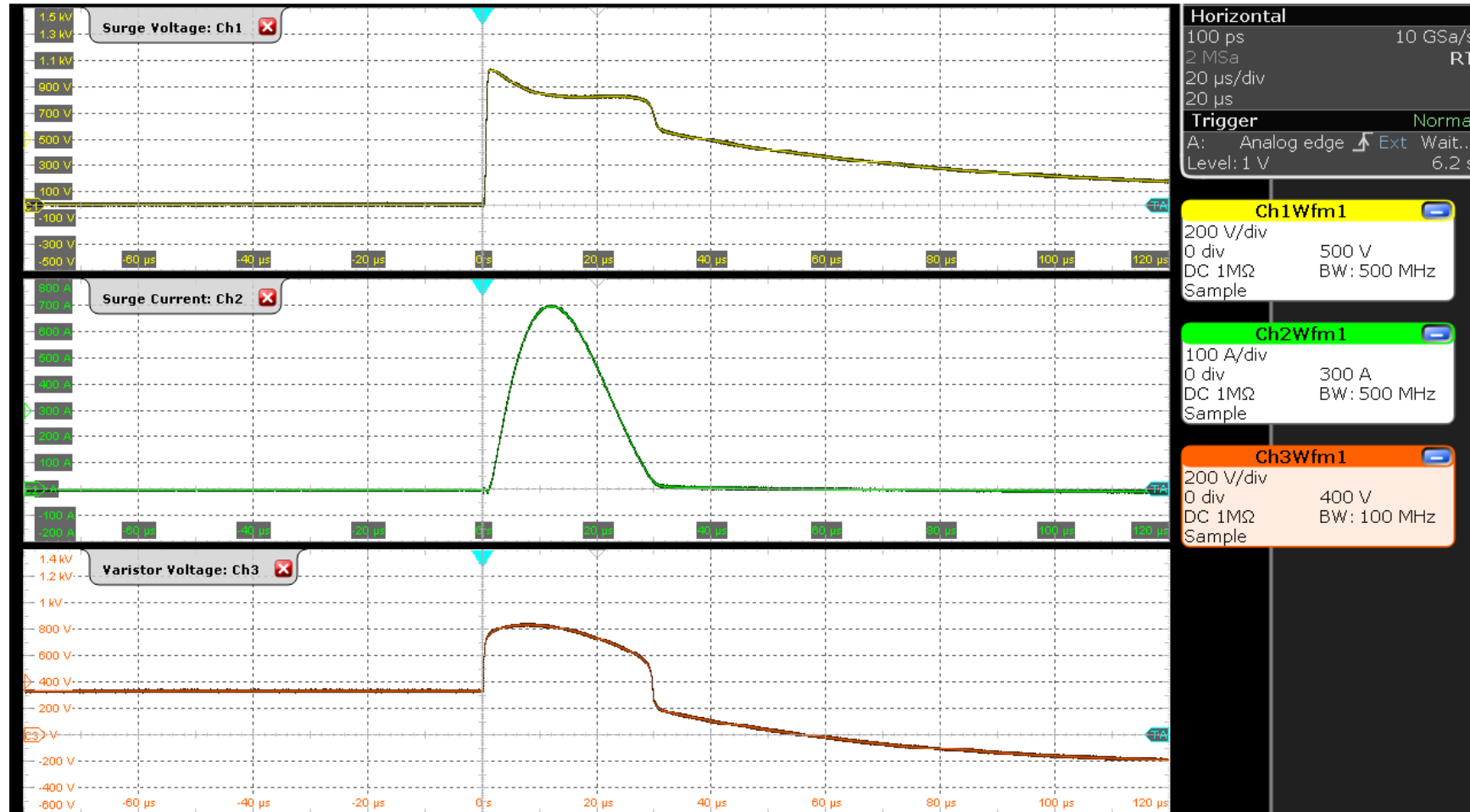
Varistor @ 2kV surge at ohmic load (lightbulb)



Lab Measurement Overview Zoom

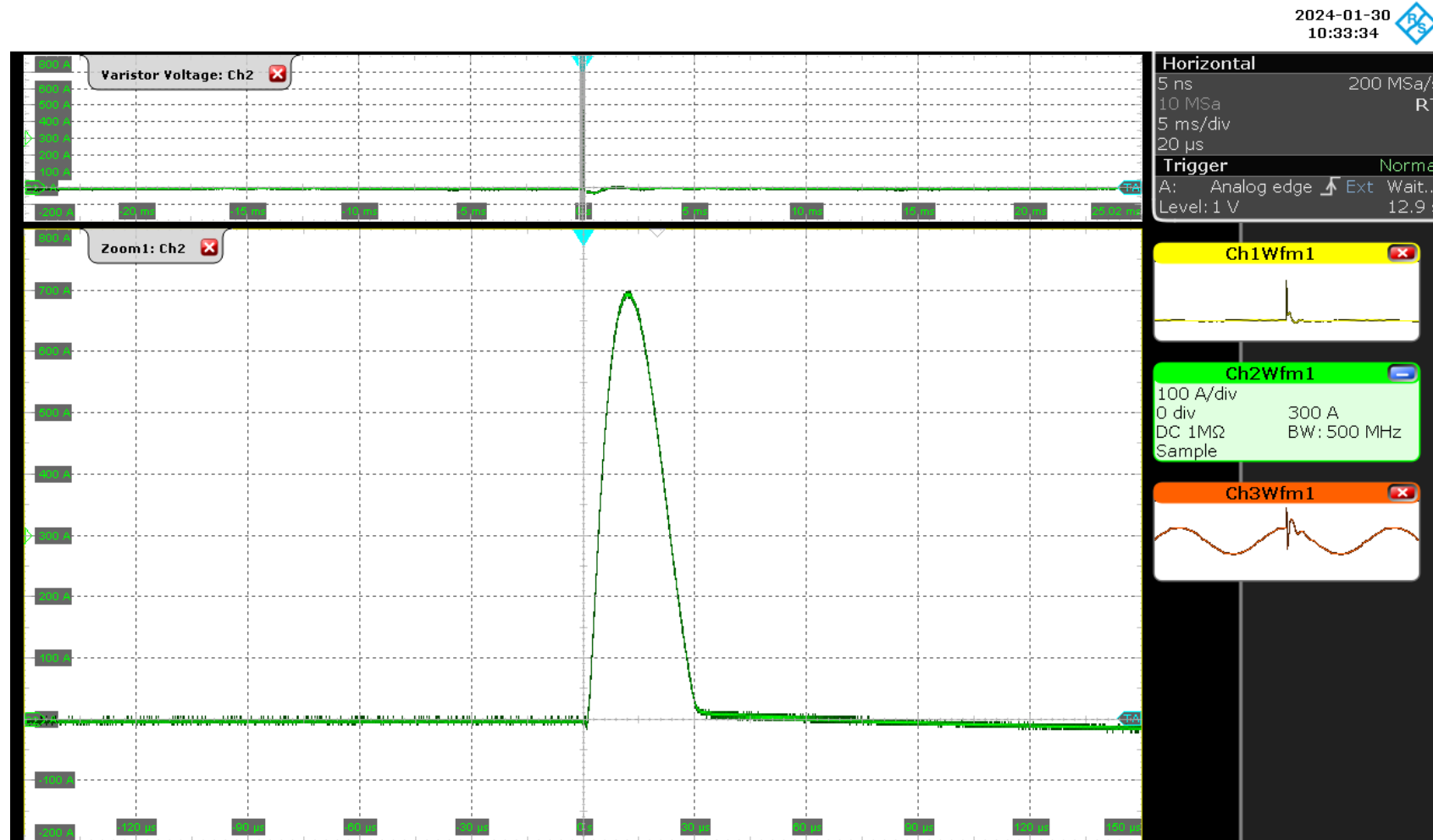
Varistor @ 2kV surge at ohmic load (lightbulb)

2024-01-30
10:23:16



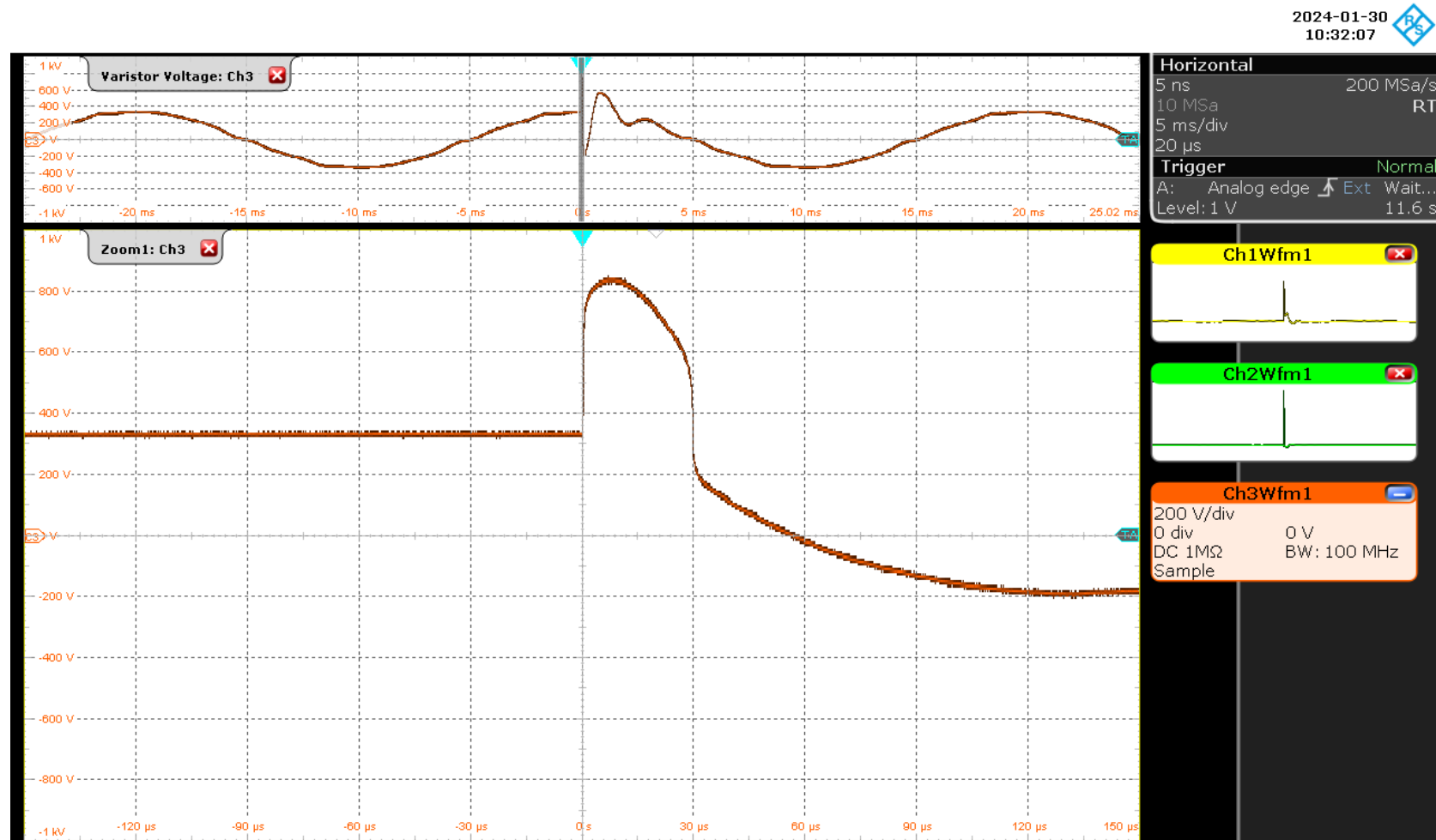
Lab Measurement: Zoom Varistorcurrent

Varistor @ 2kV surge at ohmic load (lightbulb) -> 700A max.



Lab Measurement: Zoom Varistorvoltage

Varistor @ 2kV surge at ohmic load (lightbulb) -> 850V max.



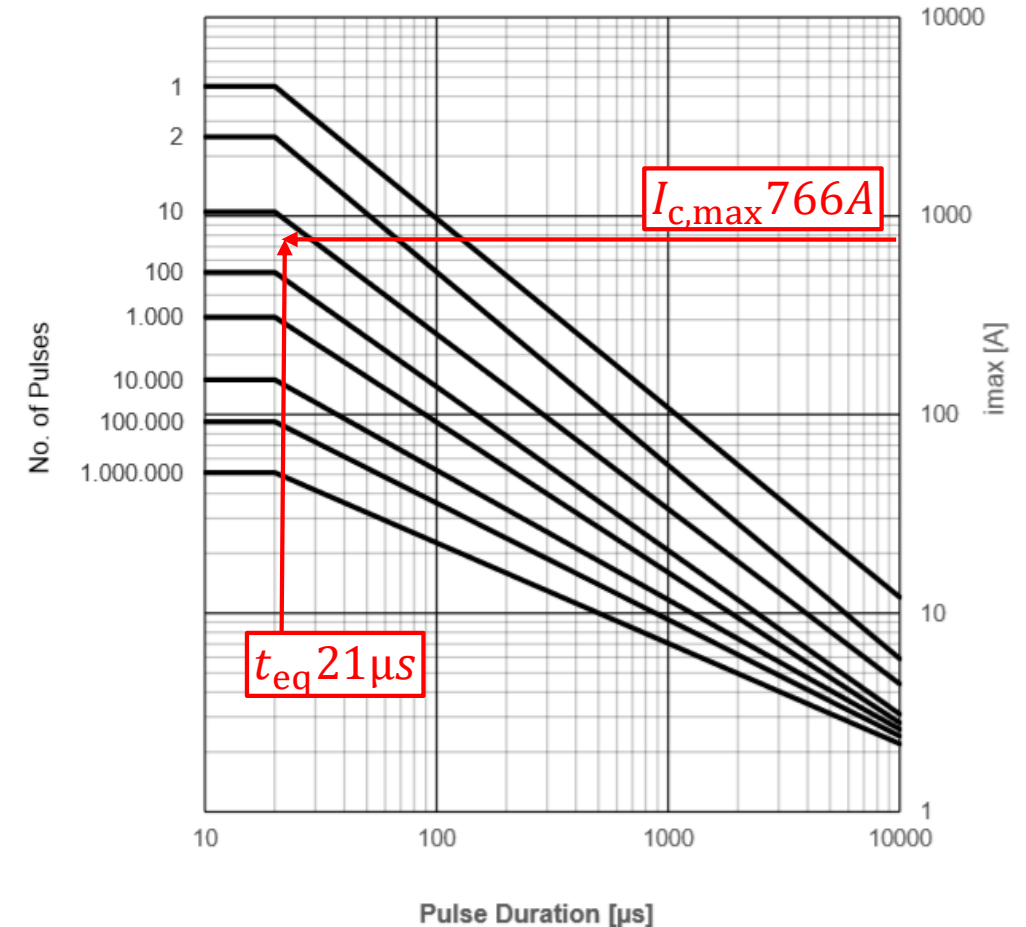
Varistor

Typical 1-Phase, 2kV Surge Test

- Check how many 8/20μs current pulses the varistor can withstand
- Determining the equivalent width of the square-wave current pulse

$$\begin{aligned}t_{eq,8/20} &= 0,5 \cdot T_1 + 1,43 \cdot (T_2 - T_1) \\&= 0,5 \cdot 8\mu s + 1,43 \cdot 12\mu s \\&= 21,16\mu s\end{aligned}$$

- The varistor can withstand approx. 30 @ 21μs pulses with 766A
- What is the maximum energy occurring in the varistor during the test? → 40 pulses → 14mm **High Surge** alternative WE-VD 820443011E (500 pulses @ 21μs @ 766A)



Varistor

Typical 1-Phase, 2kV Surge Test

- Maximum energy occurring in the varistor:

$$W_V = I_{\text{clamp,max}} \cdot U_{\text{clamp,min}} \cdot t_{\text{eq,8/20}} = 766\text{A} \cdot 794\text{V} \cdot 21,16\mu\text{s} = 12.9\text{Ws}$$

- Datasheet: $W_{\text{max}} = 140\text{J}$ ✓
- The maximum power loss is specified as $P_{\text{max}}=600\text{mW}$. The minimum necessary recovery time (cooling of the component) is therefore given as:

$$\bullet T > \frac{W_V}{P_{\text{max}}} = \frac{12.9\text{Ws}}{0,6\text{W}} = 21,4\text{s} \checkmark$$

- The basic standard DIN EN 61000-4-5 requires a maximum recovery time of 60 seconds. The varistor fulfills this criterion.

Varistor – Beispiel

- If the maximum clamping voltage $U_{(c,max)}=971V$ exceeds the dielectric strength of the circuit to be protected, a varistor with a larger diameter must be selected, e.g. WE-VD 820523011 ($D=20mm$). The current density is then lower and thus the voltage drop across the component.
- Why should a disk varistor with $D<14mm$ not be used for mains-powered devices?



- The energy compatibility is significantly lower than with a 14 mm disk varistor.
- The maximum clamping voltage is higher for the same operating voltage.
- For the same pulse current, the number of permitted overvoltage events and therefore the service life is significantly lower.

TVSP CALCULATION FOR DC SUPPLY

TVSP Calculation

Typical DC Supply – **500V** / **2Ω** Surge Test

- 24VDC supply rail
- Protection level max. 40VDC
- Chosen TVSP: 824550241 / 3000W

- $$I_C = \frac{\hat{V}_G - V_{C,max}(\hat{I}_{G,sc})}{Z_G} = \frac{500V - 38,9V}{2\Omega} = 231A$$



Electrical Properties:

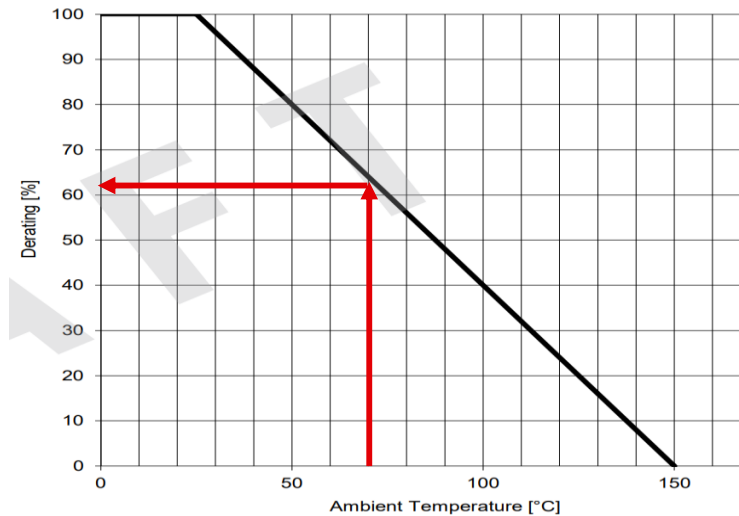
Properties		Test conditions	Value	Unit	Tol.
DC Operating Voltage	V _{DC}		24	V	max.
(Reverse) Breakdown Voltage	V _{BR}	1 mA	28.1	V	±5%
Clamping Voltage	V _{Clamp}	I _{PEAK}	38.9	V	max.
(Reverse) Peak Pulse Current	I _{Peak}	10/1000 μs	77.1	A	max.
(Forward) Peak Pulse Current ¹⁾	I _{Peak}		300	A	max.
Leakage Current	I _{Leak}	V _{DC}	2	μA	max.
Steady State Power Dissipation	P _{Diss}	T _A = 50 °C	6.5	W	max.
Power Dissipation ²⁾	P _{Diss}	10/1000 μs	3000	W	max.
Polarity	Unidirectional				

TVSP Calculation

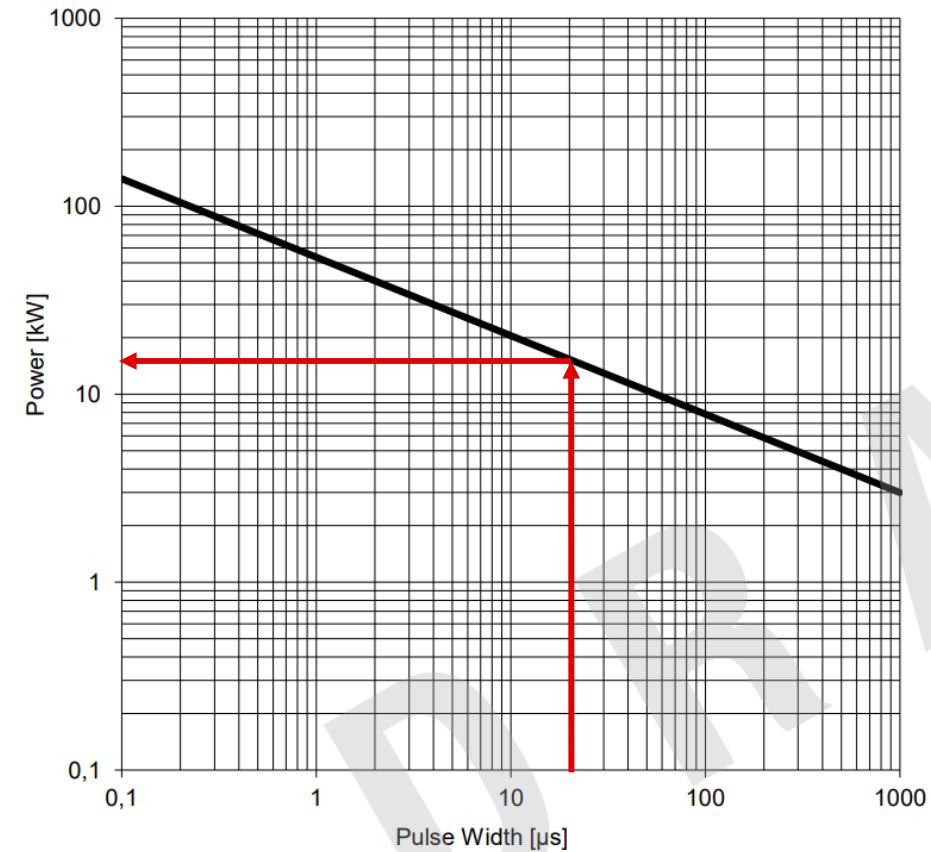
Typical DC Supply – 500V / 2Ω Surge Test

- $P_{peak} 25^{\circ}C @ 21\mu s = 18kW$
- $P_{peak} 70^{\circ}C @ 21\mu s = 0,62 \cdot 18kW = 11,2kW$
- $P_{max21\mu s} = 38,9V \cdot 231A = 8,99kW$

Peak Pulse Power and Current Derating:



Maximum Peak Pulse Power Rating:

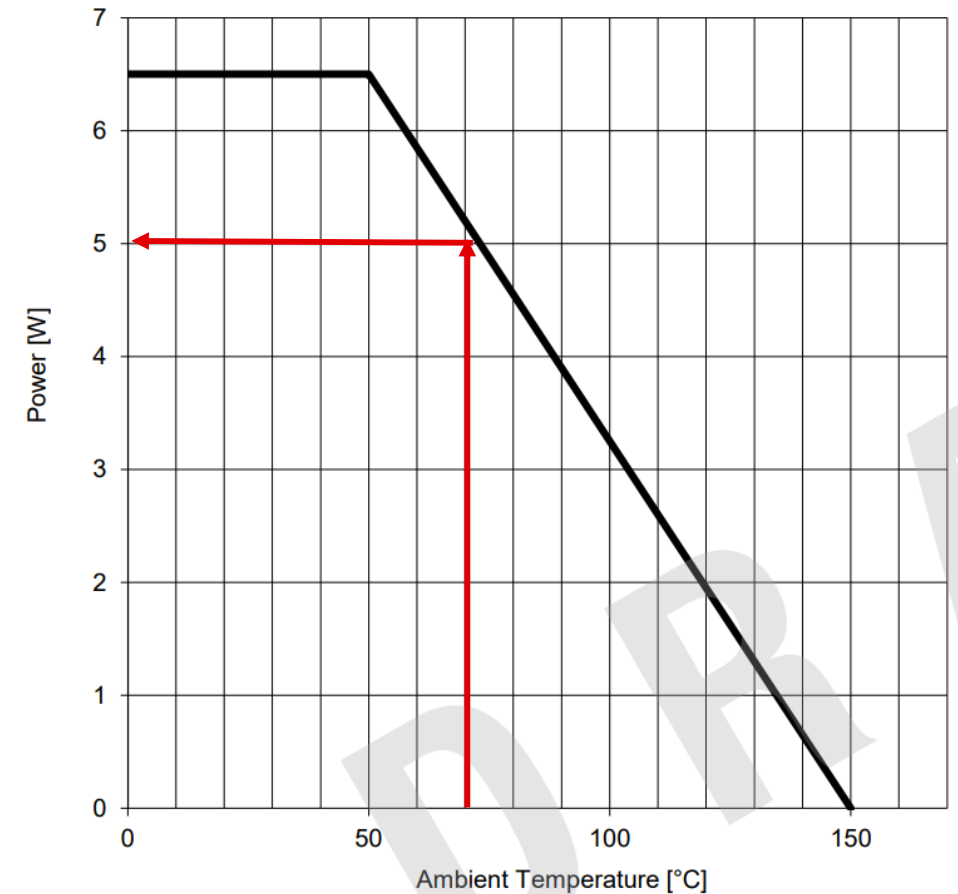


TVSP Calculation

Typical DC Supply – 500V / 2Ω Surge Test

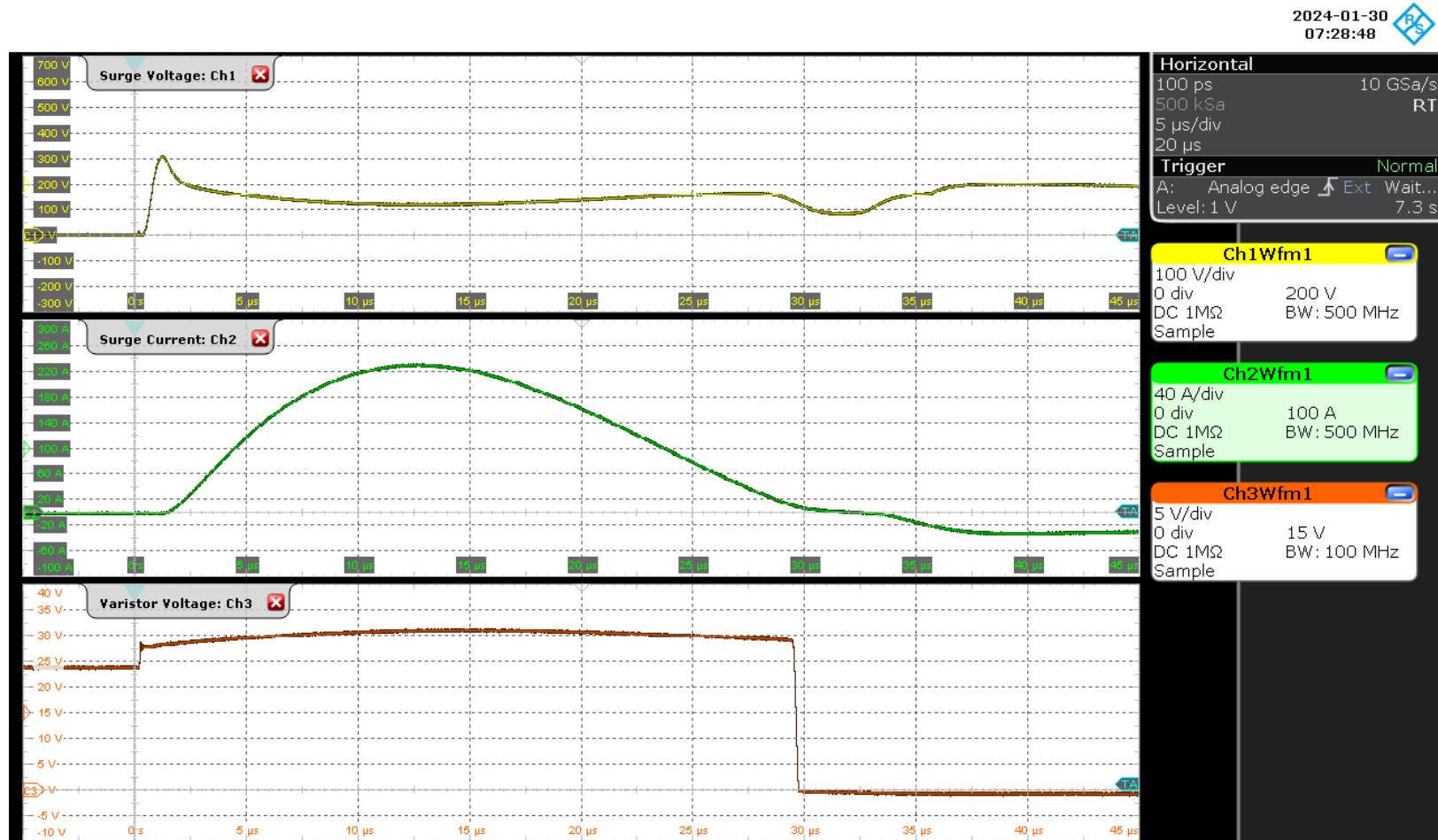
- $P_{Vmax}@70^{\circ}C = 5W$
- $W_V = I_c \cdot U_c \cdot t_{eq8,20\mu s} = 231A \cdot 38,9V \cdot 21,16\mu s = 0,19Ws$
- $T > \frac{W_V}{P_{Vmax}} = \frac{0,19Ws}{5W} = 0,04s$
- Surge puls repetition = 60s, so 0,04s recovery time is sufficient

Steady State Power Dissipation Derating:



TVSP Measurement 24VDC

500V / 2 Ω Surge Test with Ohmic Load $\rightarrow$ 230A_clamp max. / 31V_clamp max.



TVSP Calculation

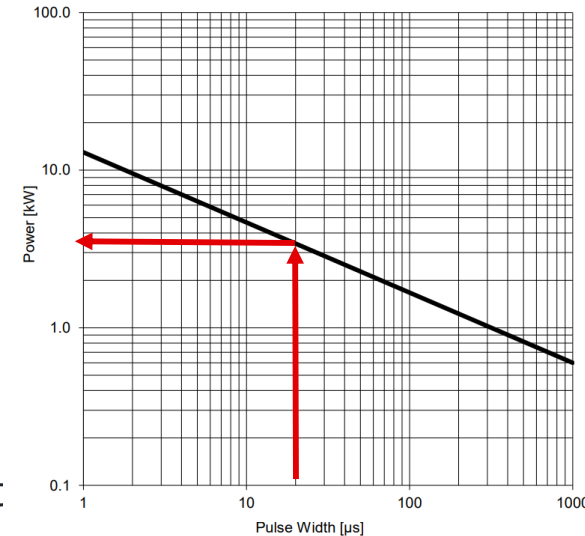
Typical DC Supply – **500V** / **42Ω** Surge Test

- TVSP **600W** 824520241
- $$I_c = \frac{\hat{V}_G - V_{c,max}(\hat{I}_{G,sc})}{Z_G} = \frac{500V - 38,9V}{42\Omega} = 11A$$
- $$P_{peak} 25^\circ C @ 21\mu s = 3.5kW$$
- $$P_{peak} 70^\circ C @ 21\mu s = 0,62 \cdot 3.5kW = 2.17kW$$
- $$P_{max21\mu s} = 38,9V \cdot 11A = 0.43kW$$
- $$P_{Vmax} @ 70^\circ C = 4W$$
- $$W_V = I_c \cdot U_c \cdot t_{eq8,20\mu s} = 11A \cdot 38,9V \cdot 21,16\mu s = 0,009Ws$$
- $$T > \frac{W_V}{P_{Vmax}} = \frac{0,009Ws}{4W} = 0,002s$$
- Surge puls repetition = 60s, so 0,002s recovery time is sufficient

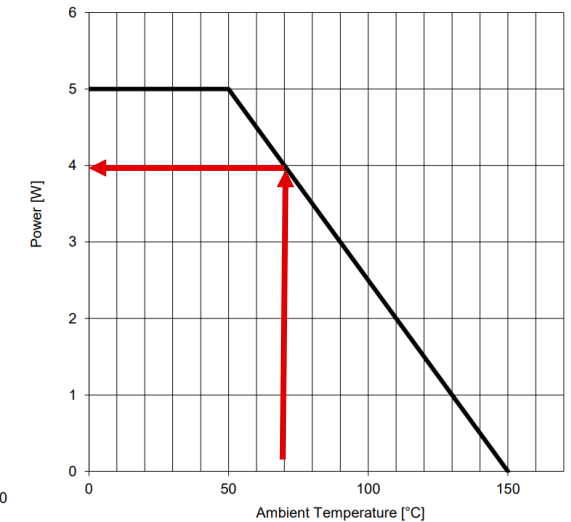
Electrical Properties:

Properties		Test conditions	Value	Unit	Tol.
DC Operating Voltage	V_{DC}		24	V	max.
(Reverse) Breakdown Voltage	V_{BR}	1 mA	28.1	V	±5%
Clamping Voltage	V_{Clamp}	I_{PEAK}	38.9	V	max.
(Reverse) Peak Pulse Current	I_{Peak}	10/1000 μs	15.5	A	max.
(Forward) Peak Pulse Current 1)	I_{Peak}		100	A	max.
Leakage Current	I_{Leak}	V_{DC}	1	μA	max.
Steady State Power Dissipation	P_{Diss}	$T_A = 50^\circ C$	5	W	max.
Power Dissipation 2)	P_{Diss}	10/1000 μs	600	W	max.
Polarity	Unidirectional				

Maximum Peak Pulse Power Rating:



Steady State Power Dissipation Derating:



Filter Design Kit 744998

Contains: PCBs, connectors, CMCs, X/Y caps, spacers, discharge resistors



